

COMPREHENSIVE NETWORK SECURITY DESIGN AND IMPLEMENTATION FOR AVALANCH COOPERATIONS

Prepared by
Yesmin Jalaldeen



Executive Summary

This report is a detailed plan and execution of a secure hierarchical network infrastructure to be used by the new branch of the Avalanche Cooperations in Negombo, Sri Lanka. The project was aimed at linking 358 hosts in eight different departments with the Main Branch through a secure Site-to-Site IPSec VPN, by which data confidentiality and integrity would be ensured. The network design is based on the high availability, scalability, and high security with the usage of redundancy in multilayer switches, rigorous VLAN segmentation and Access Control Lists (ACLs) to enforce isolation of sensitive Finance and HR activities. The integration of centralized management services was done, such as AAA, DHCP, and Syslog, so that the administration becomes easier and the same security policies apply across the organization. The last design addresses the need to achieve 99.9% uptime and offers a scale base to expand in the future.

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1. Introduction

Avalanch Cooperations is a technology services provider that deals with IT consulting, network design and deployment, security solutions, cloud services, and network monitoring, and has recently opened a new branch in Negombo, Sri Lanka. The new location will require the implementation of efficient and effective network infrastructure that will be able to sustain eight different departments with their respective operation needs [2].

1.1. Project Objectives

1. To develop the hierarchical network infrastructure to support both the present and future growth.
2. To introduce effective IP address schemes based on VLSM to maximize the use of address spaces.
3. To implement safe connectivity amid the chief branch and the new one using VPN technology.
4. To implement the segmentation of the network through VLANs to improve security and performance.
5. To centralize network services, such as DHCP, NTP, Syslog and AAA.
6. To implement security policy using ACLs, SSH, and port-security.
7. To provide high availability using redundancy.

1.2. Scope of Work

- Network topology design and documentation
- VLSM-based IP addressing scheme for efficient address allocation
- Cisco Packet Tracer implementation of the complete network infrastructure
- Configuration of eight departmental VLANs with appropriate security controls
- VPN tunnel establishment for secure inter-branch communication
- Implementation of routing protocols for connectivity
- Security mechanism deployment (ACLs, SSH, port security)
- Comprehensive testing and verification procedures
- Detailed documentation of all configurations and implementations

1.3. Organizational Context

Negombo branch should have 358 hosts serviced by eight departments, each with different security need. Finance and HR department handle very sensitive information and hence require insulation with other departments and yet need to be connected to other vital services [3]. The network design should thus be able to attain scalability, reliability and security without interfering with the optimum performance.

2. Network Requirements Analysis

2.1. Departmental Requirements

The new branch has eight departments, which have unique host requirements and security needs.

Department	Host Count	Primary Function	Security Level
WA1	130	Workgroup Area 1	Standard
WA2	72	Workgroup Area 2	Standard
IT	60	Technology Infrastructure	Elevated
C&C	35	Command & Control	Standard
Finance	18	Financial Operations	High (Isolated)
HR	18	Human Resources	High (Isolated)
SOC	15	Security Operations Center	Elevated
Admin	10	Administrative Functions	Standard

Table 1: Department Distribution of Avalanche Negombo Branch

2.2. Security Requirements

The following security requirements have been determined according to best practice policies in the organization and the industry [4]

- 1. Network Segmentation:** The different departments need to work with disconnected broadcast domains (VLANs) to restrict the attack surface and restrict a possible security attack.
- 2. Access Control:** The Finance and HR departments need to be highly isolated ensuring that
 - No inter-departmental communication.
 - No contact with other departments.

- Only the main-branch and internet access.
- 3. Secure Communication:** VPN technology should be used to encrypt all the inter-branch communications with a view of ensuring confidentiality and integrity of data as it passes [5].
- 4. Authentication:** To control administrative access, administrative access must be authenticated centrally with the help of AAA servers.
- 5. Physical Security:** Access switches have physical security that is used to prevent unauthorized connections to devices.
- 6. Security:** SSH; all SSH access is restricted to security reasons [6].

2.3. Connectivity Requirements

- 1. Inter Branch Connectivity:** There needs to be a secure and reliable connection between the main branch (192.168.1.0/24) and the new branch (10.10.0.0/16).
- 2. Internet Access:** The internet access will be controlled by allowing all the departments to use the main-branch router.
- 3. Service Access:** The departments of the new branch should be able to access centralized services (AAA, DHCP, NTP, Syslog) that are hosted in the primary branch.
- 4. Inter-VLAN Routing:** Communication between departments should be provided in a controlled way with exceptions being the restriction applied to Finance and HR.
- 5. Redundancy:** There should also be backup connections and redundancy paths so that the availability is 99.9% [7].

2.4. Performance Requirements

According to organizational criteria and industry standards:

- **Latency:** Inter-branch communication with a latency of less than 50ms.
- **Availability:** 99.9% uptime, which is up to 8.76 hours of downtime per year.
- **Throughput:** 500Mbps to desktop connections with 1Gbps backbone.
- **Scalability:** 50% growth (2 more hosts) without significant redesign.

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3. Network Design and Architecture

3.1. Topology Selection and Justification

3.1.1. Hierarchical Network Model

The network follows the hierarchical three layers design model of cisco with its hierarchical structure providing a modularity, scalability, and simplified troubleshooting [8].

This architecture is made up of:

1. Core Layer:

- High-speed packet switching and routing;
- branch router and main branch router provide WAN connectivity;
- inter-distribution layer connectivity;
- no security policies or packet manipulation to satisfy the best performance.

2. Distribution Layer:

- Main Branch: single multilayer switch (Cisco 3560) to offer core switching and connection to servers;
- New Branch: two multilayer switches (Cisco 3560) - Neg-Core-MLS01 and Neg-Core-MLS02 - to support inter-VLAN routing and redundancy;
- HSRP/VRRP set up between the two core switches to provide first-hop redundancy;
- Policy-based connectivity control
- Access layer switches include the access layer connect to the core switch.

3. Access Layer:

- End user connectivity through access switches (Cisco 2960);
- 8 departmental switches within her new branch (one switch per

department);

- VLAN tagging and port security;
- optional Quality of Service (QoS) tagging;
- Power over Ethernet (POE) option of expansion.

3.1.2. Improved Redundancy Design in Negombo Branch:

The new branch adopts dual-core architecture with two multilayer switches in active standby configuration.

Primary Core Switch (NegcoreMLS01):

- Routes all inter-VLAN routing traffic in the active mode
- HSRP priority 110 (active router)
- Connected to all eight access switches through trunk connections
- Connected with the second core switch through Gi0/22 -24 on redundancy basis.

Secondary Core Switch (Neg -Core -MLS02):

- Standby mode can be instantly failed over
- HSRP priority 90 (meaning the standby mode)
- Configuration and routing tables are synchronized
- Takes over the previous state within 3 seconds when the primary goes offline.

Interconnection of Main Switches

- Redundant connections between the two core switches (represented as dashed red lines)
- High-speed interconnection is done using EtherChannel/StackWise technology

- Spanning Tree Protocol (Rapid PVST+) is used to avoid loops but leave redundant paths.

Benefits of Dual-Core Design:

1. **High Availability:** In case Neg-Core-MLS01 is unavailable, Neg-Core-MLS02 automatically goes online and minimally disruptive (less than 3 seconds of downtime) [7]
2. **Load Balancing:** Specific VLAN can be configured to use a different switch as the primary gateway, and this is used to load balance traffic
3. **Maintenance Flexibility:** one switch can be brought offline to allow maintenance, and the other switch continues, which is critical to business survival
4. **Fault Tolerance:** a single point of failure at the distribution layer is removed
5. **Scalability:** Additional switches can be added as network grows

This increased redundancy within the distribution layer to guarantee adherence to a 99.9% uptime specification (maximum 8.76 hours of downtime per year), so that the network is now business critical [82].

3.1.3. Design Rationale

The hierarchical model was chosen due to a number of strong reasons:

1. **Scalability:** It is possible to add new departments or expand on the existing ones without affecting the basic infrastructure [9].
2. **Performance:** Traffic localization at the access layer to the network minimizes unnecessary core traffic, which is a better way of enhancing network performance.
3. **Redundancy:** There are several paths between layers which offer high availability and load balancing.

4. **Security:** There is good delimitation of points which makes it possible to enforce security policy at the distribution layer.
5. **Manageability:** Logical segmentation and distinct traffic paths result in simplified troubleshooting [10].
6. **Cost -Efficiency:** The location of equipment optimizes spending, such that expensive multilayer switches are deployed where needed.

3.1.4. Topology Type

Physical topology is a star topology where the access layer- devices are linked to central switches- whereas in the distribution and core layers, a partial-mesh topology is used to provide redundancy. This is a combination of cost and performance in this hybrid method coupled with reliability [11]. The benefits of the selected topology are the;

- Single points of failure are eliminated by redundancy.
- Easier trouble Shooting and fault isolation.
- Effective bandwidth usage.
- Assistance with the implementation of QoS.
- Future wireless integration accommodation.

3.2. VLSM Addressing Scheme Design

3.2.1. Address Space Allocation

The following address space has been allocated to the organization:

1. Main Branch: 192.168.1.0/24 (254 addresses in use).
2. New Branch: 10.10.0.0/16 (65,534 usable addresses).
3. WAN Link: 200.200.200.0/30 (2 usable addresses).

The 10.10.0.0/16 address space is spacious enough and efficient address utilization and VLSM are achieved [12].

3.2.2. VLSM Calculation Methodology

Variable Length Subnet Masking (VLSM) allows effective assignment of IP addresses by creating subnets of different sizes according to real needs, and in this way reducing the amount of wasted addresses [13].

Calculation Process:

1. Arrange sort departments in order of the number of hosts (most to the fewest).
2. Divide the number of required addresses by formula $2^n \geq (\text{hosts} + 2)$ where the two extra addresses compensate for network address and broadcast address.
3. Give subnet masks according to the requirements calculated.
4. Assign adjacent blocks of addresses so that route summarization can be done.

Department Sorting (Largest to Smallest):

1. WA1: 130 hosts $\rightarrow 2^8 = 256$ addresses $\rightarrow /24$ (254 usable)
2. WA2: 72 hosts $\rightarrow 2^7 = 128$ addresses $\rightarrow /25$ (126 usable)
3. IT: 60 hosts $\rightarrow 2^6 = 64$ addresses $\rightarrow /26$ (62 usable)
4. C&C: 35 hosts $\rightarrow 2^6 = 64$ addresses $\rightarrow /26$ (62 usable)
5. Finance: 18 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
6. HR: 18 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
7. SOC: 15 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
8. Admin: 10 hosts $\rightarrow 2^4 = 16$ addresses $\rightarrow /28$ (14 usable)

3.2.3. Complete VLSM Table

Department	Hosts	Network Address	Subnet Mask	CI DR	Usable Range	Default Gateway	Broadcast	VLAN
WA1	130	10.10.0.0	255.255.255.0	/24	10.10.0.1 - 10.10.0.254	10.10.0.1	10.10.0.255	10
WA2	72	10.10.1.0	255.255.255.128	/25	10.10.1.1 - 10.10.1.126	10.10.1.1	10.10.1.127	20
IT	60	10.10.1.128	255.255.255.192	/26	10.10.1.129 - 10.10.1.190	10.10.1.129	10.10.1.191	30
C&C	35	10.10.1.192	255.255.255.192	/26	10.10.1.193 - 10.10.1.254	10.10.1.193	10.10.1.255	40
Finance	18	10.10.2.0	255.255.255.224	/27	10.10.2.1 - 10.10.2.30	10.10.2.1	10.10.2.31	50
HR	18	10.10.2.32	255.255.255.224	/27	10.10.2.33 - 10.10.2.62	10.10.2.33	10.10.2.63	60
SOC	15	10.10.2.64	255.255.255.224	/27	10.10.2.65 - 10.10.2.94	10.10.2.65	10.10.2.95	70
Admin	10	10.10.2.96	255.255.255.240	/28	10.10.2.97 - 10.10.2.110	10.10.2.97	10.10.2.111	80

Table 2: Complete VLSM Table for Avalanch Corporations

Additional Subnets:

Purpose	Network Address	Subnet Mask	CIDR	Usable Range	Usage
Management	10.10.3.0	255.255.255.240	/28	10.10.3.1 - 10.10.3.14	Switch management interfaces
WAN Link	200.200.200.0	255.255.255.252	/30	200.200.200.1 - 200.200.200.2	Router-to-router connection
Main Branch	192.168.1.0	255.255.255.0	/24	192.168.1.1 - 192.168.1.254	Main branch LAN

Table 3: Additional Subnets for the Avalanch Corporations

3.2.4. Address Space Efficiency Analysis

- Total Addresses Allocated: 638 addresses
- Total Addresses Used: 358 hosts
- Efficiency: 56.1%
- Growth Capacity: 280 additional hosts (78.2% expansion possible)

This efficiency ratio is acceptable as it provides significant growth capacity while maintaining hierarchical address structure for route summarization [14].

3.2.5. Route Summarization Opportunities

The nearby allocation of addresses allows effective summarization of routes:

- Summary Route for New Branch: 10.10.0.0/16
- Advertised to Main Branch: One route that will be used to represent all the eight departments.
- Advantage: Smaller routing table, faster convergence and easier routing [15].

3.3. Equipment Selection

3.3.1. Router Selection: Cisco 2911

Both branches have been selected as having Cisco 2911 Integrated Services Router on the following ground [16]:

- **VPN Capabilities:** Hardware-accelerated IPsec with encryption (AES-256) with a throughput of 100Mbps.
- **Port Density:** WAN and LAN Multiple Gigabit Ethernet interfaces.
- **Security Features:** VPN concentrator, IPS features, and firewall built-in.
- **Scalability:** Modular construction that can be expanded to large interface cards in the future.
- **Reliability:** Enterprise-grade hardware with MTBF of over 200,000 hours.
- **Cost-Effectiveness:** Best tradeoff between functionality and financial limitations.

Technical Specifications:

- Throughput: 300Mbps (no encryption), 100Mbps (IPsec encrypted)
- Interfaces: 3 Gigabit Ethernet (built-in) and modules extendable.
- Memory: 512 MB DRAM, 256MB Flash (default)
- Routing Protocols RIPv2, EIGRP, OSPF, BGP.
- Security: IPsec, IPS VPN, Firewall, IPS.

3.3.2. Multilayer Switch Selection: Cisco 3650-24PS

Core/distribution switch The Cisco Catalyst 3650 24PS was chosen as the core/distribution switch in the two branches [17]:

- **Layer3 Routing:** Hardware routing that supports inter-VLAN routing at wire rate.
- **Performance:** 32GB switching, sufficient to support departmental traffic.
- **VLAN Support:** 1,024 VLANs (which is much more than eight VLANs, as is needed)
- **Routing Protocols:** Complete support of OSPF and EIGRP as inter-branch routing.
- **Security Features:** Port security, 802.1X authentication, ACL support.
- **PoE Support:** Power over ethernet in the future to be used with wireless AP.
- **Stackability:** StackWise technology in case of future redundancy and expansion.

Technical Specifications:

- Switching Capacity: 32 Gbps
- Forwarding Rate: 23.8 Mpps
- Port Configuration: 24x 10/100+ 2x uplinks SFP.
- Routing Performance: 8.8Mpps (Layer 3)
- VLAN Support: 1,024 VLANs, 4,096 VLAN IDs
- ACL Capacity: 1,024 entries

The common router-on-a-stick networks use a router interface to route all inter-VLAN traffic through one router, thus forming a performance bottleneck [18]. Multilayer switch approach has the following benefits:

- **Better Performance:** Hardware vs. Software Routing.
- **Reduced Latency:** Traffic exchanged within a local area without the participation of the router.
- **Scalability:** It is less difficult to add VLANs without compromising performance.
- **Reduced Configuration:** Switch Virtual Interfaces (SVIs) less complex to do.
- **Economical:** The switching is done by one device that is also used as a router.

3.3.3. Access Switch Selection: Cisco 2960-24TT

Cisco Catalyst 2960-24TT has been chosen to be installed on access-layers [19]:

- **Cost-Effective:** A suitable feature configuration at the access layer without allowing extraneous cost.
- **Port Density:** 24 Fast Ethernet ports to meet the needs of departments.
- **VLAN Support:** Complete 802.1Q VLAN support of segmentation.
- **Security Features:** Port security, DHCP snooping, DAI (Dynamic ARP Inspection).
- **Reliability:** Established platform under 10 failure reasons.
- **Energy Efficiency:** Low power consumption which is appropriate to stay on constantly.

Technical Specifications:

- Switching Capacity: 16 Gbps
- Port Type: 24×10/100 + 2x 10/100/1000T uplinks
- VLAN Support: 255 active VLANs
- MAC Address Table: 8,192 entries

3.3.4. Server Specifications

1. AAA Server : Unified network device access control.

Software: Cisco ISE or RADIUS compatible.

Protocols: RADIUS, TACACS+

2. DHCP Server: Automation of IP address assignment to every VLAN.

Capacity: 1,000+ DHCP leases.

Capabilities Scope management, lease tracking, conflict detection.

3. NTP Server: Time synchronization on a network-wide basis.

Accuracy: Stratum 2 or better (within 50ms of UTC)

Significance: Crucial to the log correlation and security event analysis [20].

4. Syslog Server: Observation and evaluation of logs.

Capacity: 10000 messages/ sec. and above.

Storage: 500GB is required to store a 90D retention.

Functionality: Log filtering, alerting, reporting functionality.

3.3.5. Complete Equipment List

Main Branch:

Device	Model	Quantity	Purpose	IP Address
Router	Cisco 2911	1	Edge routing, VPN termination, internet gateway	192.168.1.1
Main MLS	Cisco 3650-24PS	1	Core switching, inter-VLAN routing	192.168.1.254
NTP Server	Generic Server	1	Network time synchronization	192.168.1.5
Syslog Server	Generic Server	1	Centralized log management	192.168.1.8
AAA Server	Generic Server	1	Authentication, Authorization, Accounting	192.168.1.7
Backup Server	Generic Server	1	TFTP backup storage for configurations	192.168.1.20
End Devices	Generic PC	2	Administrative workstations (static IP)	192.168.1.0/24

Table 4: Equipment List for Avalanch Main Branch

New Branch (Negombo)

Device	Model	Quantity	Purpose	Location/VLAN
Router	Cisco 2911	1	Branch routing, VPN termination	10.10.3.2
Main MLS (Primary)	Cisco 3650- 24PS	1	Inter-VLAN routing, HSRP Active, Root Bridge	Neg-MLS01
Main MLS (Secondary)	Cisco 3650- 24PS	1	Inter-VLAN routing, HSRP Standby	Neg-MLS02
Access Switch - WA1	Cisco 2960- 24TT	1	Department access layer	VLAN 10
Access Switch - WA2	Cisco 2960- 24TT	1	Department access layer	VLAN 20
Access Switch - IT	Cisco 2960- 24TT	1	Department access layer	VLAN 30
Access Switch - C&C	Cisco 2960- 24TT	1	Department access layer	VLAN 40
Access Switch - Finance	Cisco 2960- 24TT	1	Department access layer (ISOLATED)	VLAN 50
Access Switch - HR	Cisco 2960- 24TT	1	Department access layer (ISOLATED)	VLAN 60
Access Switch - SOC	Cisco 2960- 24TT	1	Department access layer	VLAN 70
Access Switch - Admin	Cisco 2960- 24TT	1	Department access layer	VLAN 80
DHCP Server	Generic Server	1	Local DHCP services for Negombo branch	VLAN 30 (IT)
End Devices	Generic PC	358	Departmental workstations	Distributed across 8 VLANs

Table 5: Equipment List for Avalanch Negombo Branch

3.4. Network Diagram

The comprehensive network diagram illustrates a hierarchical three-layer architecture connecting the Main Branch and Negombo Branch via an encrypted IPSec VPN tunnel.

Key elements include:

1. Main Branch (Left Section):

- Edge router (Main-Router) that is linked to the internet/WAN cloud.
- Main Core switching (Main-Sw1) offers switching services.
- Three central servers (AAA,NTP, Syslog)
- Additional Backup server with TFTP feature on
- Testing and administrative end-user workstations.

2. WAN Connection (Top Section):

- Internet cloud which represents the public network.
- IPSec VPN tunnel (encrypted tunnel) between the two routers.
- Serial WAN connections.
- GRE Tunnel (172.16.1.0/30) for additional VPN redundancy
- Addressing: 200.200.200.0/30

3. New Branch- Negombo (Right Section):

- VPN termination Edge router (Negombo-Router)
- Dual core Architecture in Multi-Layer Switches
 - Primary Core Switch - Neg-MLS01:
 - Active HSRP router for VLANs 10, 20, 30, 40
 - Root Bridge for Spanning Tree Protocol
 - Primary gateway for departmental traffic
 - Connected to all eight access switches via trunk links

- Secondary Core Switch - Neg-MLS02:
 - Standby HSRP router for VLANs 50, 60, 70, 80
 - Backup spanning tree bridge
 - Automatic failover capability (sub-3 second convergence)
 - Redundant connections to all access switches
- Eight access switches in two row:
 - Top Row:- WA1-SW VLAN 10 WA2-SW VLAN 20 IT-SW VLAN 30 C&C-SW VLAN 40
 - Bottom Row Finance-SW (VLAN 50), HR-SW (VLAN 60), SOC-SW (VLAN 70), Admin-SW (VLAN 80)
- Departmentally dispersed end devices.

Connection Types:

- Trunk Links: Connects between access switches and multilayer switches
- Access Links: Switches to end devices (single VLAN)
- Redundant Links : Multi layer switches interaction with HSRP + LACP +STP
- WAN Links: Green dashed line implied VPN tunnel encryption.

3.4.1. Network Segmentation and Security Zones:

Zone 1 - Main Branch

- Corporate services and internet gateway
- Centralized management servers
- Trusted network segment

Zone 2 - WAN/Internet

- Public network infrastructure
- Encrypted VPN tunnel overlay
- Untrusted network segment

Zone 3 - Negombo Branch

- Eight departmental VLANs with logical segmentation
- Finance and HR departments isolated from other departments
- Local DHCP services in IT department

3.4.2. IP Addressing Summary

Network Segment	Network Address	Purpose
Main Branch LAN	192.168.1.0/24	Main branch devices and servers
WAN Link	200.200.200.0/30	Router-to-router connection
Backup GRE Tunnel	172.16.1.0/30	Secondary WAN connection
Negombo Branch	10.10.0.0/16	All departmental networks
Management VLAN	10.10.3.0/28	Switch management interfaces

Table 6: IP addressing Summary Table

Design Notes:

- Every trunk port that is encapsulated 802.1Q.
- Access ports dedicated to certain VLANs (None of VLANs 1)
- High availability uplinks that are redundant.
- VLAN switch administration (VLAN 99).
- Finance and HR switches were pointed out to denote ACL protection.

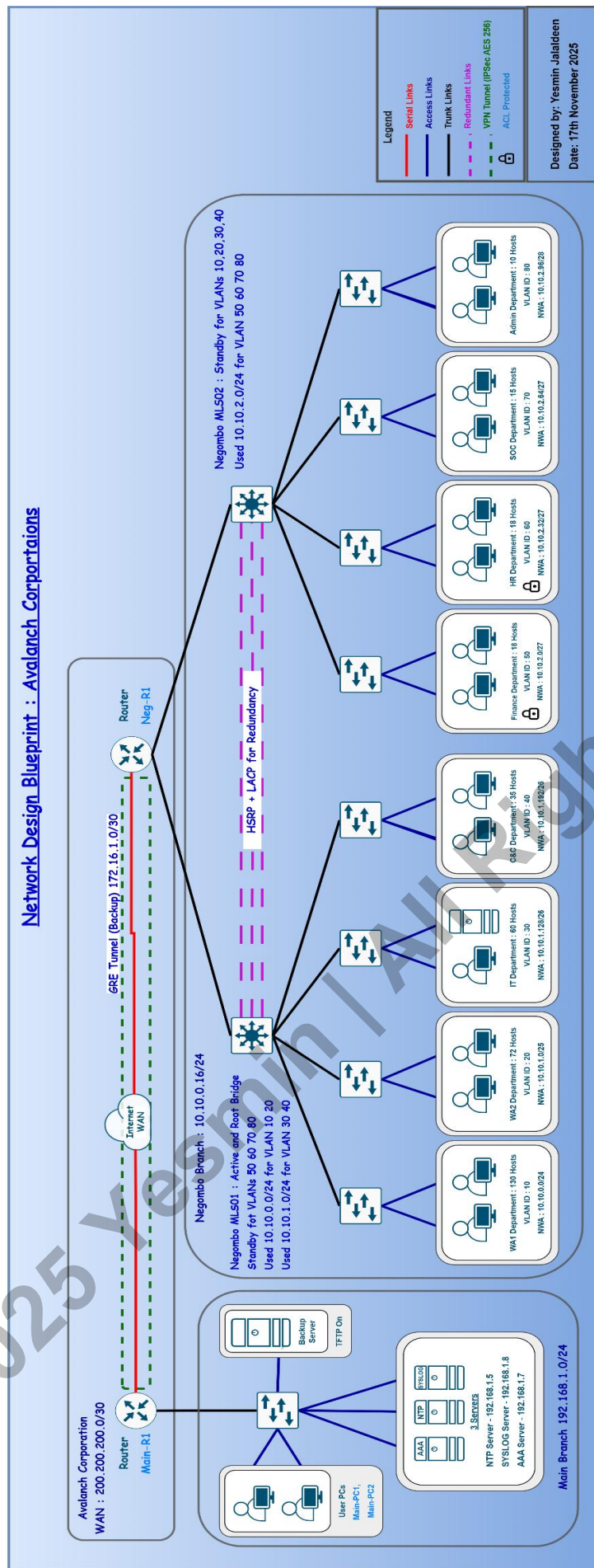


Figure 1: Network Design Blueprint for Avalanche Corporation

4. Network Design Implementation

The network topology has been designed and visualized with the draw.io tool in order to render correct and industry-standard documentation of a hierarchical design, which has given a clear logical picture. Cisco Packet Tracer Version 9 was used in practical implementation, configuration and testing of the network infrastructure. Every protocol, such as EIGRP routing, HSRP redundancy, and IPSec VPN tunnels, could be tested within this simulation environment before deploying in the real world, and, therefore, all the configurations were verified to comply with the high security and performance standards of Avalanche Cooperations.

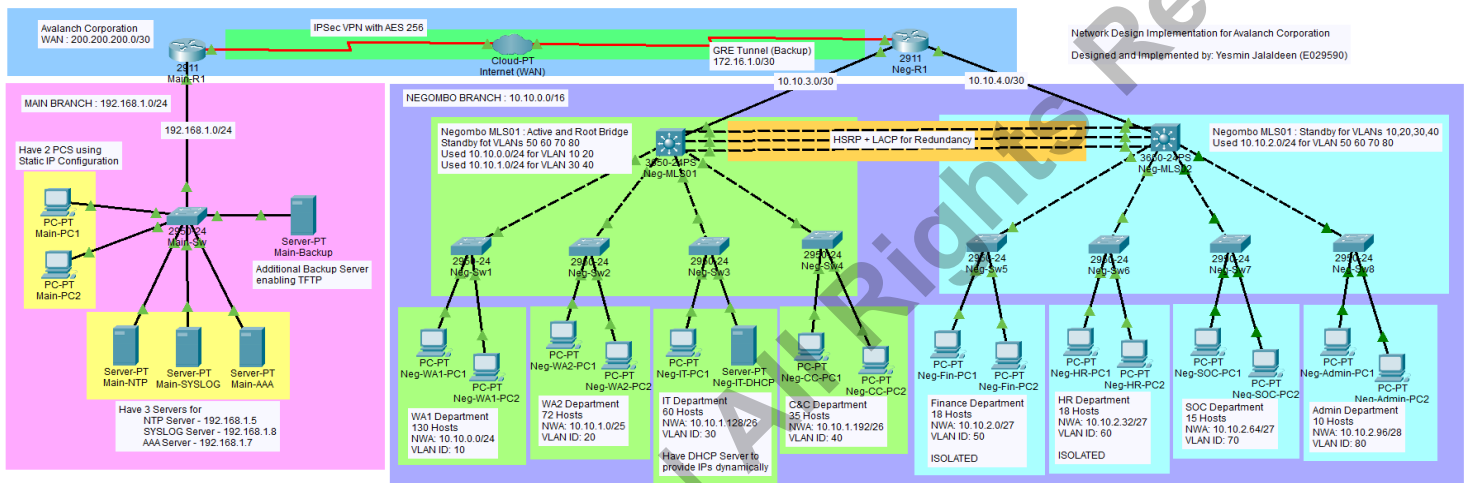


Figure 2: Labelled Network Design in PKT

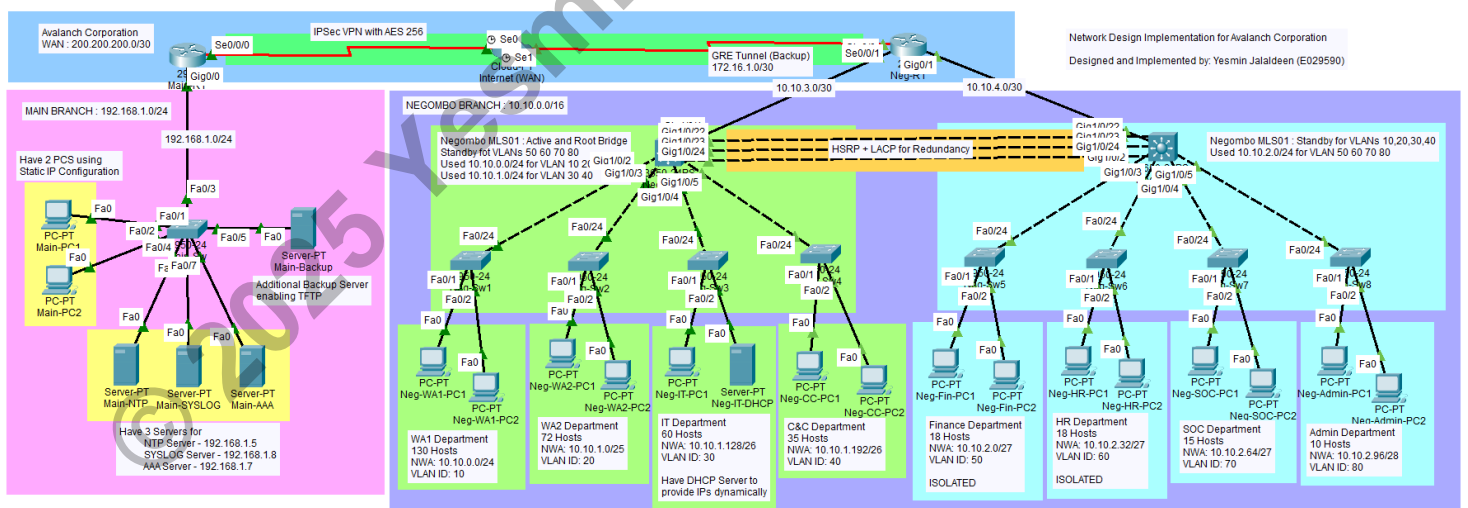


Figure 3: Labelled Network Diagram in PKT with Visible Interfaces

4.1. VLAN Design Strategy

VLANs allow logical physical networks segmentation, which has many advantages that are vital to organizational security and performance [21]:

- **Security Isolation:** There is separation of broadcast domains to reduce the threat of unauthorized cross-departmental access.
- **Traffic Management:** The decrease in broadcast traffic is associated with the enhanced performance of the whole network.
- **Eased Administration:** Logical grouping is not tied to physical location and as such, administration is made easier.
- **Policy Enforcement:** It is possible to apply various security policies on a departmental basis.
- **Scalability:** New users can be added without physical reconfiguration being required.

4.1.1. VLAN Assignment Strategy

Department VLANs (Data Plane)

VLAN ID	VLAN Name	Network	Department	Security Level
10	WA1-VLAN	10.10.0.0/24	WA1	Standard
20	WA2-VLAN	10.10.1.0/25	WA2	Standard
30	IT-VLAN	10.10.1.128/26	IT	Elevated
40	CC-VLAN	10.10.1.192/26	C&C	Standard
50	FINANCE-VLAN	10.10.2.0/27	Finance	High (Isolated)
60	HR-VLAN	10.10.2.32/27	HR	High (Isolated)
70	SOC-VLAN	10.10.2.64/27	SOC	Elevated
80	ADMIN-VLAN	10.10.2.96/28	Admin	Standard

Table 7: Department VLAN Breakdown of Avalanch Corporation

Infrastructure VLANs (Management/Control Plane):

VLAN ID	VLAN Name	Network	Purpose
99	MANAGEMENT	10.10.3.0/28	Switch management interfaces
100	NATIVE	-	802.1Q native VLAN (tagged)

Table 8: Infrastructure VLANs for Avalanch Corporation

Design Rationale:

- **VLAN 1 Avoided:** VLAN 1 was not used in order to reduce the possibility of VLAN hopping attacks [22].
- **Separate Management VLAN:** The traffic of the management was separated to user traffic and hence increased security.
- **Native VLAN Tagged:** Trunk traffic was set to be tagged to curb VLAN hopping.
- **Use of logical numbering:** VLAN identifiers were assigned as multiples of ten to help in making it easier to understand and easy to expand in future.
- **High-Security VLANs:** The Finance (50) and the HR (60) were numbered in order to enhance the ACL performance.

Additionally, Switch Management interfaces are configured with IP addresses for remote administration via SSH, following security best practices.

4.2. VLAN Configuration

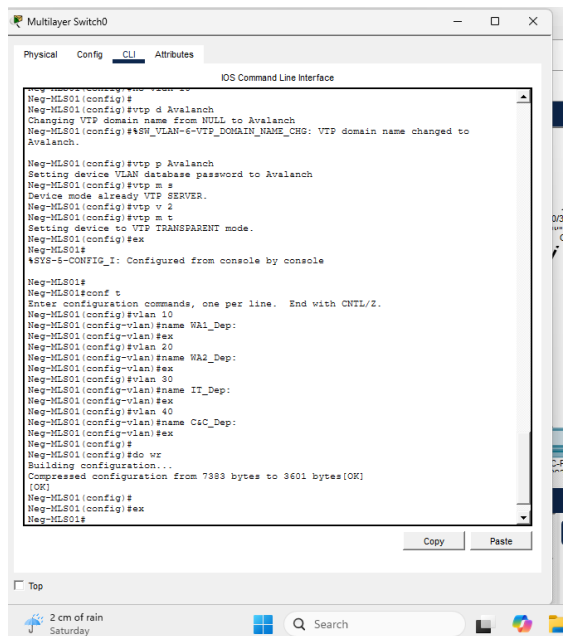


Figure 5: VLANs in Neg-MLS01

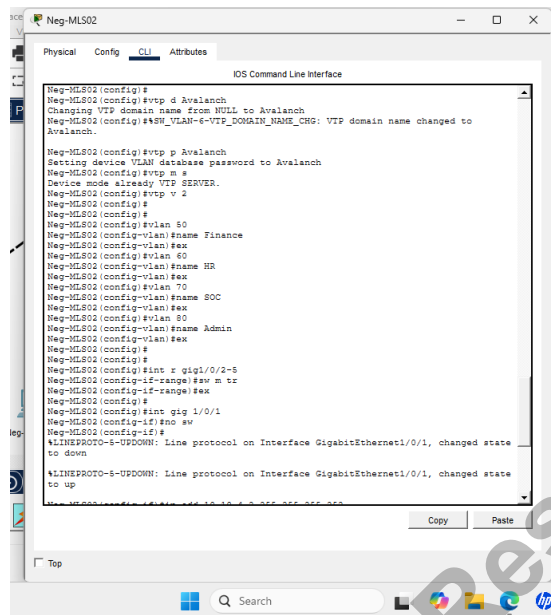


Figure 4: VLANs in Neg-MLS02

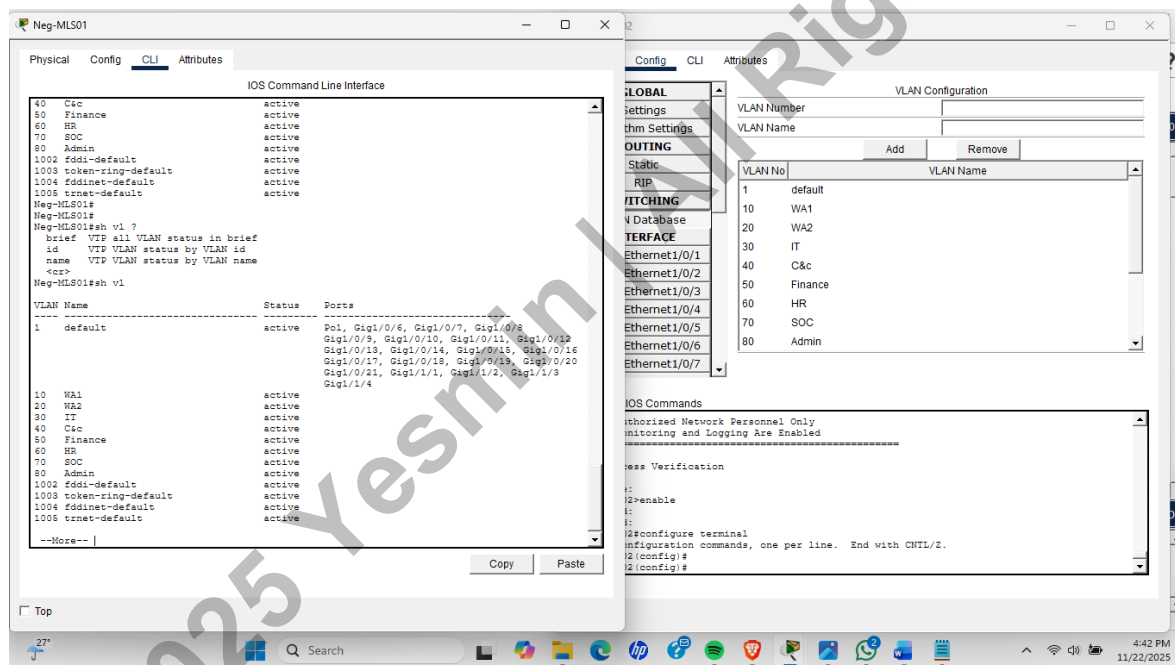


Figure 6: How VLANs are displayed in both MLSS

4.3. VTP Configuration

VLAN Trunking Protocol (VTP) is a proprietary protocol in Cisco which is used to forward VLAN configuration between switches and thus saves administrative overhead [23]. Nevertheless, VTP may pass on the misconfigurations, and this may cause network outages to spread out.

4.3.1. VTP Mode Selection

To do this implementation, the following reasons were chosen with **VTP Client/Server Mode**:

- **Safety:** The switch is not involved in VTP advertisements and therefore, there is no possibility of deleting VLANs accidentally.
- **Control:** VLAN database changes are operational either within a single switch, and therefore, it has more control.
- **Flexibility:** Custom VLAN can also be used with specific switches without interfering with others.
- **Security:** This means that the processing of VTP advertisements is not done, which decreases the attack surface [24].
- **Reliability:** The threat of conflict of VTP domain name or revision number is removed.

4.3.2. VTP Configuration Parameters

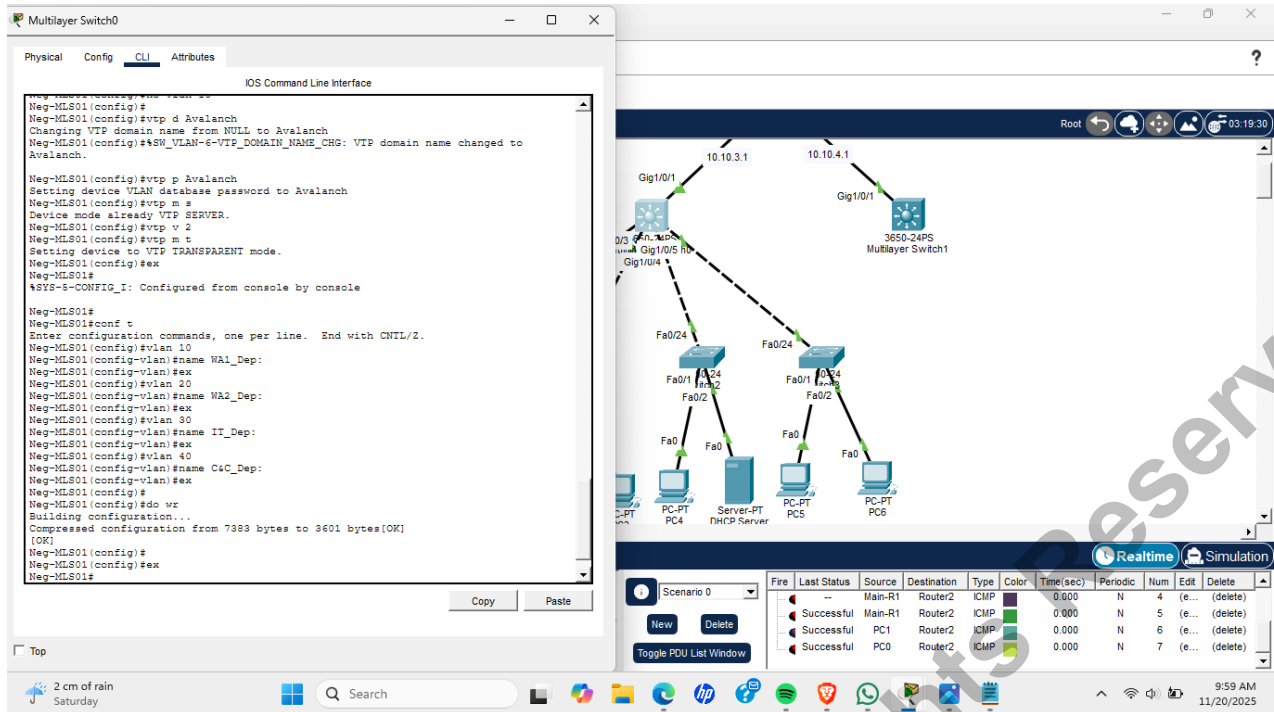


Figure 7: VTP Server and VLAN Database in Neg-MLS01

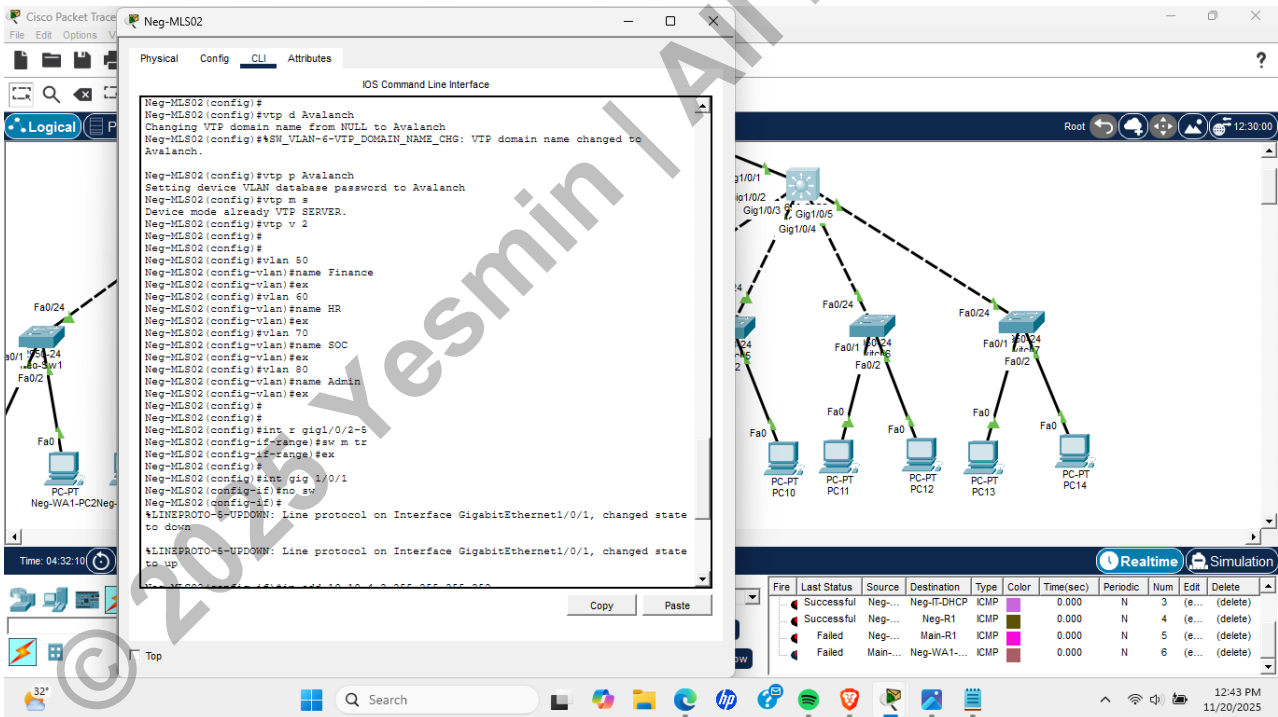


Figure 8: VTP Server and VLAN Databases in Neg-MLS02

I esmin Jalaldeen

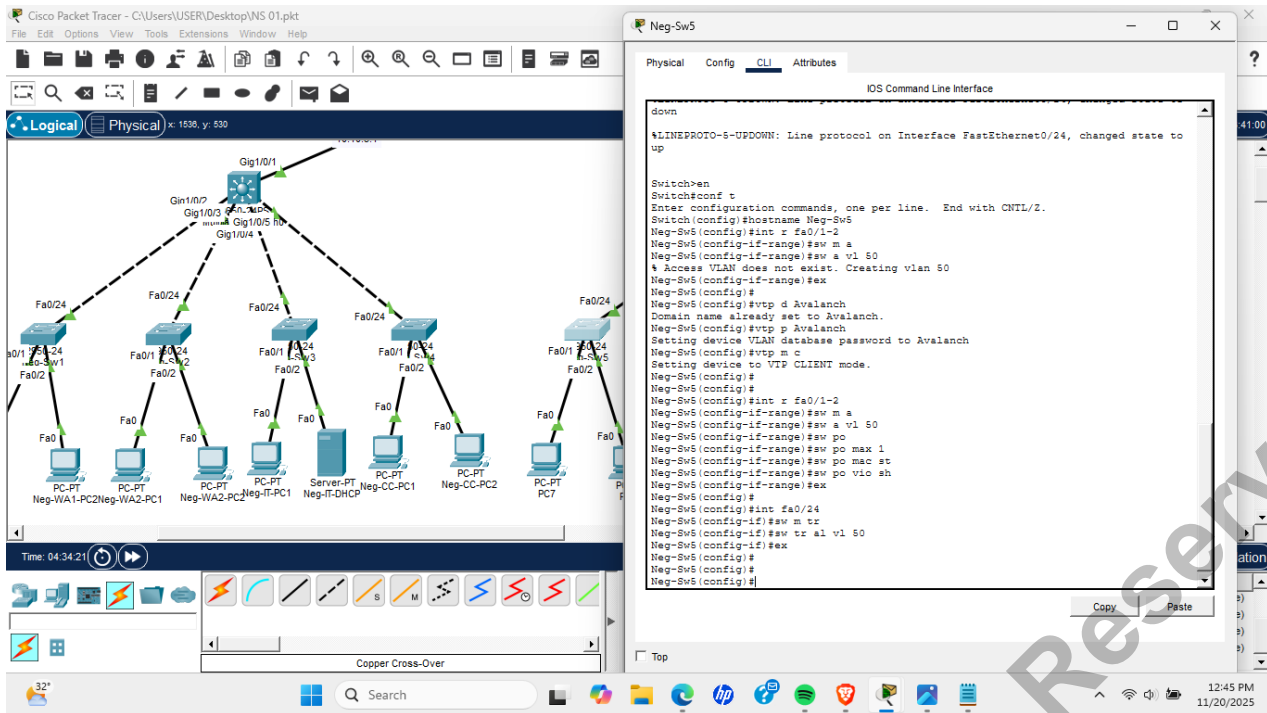


Figure 10: VTP Client in Negombo Branch

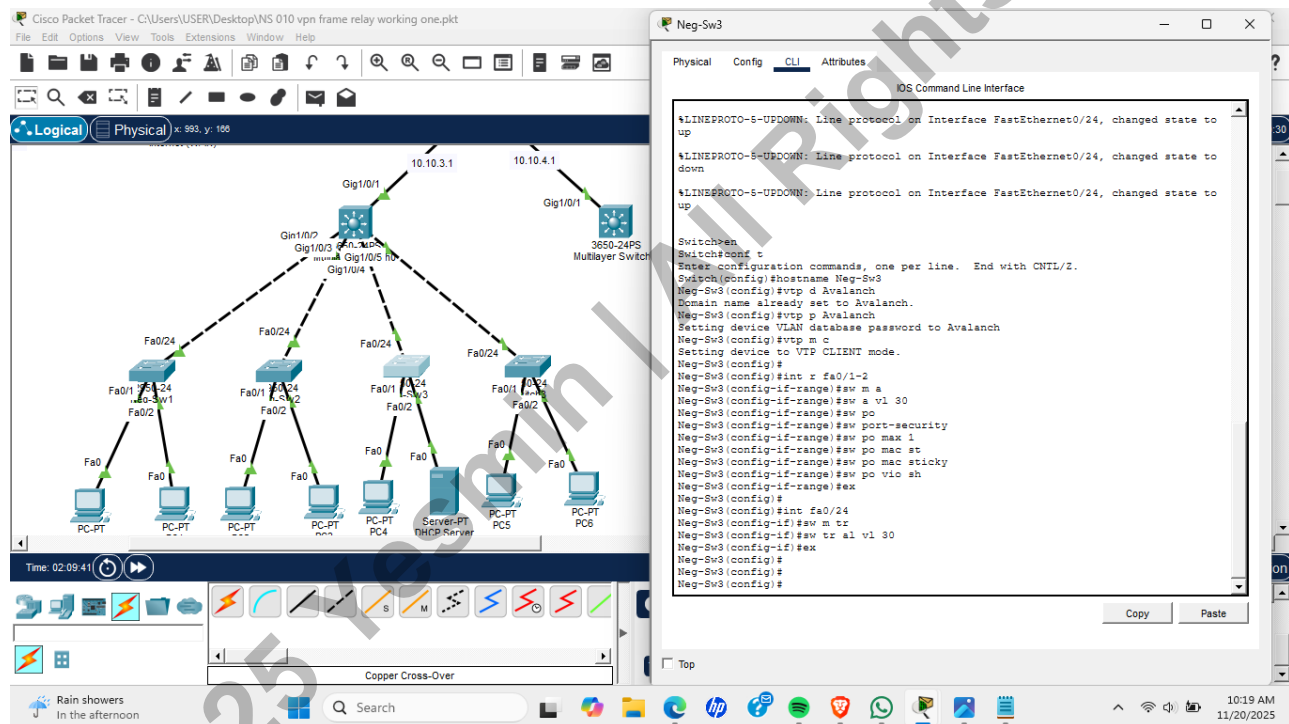


Figure 9: VTP Client in Negombo Branch

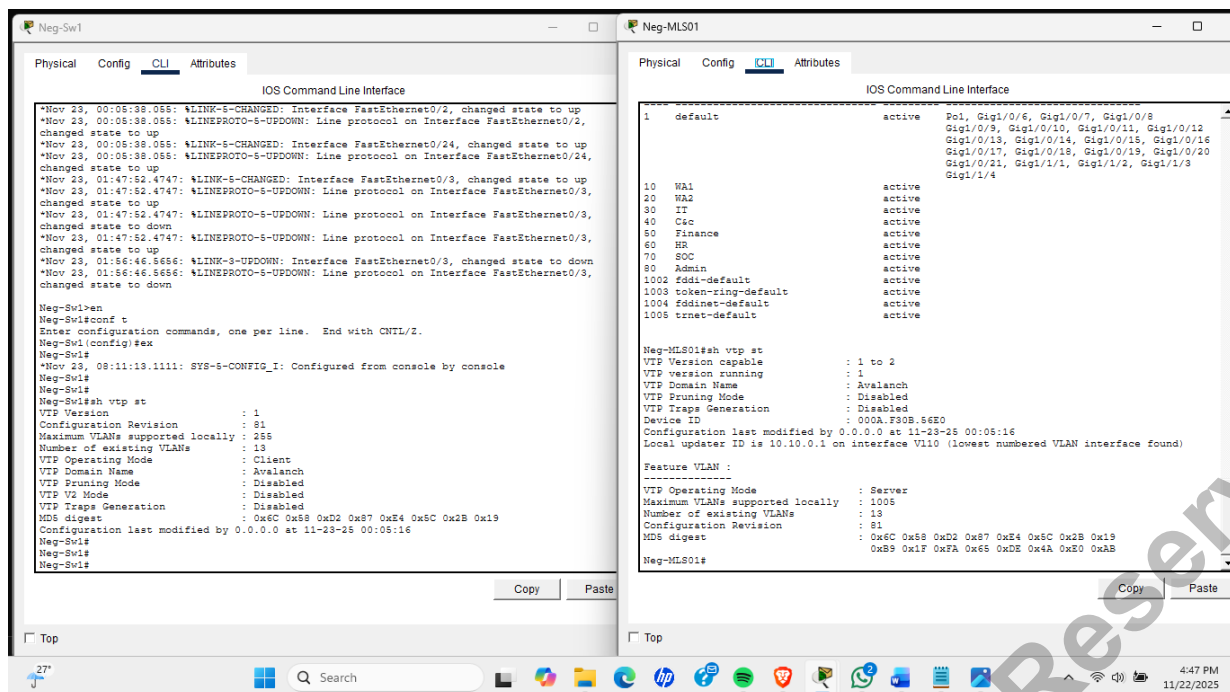


Figure 11: How VTP displays

4.4. Trunk and Access Port Configuration

4.4.1. Trunk Port Configuration

The trunk ports carry traffic across switches that are part of multiple VLANs over 802.1Q encapsulation [26]. In below, the trunk ports are allowed and assigned to relevant VLAN

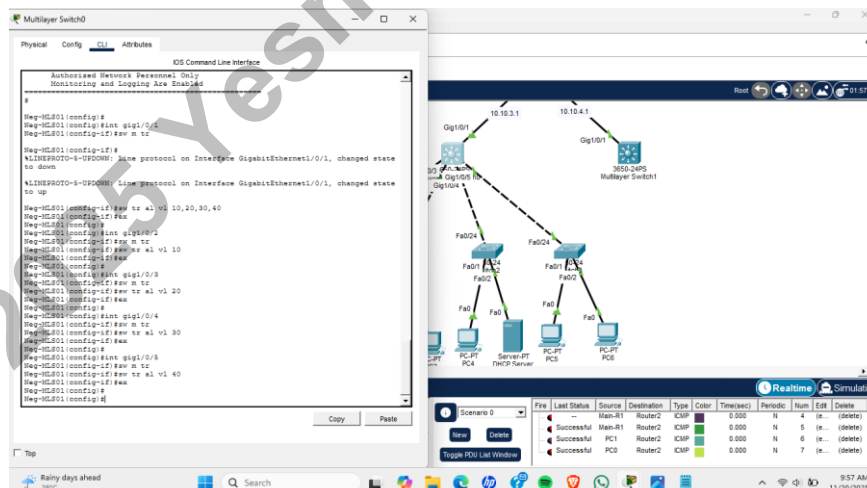


Figure 12: Trunk Ports in Neg-MLS01



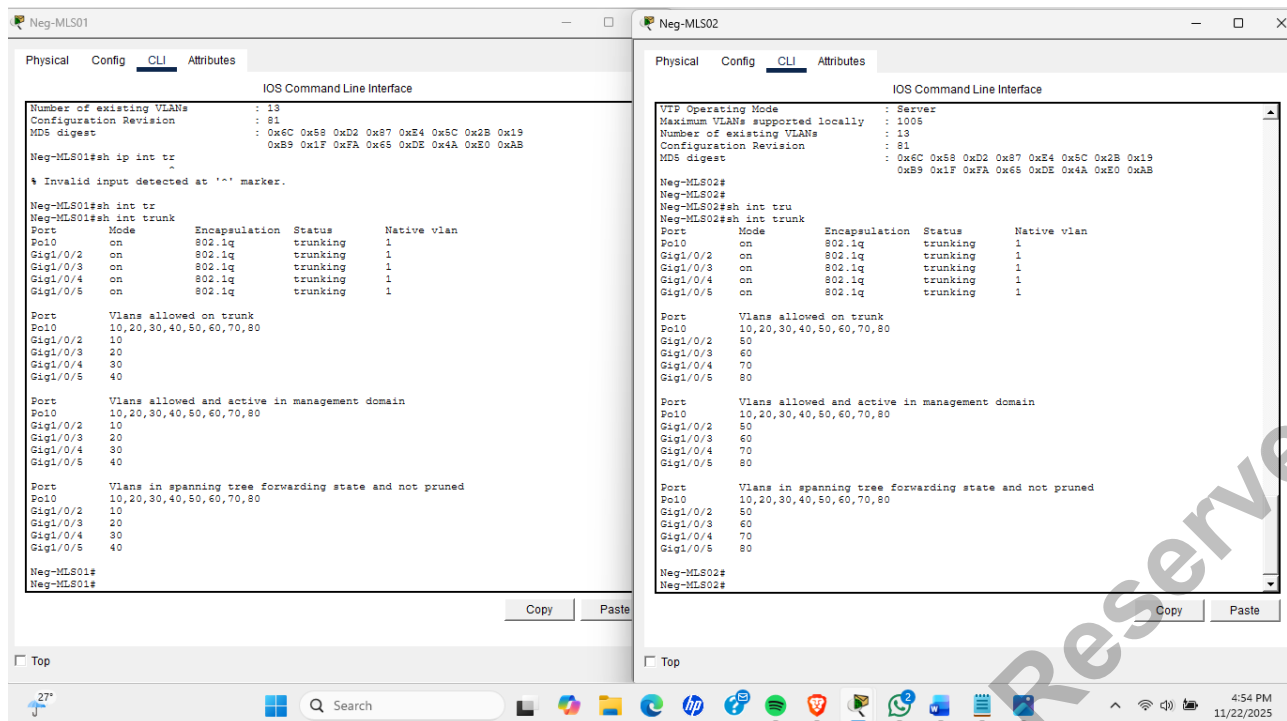


Figure 15: How Trunk Ports assigned

Security Hardening

- **Native VLAN Changed:** VLAN default (VLAN 1) is not the native VLAN to reduce VLAN-hopping attacks.
- **Native VLAN Tagged:** The `vlan dot1q tag native` command makes sure that even native VLAN is tagged.
- **Limit VLANs Limited:** VLANs Only the VLANs necessary to each service are authorized on trunk links, hence observing the principle of least privilege.
- **DTP Disabled:** Dynamic Trunking Protocol is switched off using `switchport non-negotiate` to avoid negotiation attack [27].
- **Unused Ports Disabled:** Trunk Ports that are not used are administratively disabled.

4.4.2. Access Port Configuration

Access ports are used to connect the end devices, and they are dedicated to a single VLAN [28].

4.4.3. Port Security Configuration

This port can be configured to either allow or deny entry to the host based on a predetermined set of rules

Port security limits the number of MAC addresses accepted on a port to stop unauthorized connections to the device [29].

In configuration I have done both at the same time for easiness

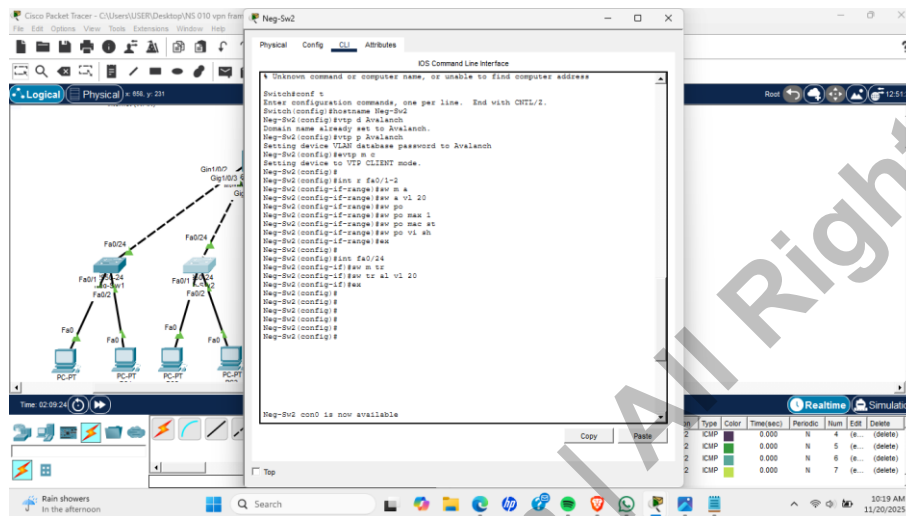


Figure 16: Access ports and Port Security in Neg-Sw2

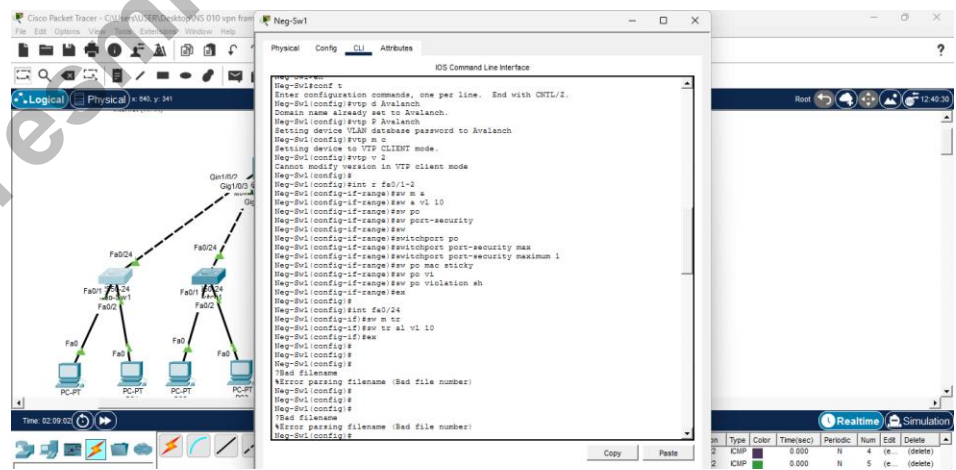


Figure 17: Access ports and Port Security In Neg-Sw1



- **Prevent MAC Flooding Attacks:** The limiting of the number of MAC addresses assigned to a port suppresses flooding.
- **Unauthorized Device Prevention:** Devices can be only connected to pre-registered devices.
- **Compliance and Audit Trail:** Port Security meets security policy standards by use of syslog; messages are caused by violations.

5. Network Devices Configuration

5.1. Router Configuration and Interconnections

5.1.1. Routing Protocol Selection

There is a need of a dynamic routing protocol to exchange route information and give automatic failover features to connect the main branch and the new branch.

EIGRP was chosen as the routing protocol because of the following reasons:

- **Fast Convergence:** Sub-second convergence in networks of small size, which is essential to sustain the VPN tunnels stability [77].
- **Low Overhead:** It does not need periodic updates as periodic updates done by RIP, but just when topology is changed.
- **Automatic Route Summarization:** VLSM and manual summarization assist in efficient routing.
- **Cisco Optimized:** Excellent performance on Cisco machines with little CPU usage.
- **Simplicity:** The OSPF configuration is easier than in two-site topology.
- **Unequal Cost Load Balancing:** Is capable of using many paths with varied metrics (extension in the future).

5.1.2. Router Configurations – Main

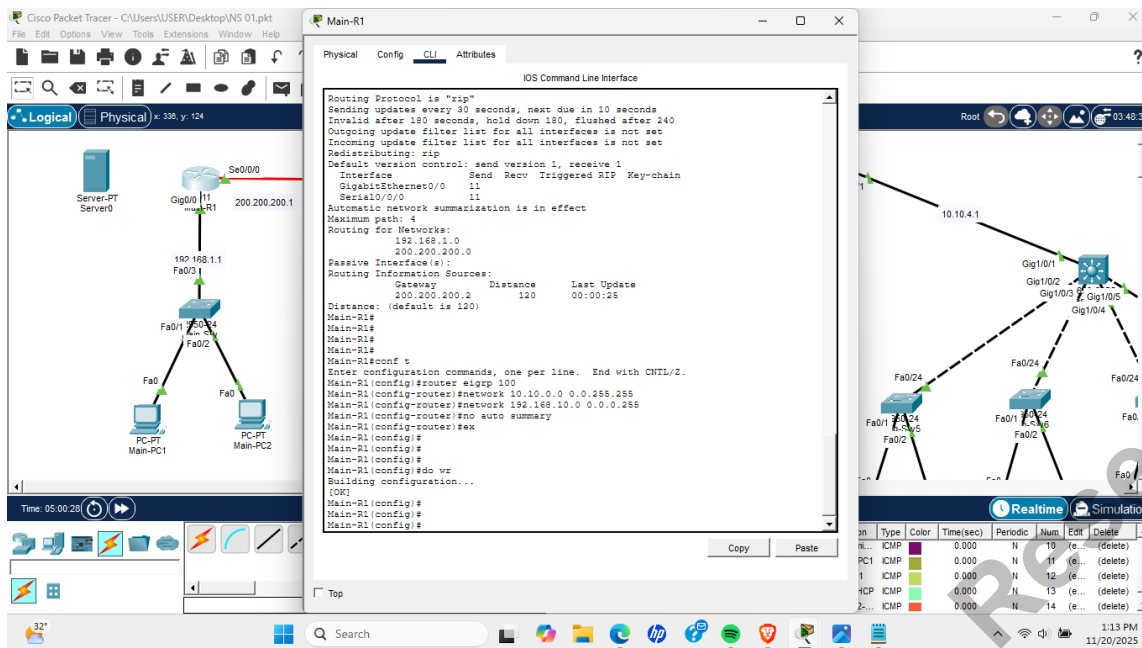


Figure 20: EIGRP in Main-R1

Configuration Explanation:

- AS Number 100: The number of an Autonomous System should be the same across all EIGRP routers.
- Network Statements: Advertise LAN (192.168.10.0/24) and WAN link (200.200.200.0/30)
- No Auto-Summary: Turns off classful summarization, which is needed by VLSM [78].
- Passive Interface Default: Disables default EIGRP sending hellos on all interfaces (security).
- No Passive on Gi0/1: this is used to allow the EIGRP neighbor formation on the WAN interface only.

Router ID: Unique identifier of EIGRP process (follows a similar format like OSPF)

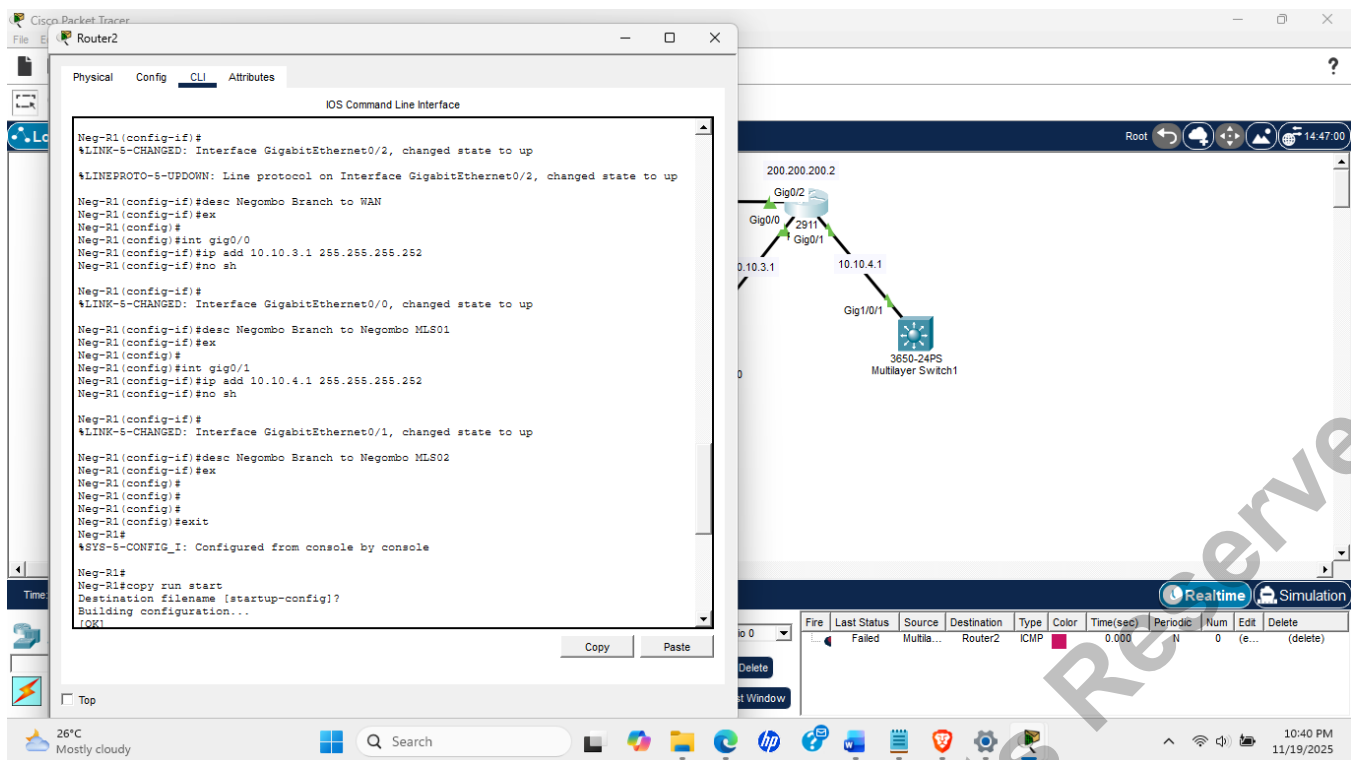


Figure 25: IP Route in Neg-R1

5.2. Multi-Layer Switch Configurations

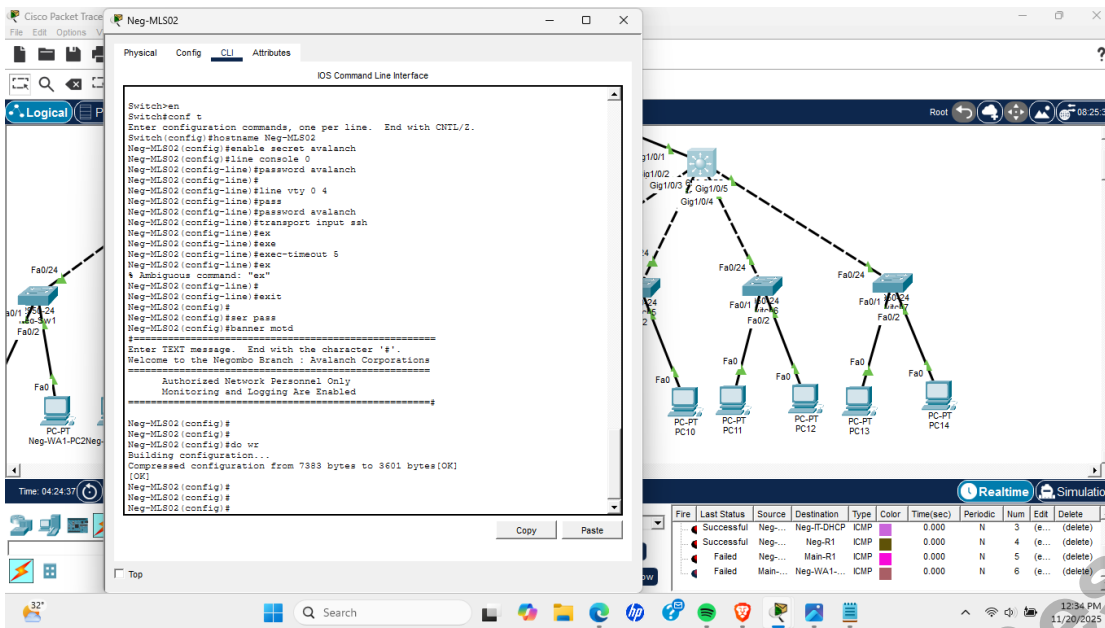


Figure 27: Basic Security in Neg-MLS02

5.2.1. Negombo Branch (Neg-MLS01 and Neg-MLS02)

5.3. Spanning Tree Implementation (STP)

STP is a network protocol (IEEE 802.1D) available in Layer 2, which can be used to eliminate loops in a network topology by blocking redundant paths when there are two or more paths between two points. It guarantees a path loop free up to the root bridge and the redundant links remain in an inactive state until an automatic activation occurs in the event of a primary path failure.

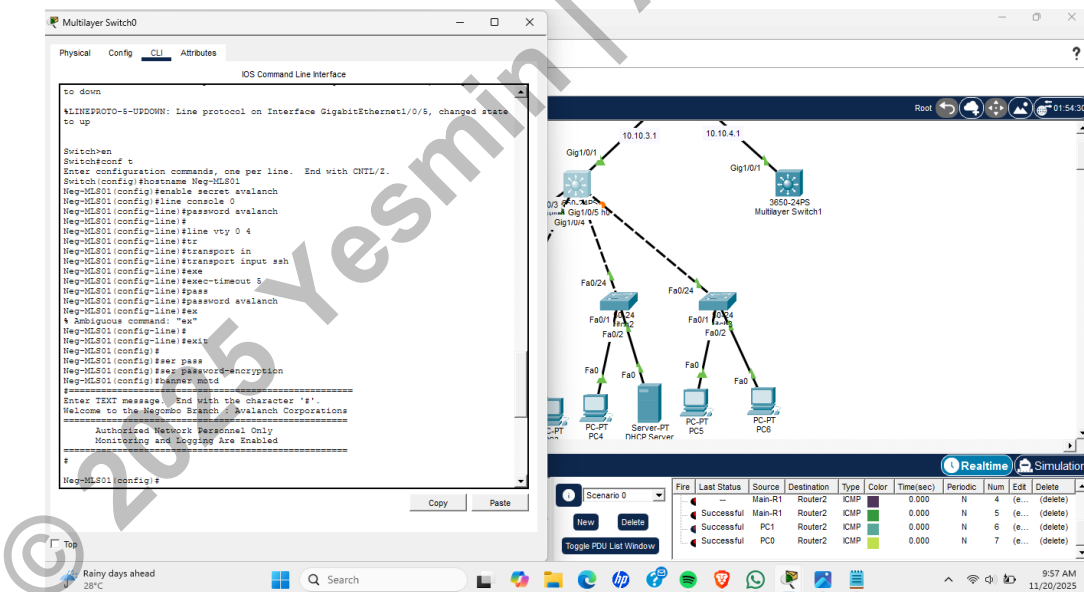


Figure 26: Basic Security in Neg-MLS01

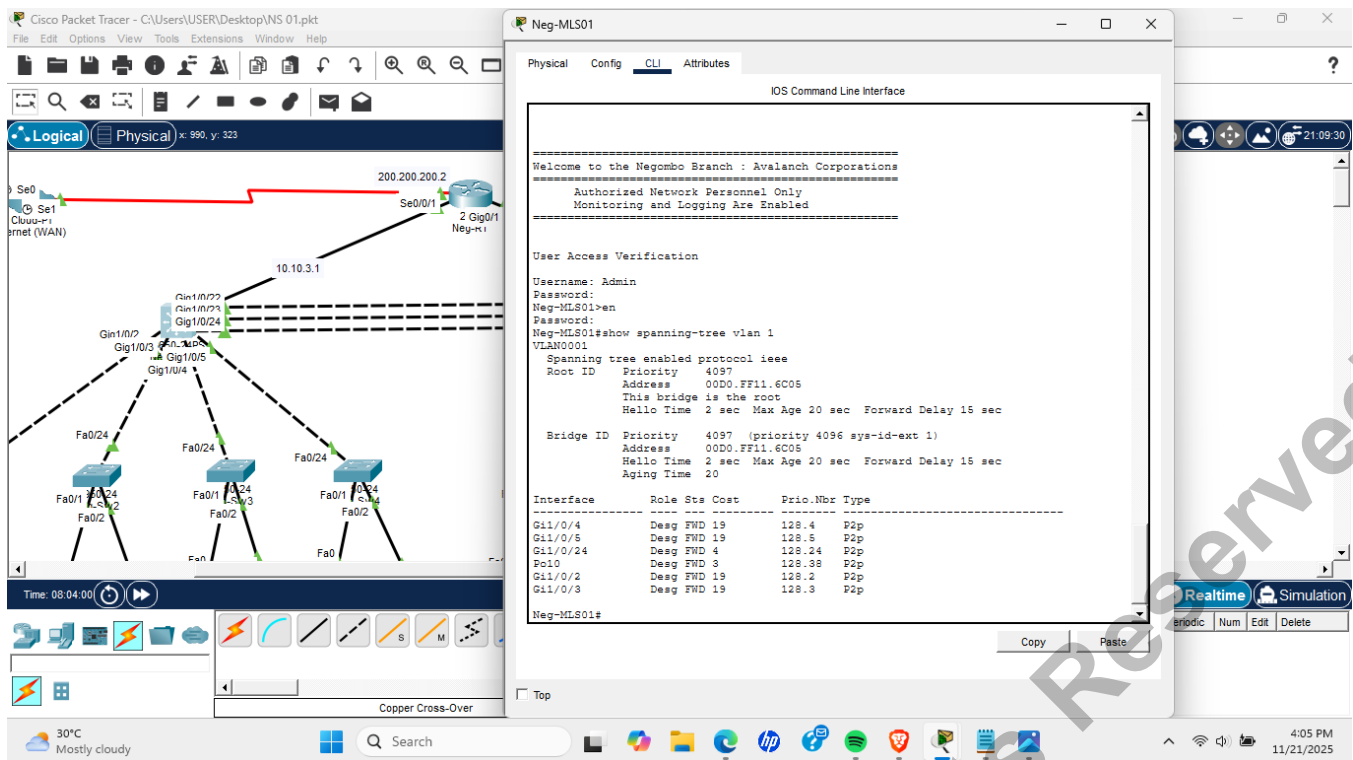


Figure 29: STP and STP Results in Neg-MLS01

5.4. Inter-VLAN Routing (SVI)

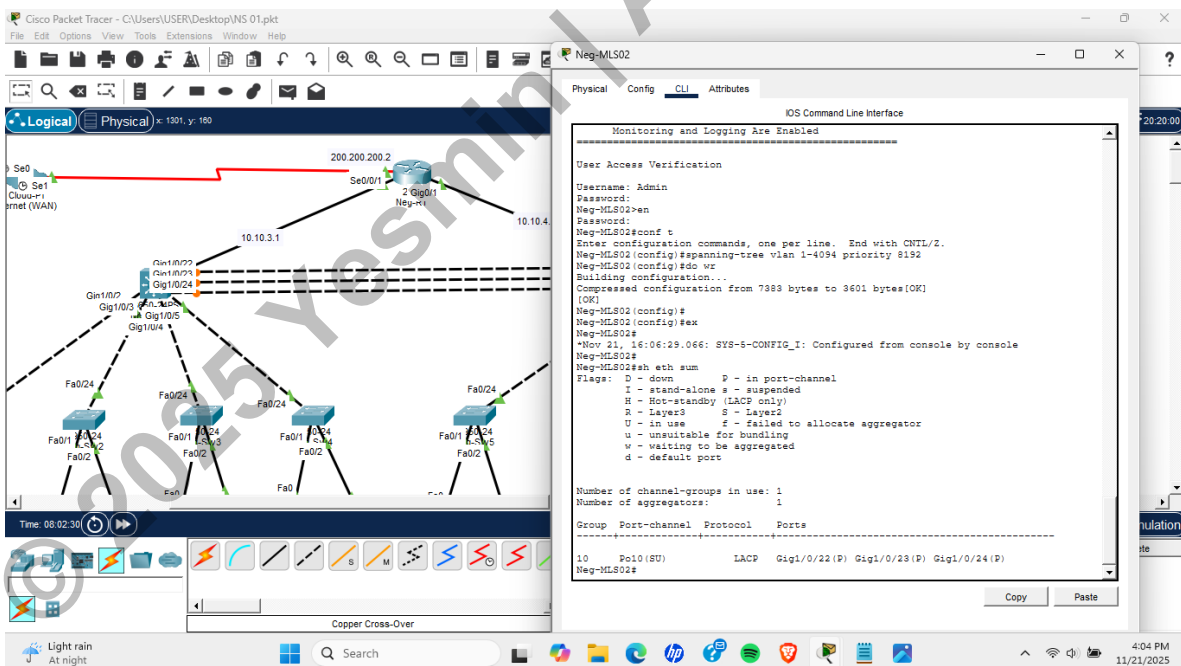


Figure 28: STP and STP Results in Neg-MLS02

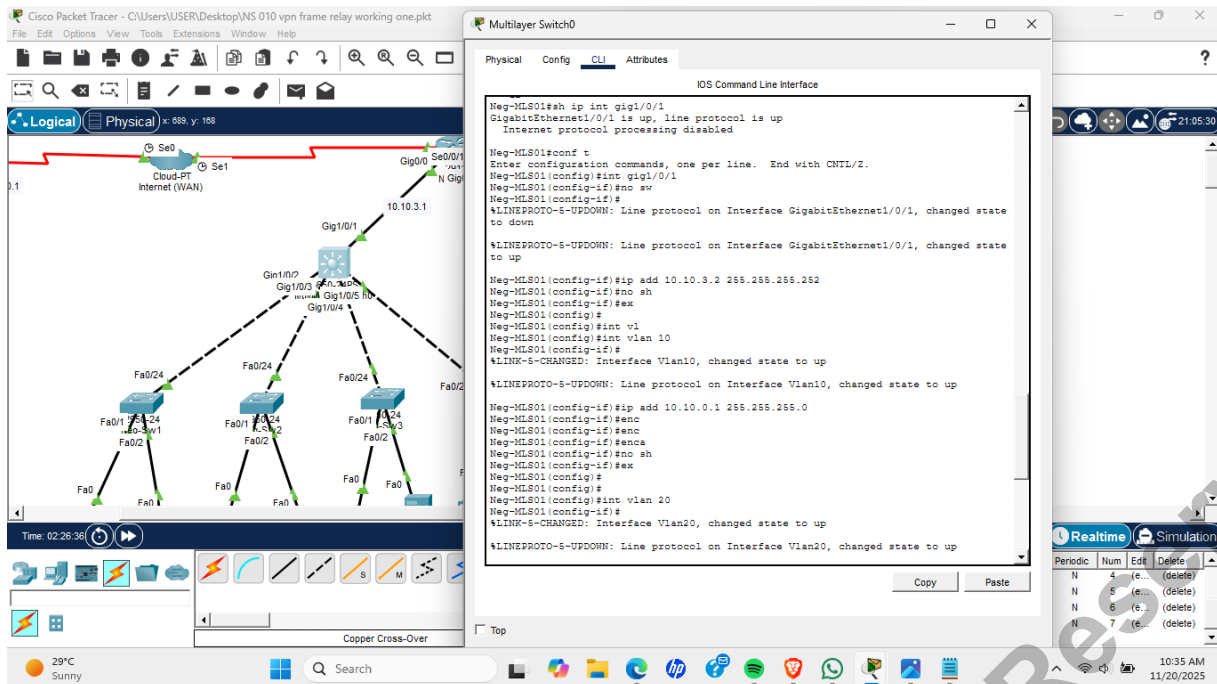


Figure 30: Inter-VLAN Neg-MLS01

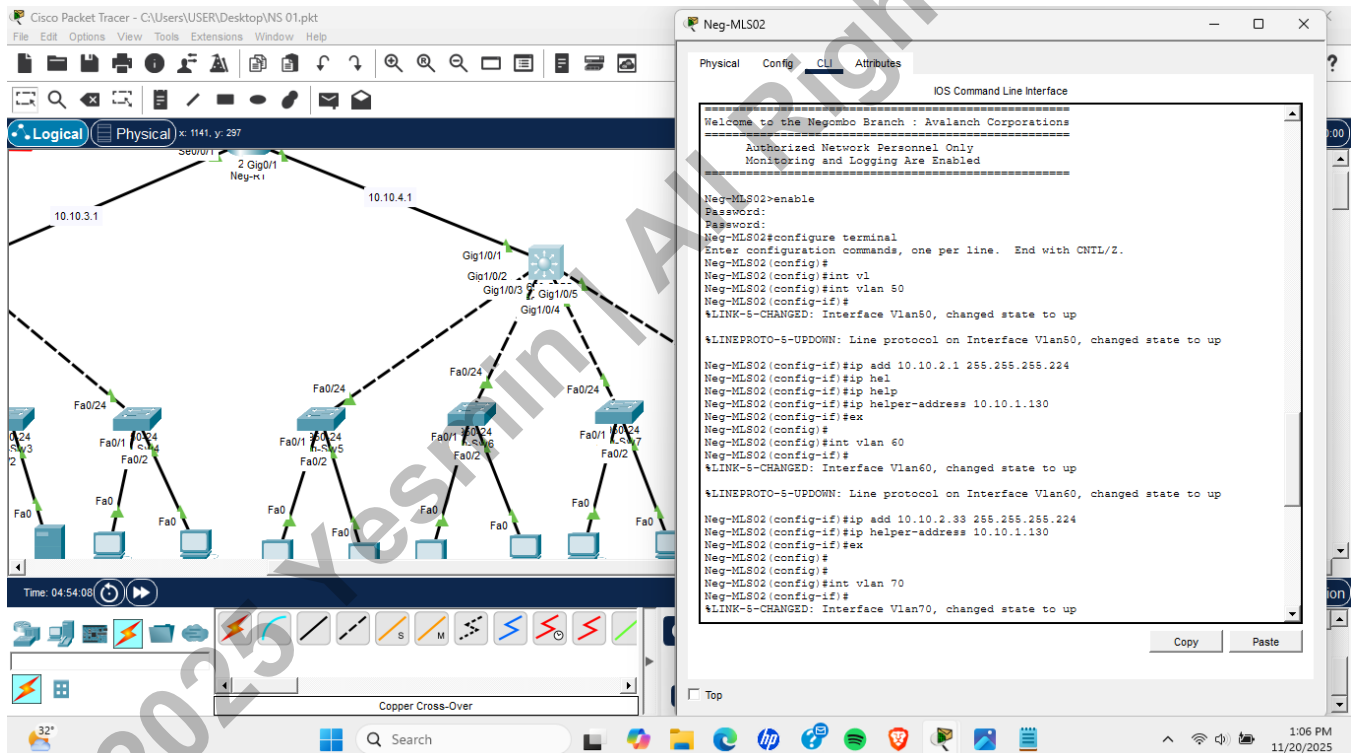


Figure 31: Inter Vlan Neg-MLS02

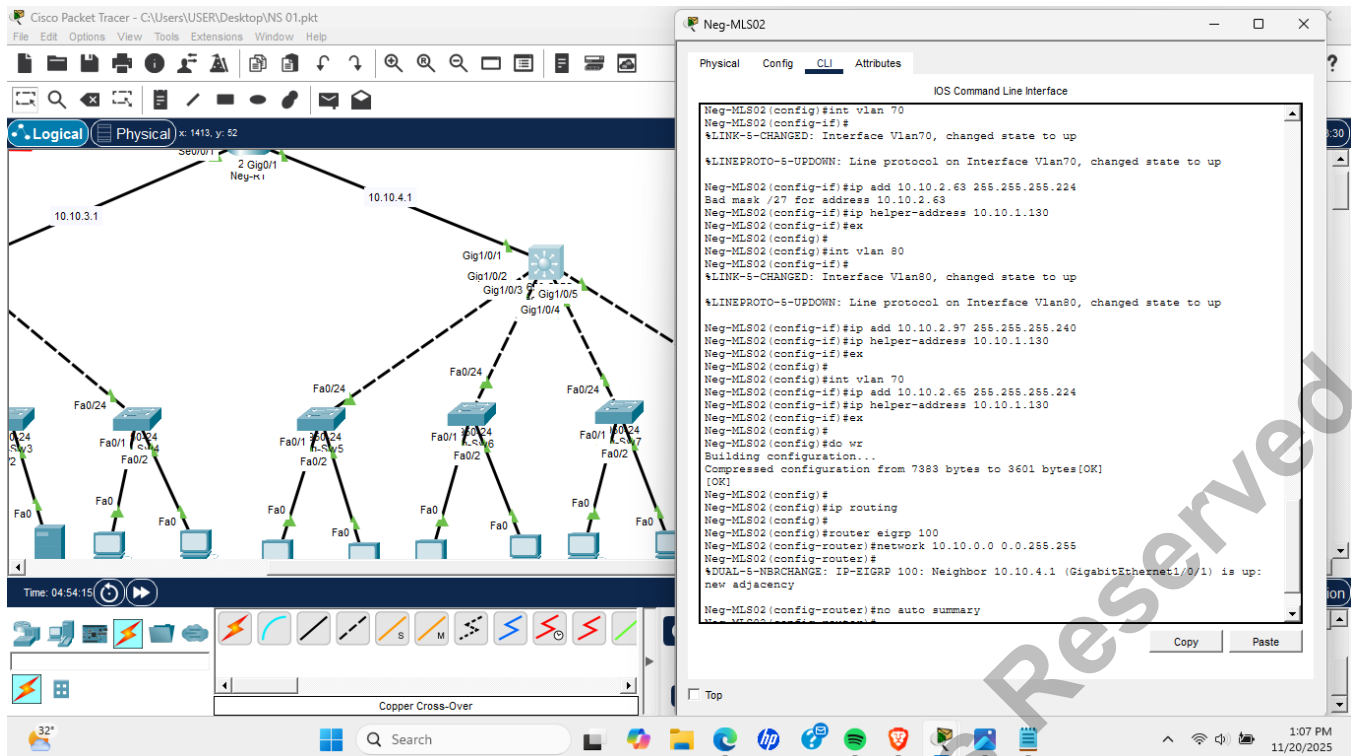


Figure 32: Inter VLAN Neg-MLS02

5.5. HSRP for Redundancy

HSRP is a redundancy protocol that is designed to group several routers into one virtual router and share an IP address. It guarantees network connectivity by assigning one router to be active and the rest to be standby so that in case primary router fails, the other router will automatically assume control without interrupting connections of the users.

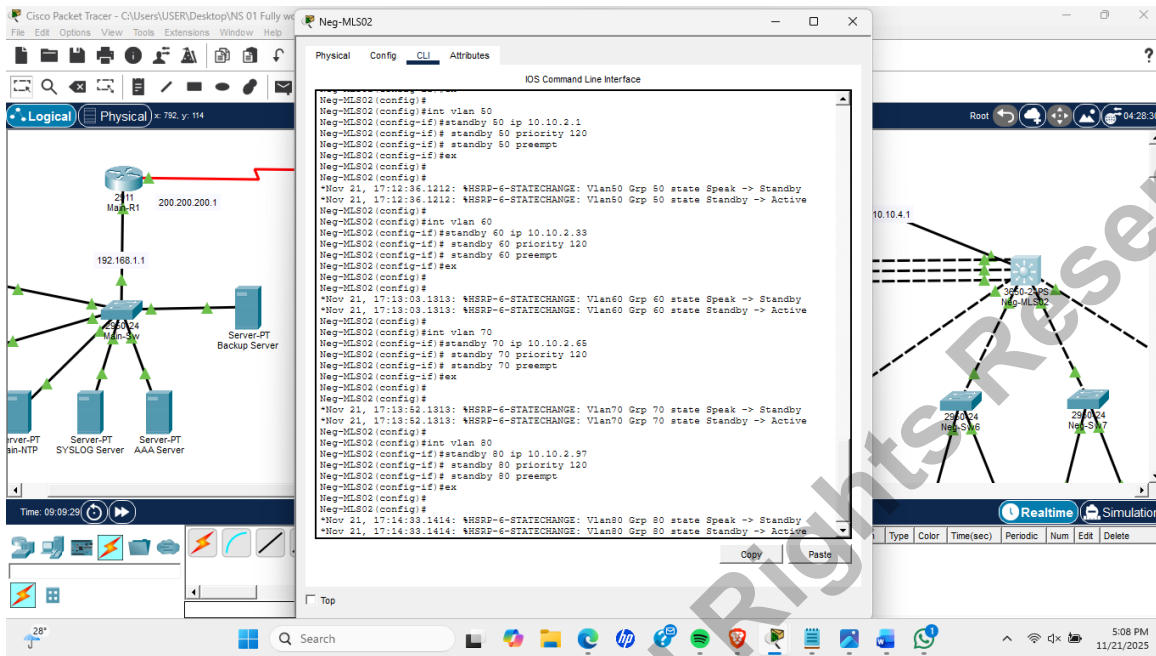


Figure 34: HSRP in Neg-MLS01

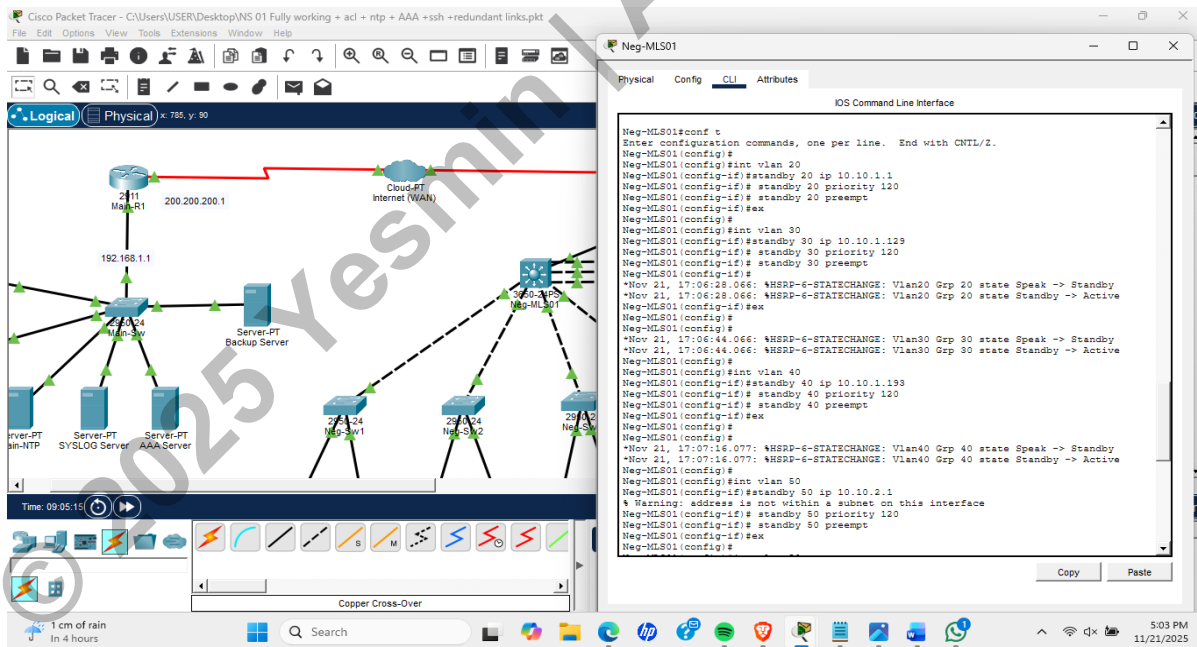


Figure 33: HSRP in Neg-MLS02

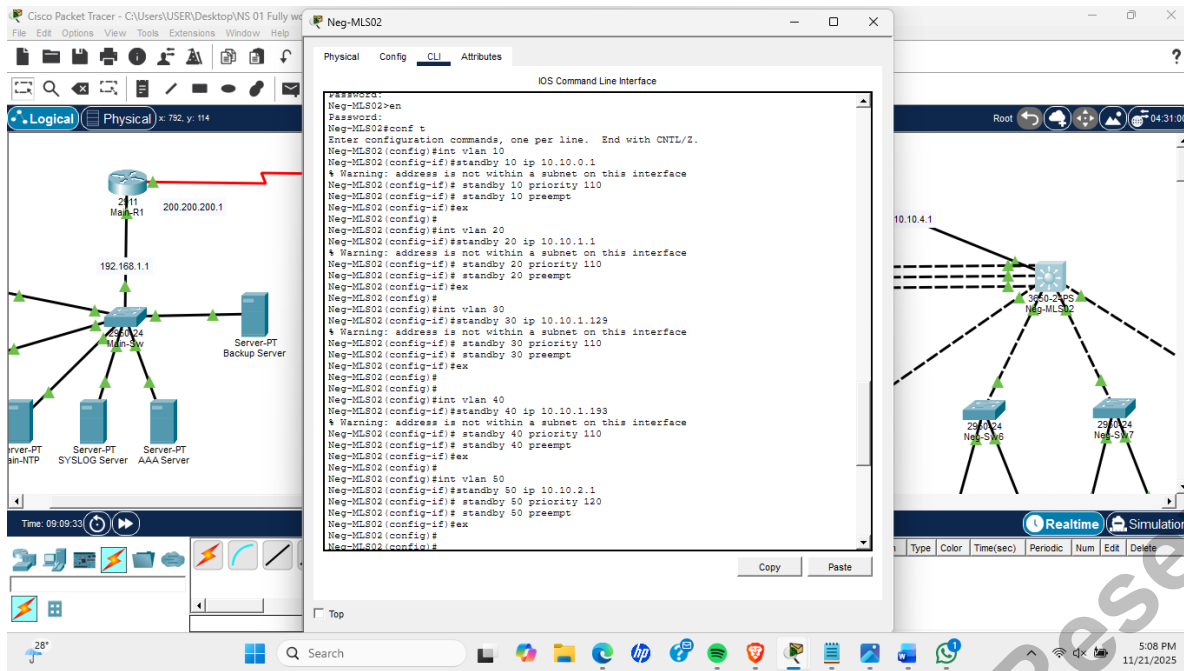


Figure 35: HSRP in Neg-MLS02

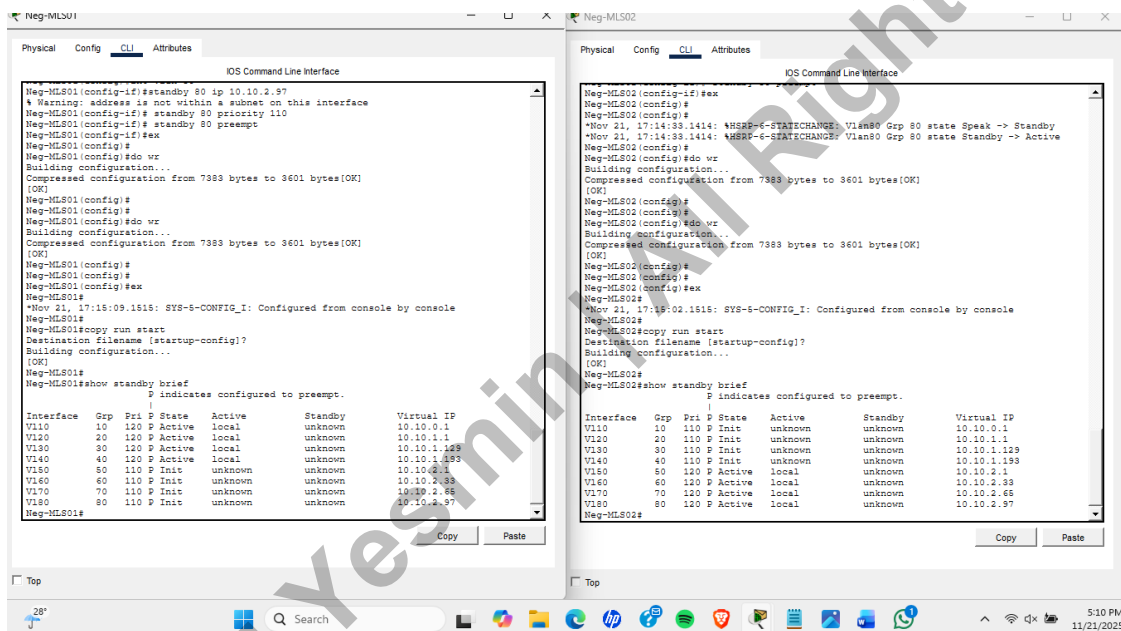


Figure 36: HSRP Results

5.6. Link Aggregation Control Protocol (LACP)

LACP is a protocol (IEEE 802.3ad) that is an open standard and is designed to provide the functionality of combining several physical switch ports into one logical channel, referred to as an Etherchannel. This routing aggregation enhances the total bandwidth and redundancy, whereby in case one physical link fails, the traffic will be routed in a different link that remains active.

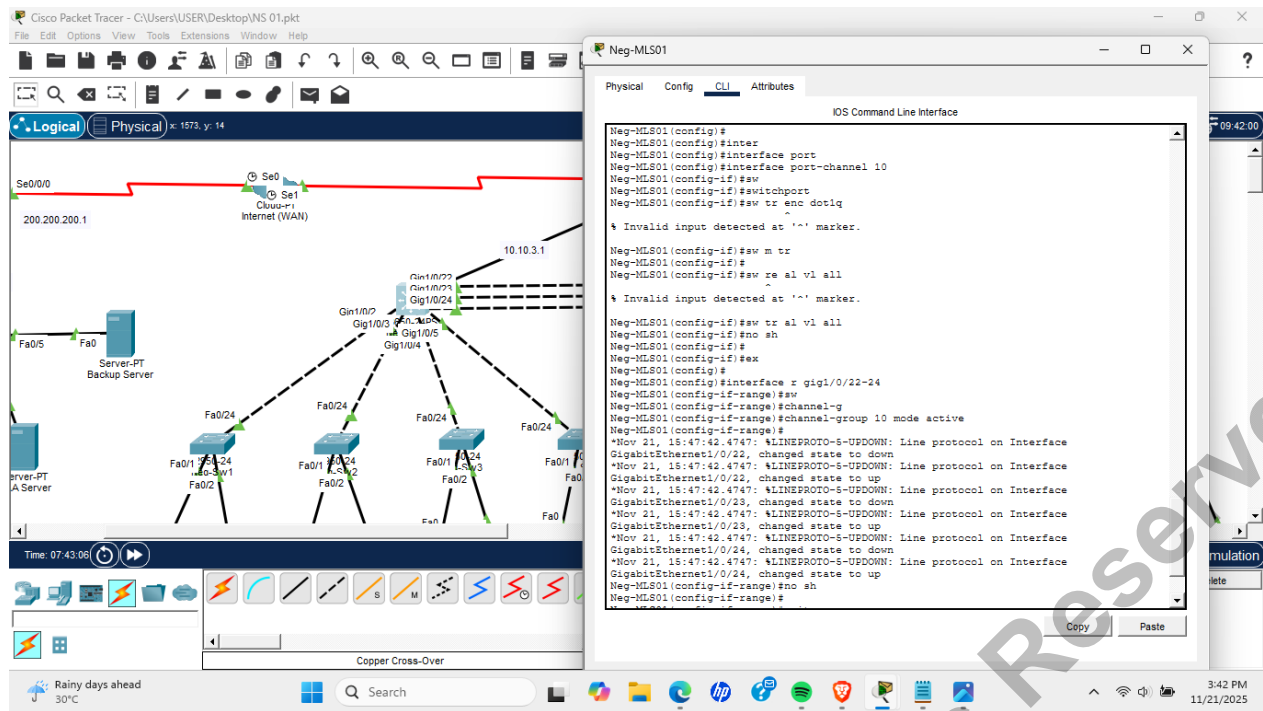


Figure 37: LACP in Neg-MLS01

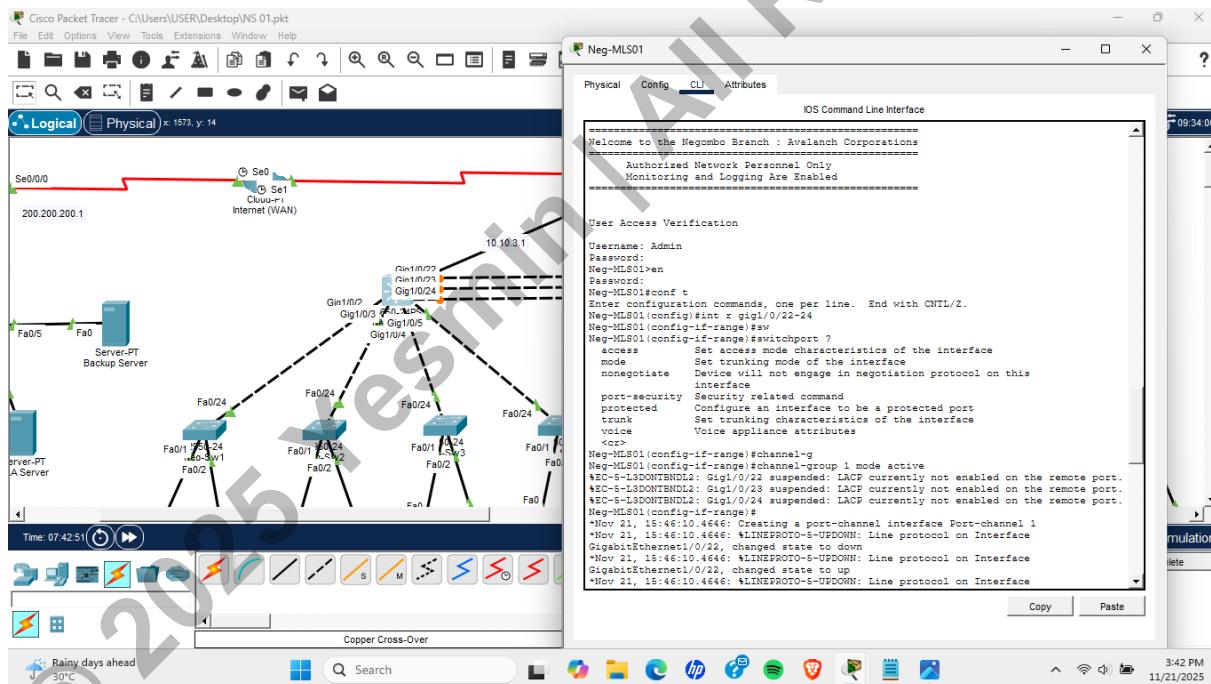


Figure 38: LACP in Neg-MLS01 pt 2

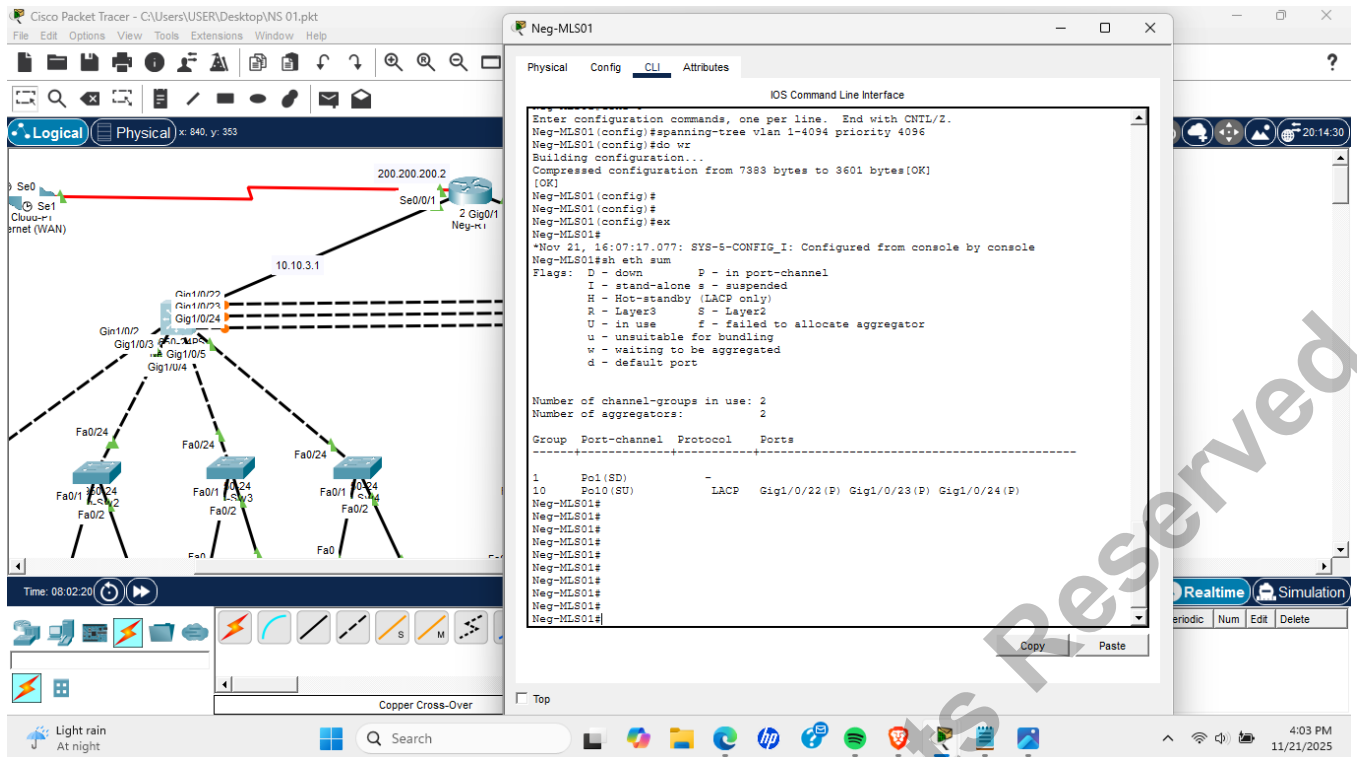


Figure 39: LACP results

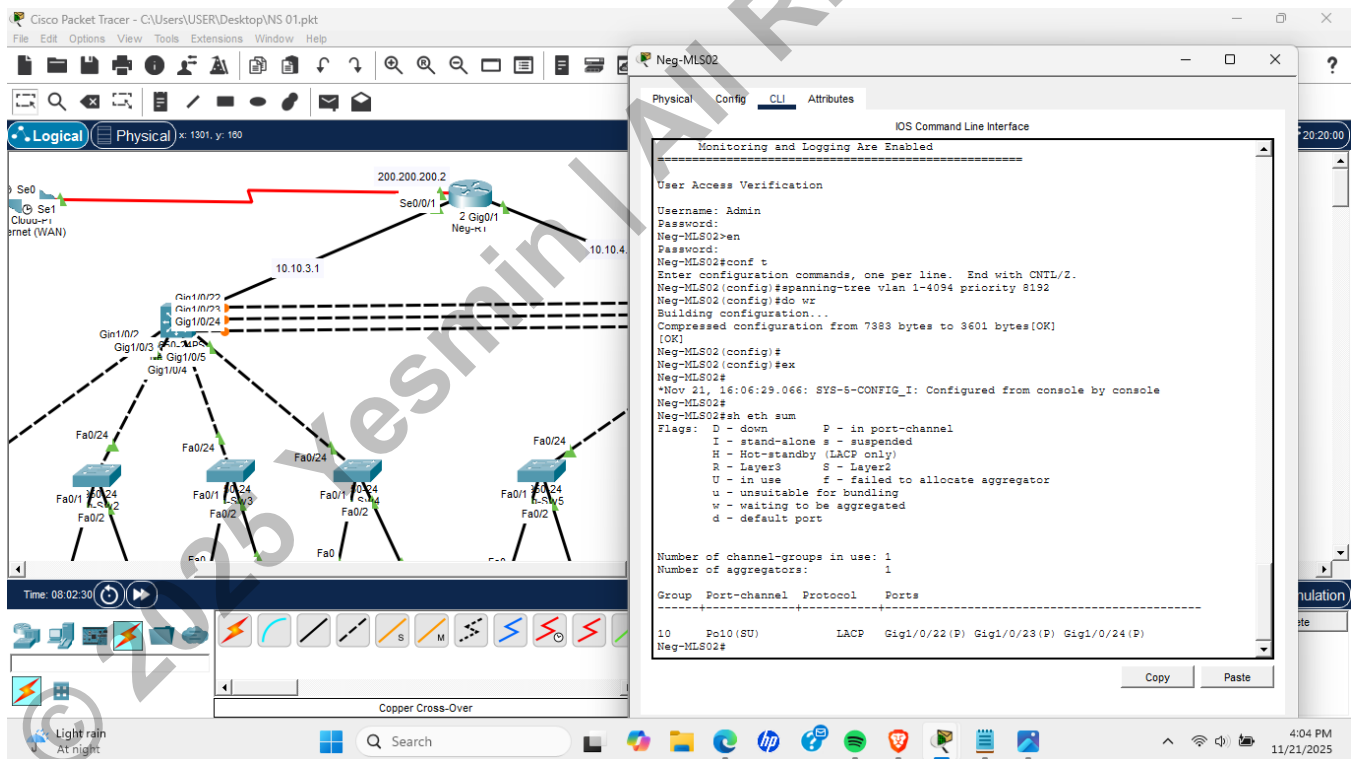


Figure 40: LACP Results in Neg-MLS02

6. Network Services Implementation

6.1. DHCP Configuration

6.1.1. DHCP Design Strategy

Dynamic Host Administration Protocol (DHCP) is used to automatically assign IP addresses, which saves the administrative work and configuration mistakes [30]. The main branch has a centralized DHCP server where issues are addressed to all the departments on the two branches.

Advantages of Centralized DHCP

- **Easy to Manage:** There is a single point of configuration that is used across all network.
- **Consistency:** There are uniform policies of address allocation which apply throughout the organization.
- **Low overhead:** Eliminates the use of DHCP servers at each site.
- **Centralized Logging:** Transactions of the DHCP are stored at a single point.
- **Cost-Efficient:** The use of a single server is effective in the management of multiple scopes [31].

6.1.2. DHCP Pool Configuration

The screenshot shows the 'Neg-IT-DHCP' configuration window. The 'Services' tab is active, displaying the DHCP configuration for the 'FastEthernet0' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '0.0.0.0'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '0' and the 'Subnet Mask' is '0'. The 'Maximum Number of Users' is '0'. The 'TFTP Server' is '0.0.0.0' and the 'WLC Address' is '0.0.0.0'. Below the configuration fields is a table of configured DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0.0.0.0	0.0.0.0
Admin Dep:	10.10.2.97	0.0.0.0	10.10.2.100	255.255.255.240	12	0.0.0.0	0.0.0.0
SOC Dep:	10.10.2.63	0.0.0.0	10.10.2.65	255.255.255.224	31	0.0.0.0	0.0.0.0
HR Dep:	10.10.2.33	0.0.0.0	10.10.2.35	255.255.255.224	29	0.0.0.0	0.0.0.0
Finance Dep:	10.10.2.1	0.0.0.0	10.10.2.5	255.255.255.224	27	0.0.0.0	0.0.0.0
C&C Dep:	10.10.1.193	0.0.0.0	10.10.1.195	255.255.255.192	61	0.0.0.0	0.0.0.0
IT Dep:	10.10.1.129	0.0.0.0	10.10.1.130	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 41: DHCP Server Pools

6.1.3. DHCP Relay Configuration

Due to the distance between the DHCP server (main branch) and the clients (new branch), a DHCP relay (IP helper-address) is required in the VLAN interfaces of the core switch [32].

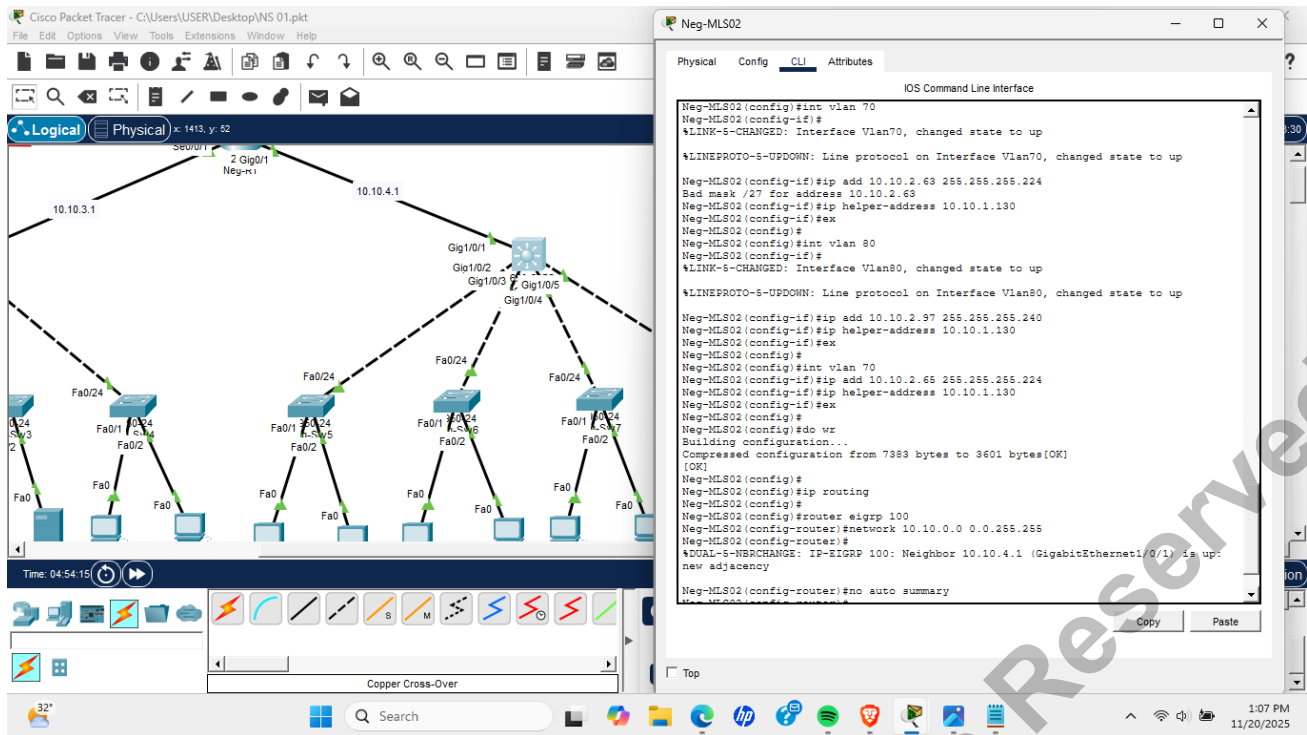


Figure 42: DHCP Relay on the Neg-MLS02

How does DHCP Relay work?

1. Client sends a DHCP DISCOVER in its local network.
2. The router/ multilayer switch accepts the broadcast and routes it as a unicast to the DHCP server at 192.168.1.10 through IP helper address.
3. The server of the DHCP responds with an address that belongs to the corresponding pool and the switch sends the answer of the server to the client [33].

6.2. NTP Configuration

Network Time Protocol (NTP) keeps devices on network in synchronization of time which is necessary in:

- **Security Event Correlation:** Proper correlation of security events is possible with proper timestamps.
- **Log Analysis:** This service brings sequential sorting of events across devices.
- **Authentication:** Time based authentication protocols (Kerberos, certificate

validation).

- **Compliance:** Met regulatory requirements to have accurate audit trails.
- **Troubleshooting:** Provides proper time sequencing of network troubleshooting.

Time Discrepancy Effect : Studies have determined that a ten seconds drift is enough to make security event correlation futile [37]. Therefore, security operations need less than a second accuracy.

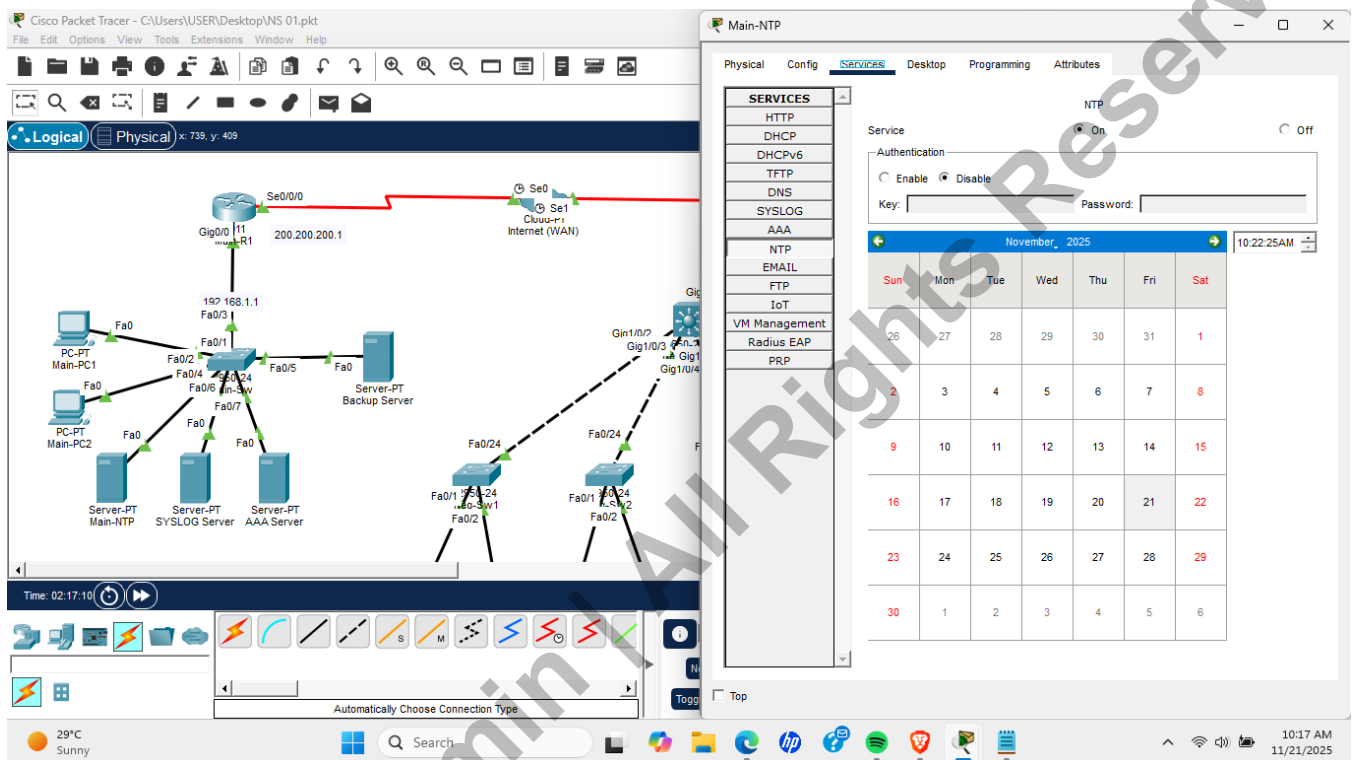


Figure 43: NTP Server

Every network equipment (switch and router) is synchronized to the configured NTP sources

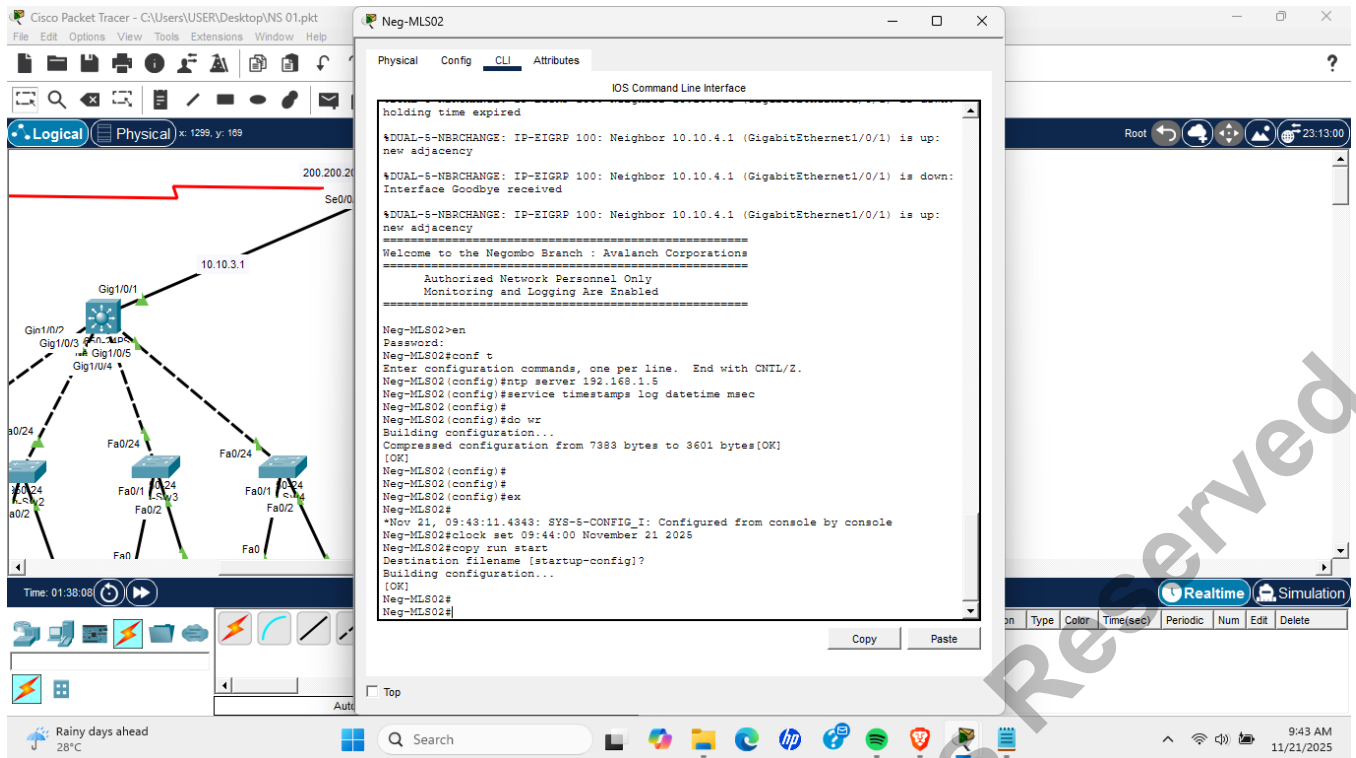


Figure 46: NTP Configuration Nrg-MLS02

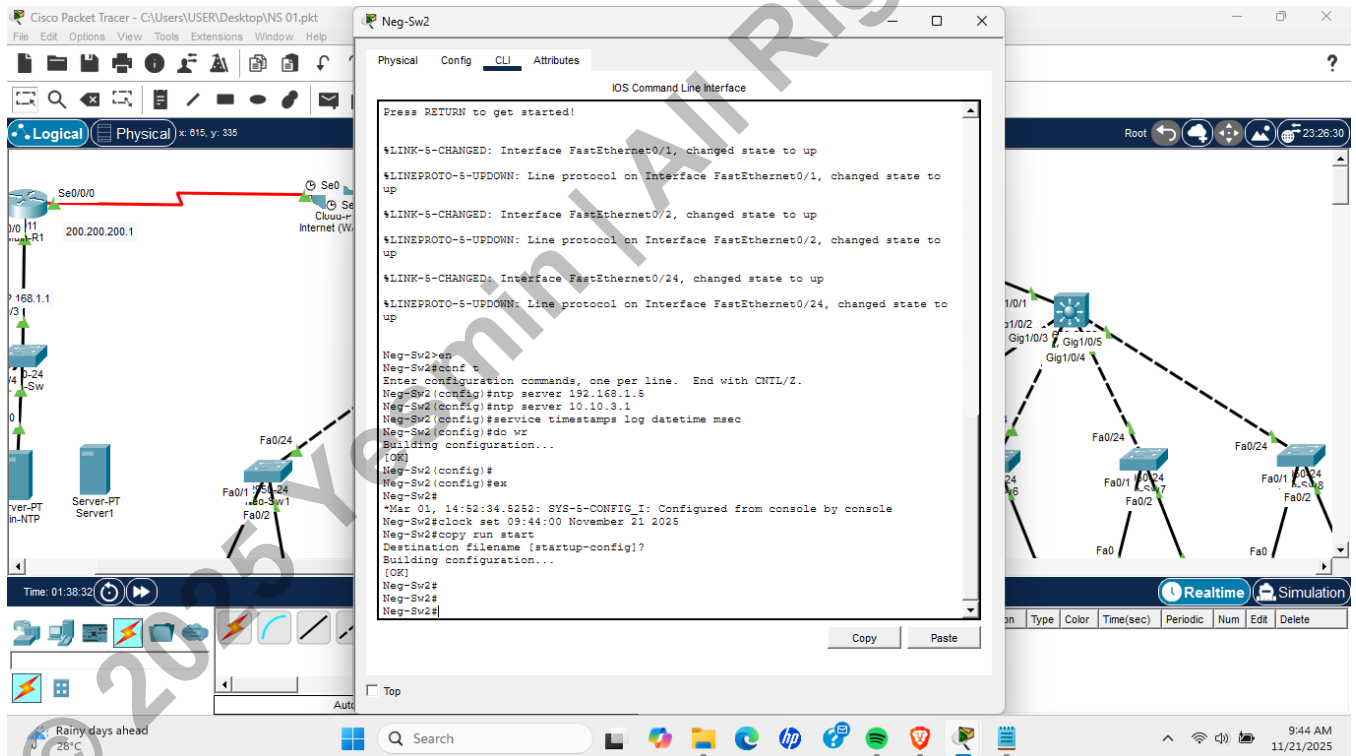


Figure 47: NTP Configuration Neg-Sw2

6.2.1. NTP Verification and Monitoring

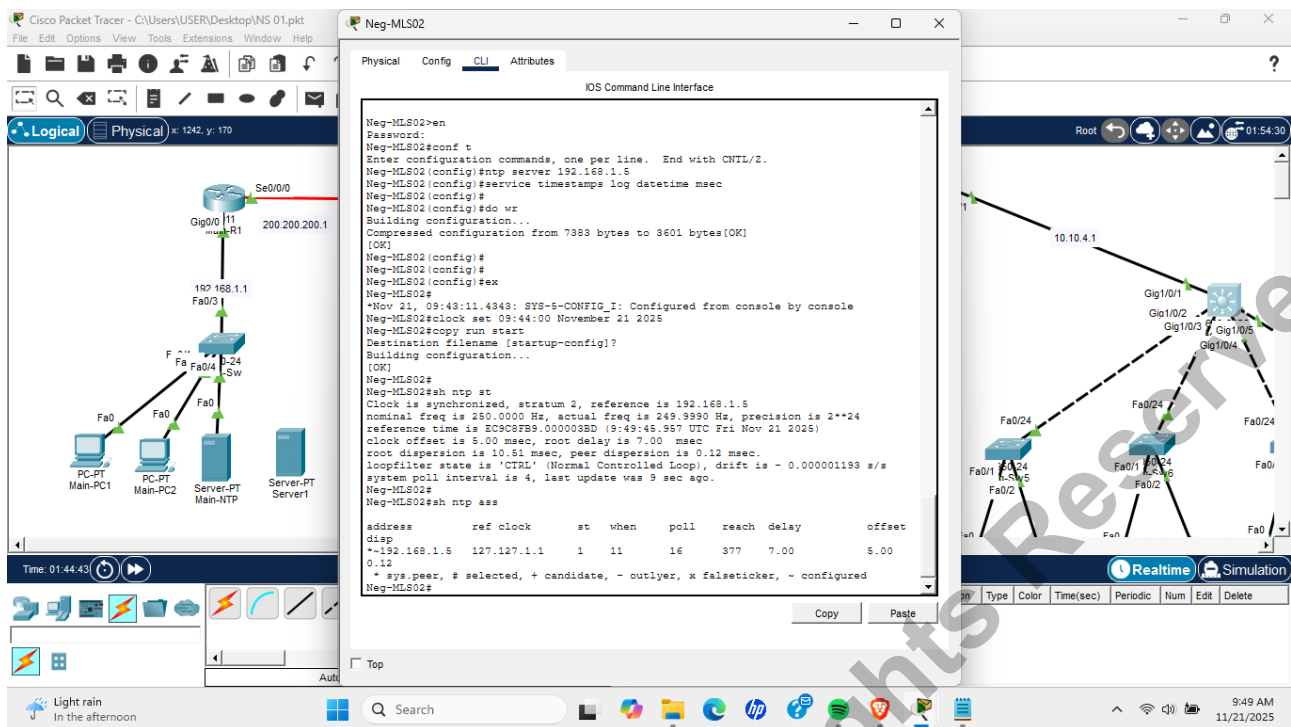


Figure 48: NTP Statistics Neg-MLS02

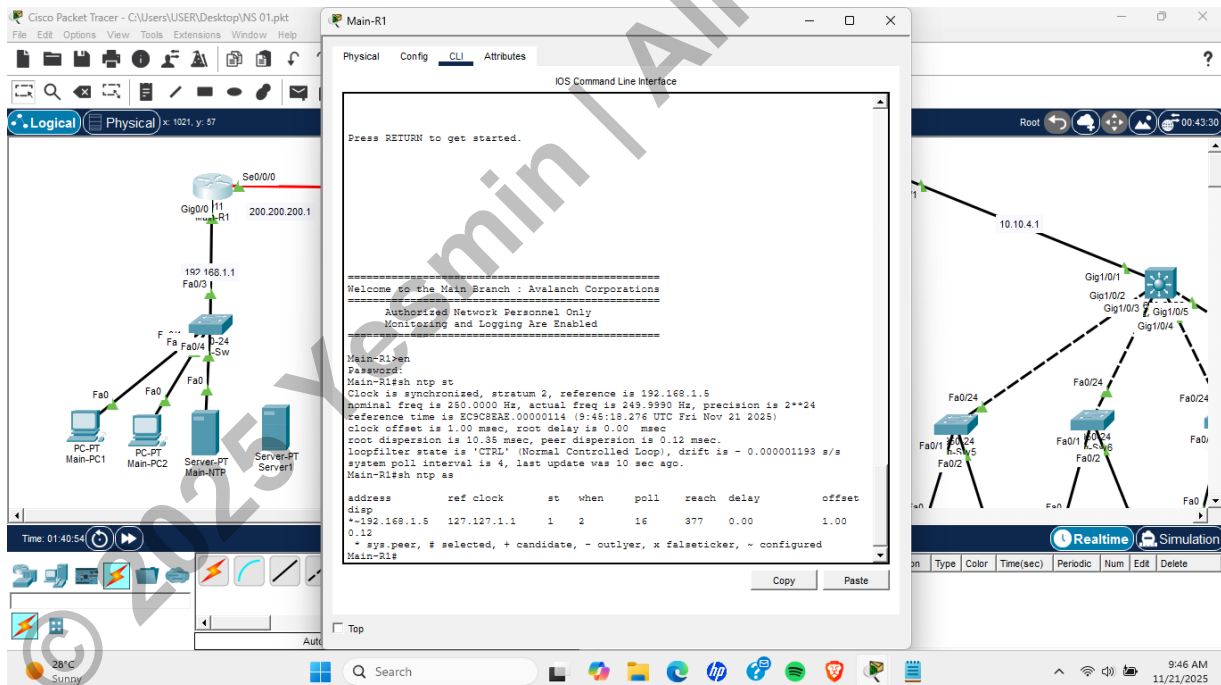


Figure 49: NTP Result Main-R1

Syslog can use all messages that are sent by network devices and logs them under a single repository, therefore, supporting:

- **Security Monitoring:** Allows real time security event detection.
- **Troubleshooting:** It is a historical record of what has happened to the network.
- **Compliance:** Provides an audit trail which is mandated by regulatory frameworks (ISO 27001, PCI DSS).
- **Capacity Planning:** Enables us to analyze the trends of network usage.
- **Forensic Analysis:** Aids capabilities of post-incident investigation.

Regulatory Requirements

- The minimum length of log retention of PCI DSS is 90 days.
- HIPAA requires a period of six years, whereas SOX takes seven years.
- The GDPR retention times take different values based on the kind of data [40].

6.3.1. Syslog Server Implementation

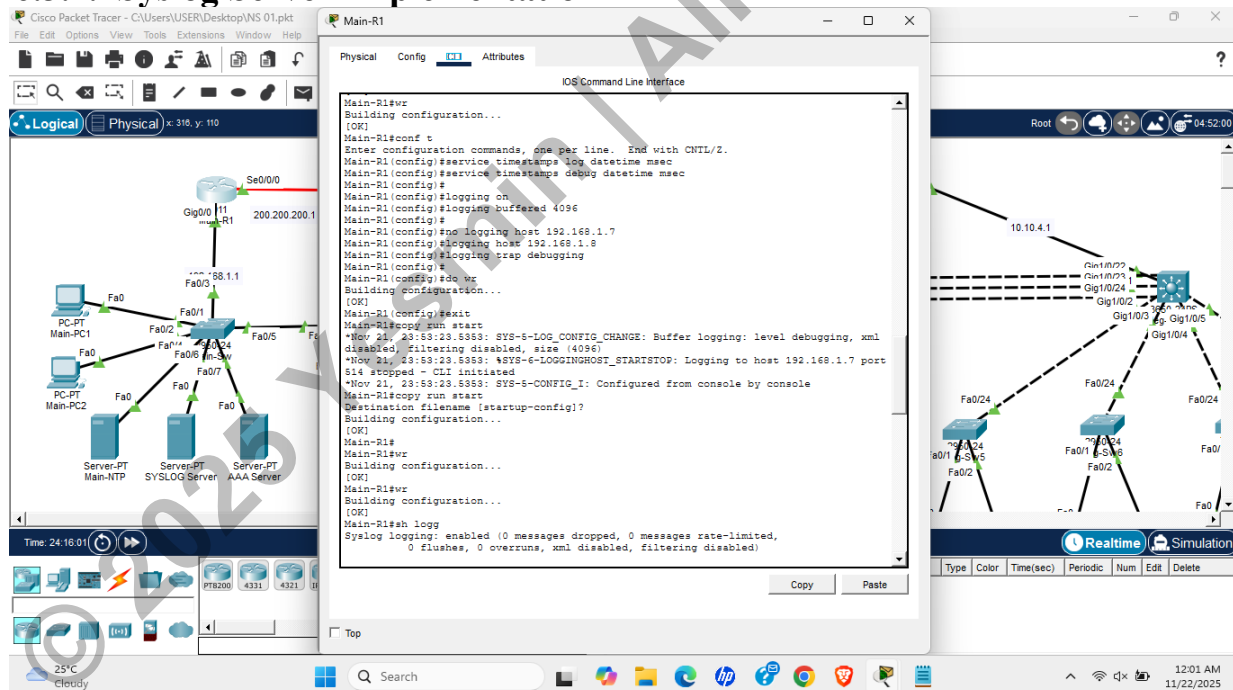


Figure 52: Syslog Config Main-RL

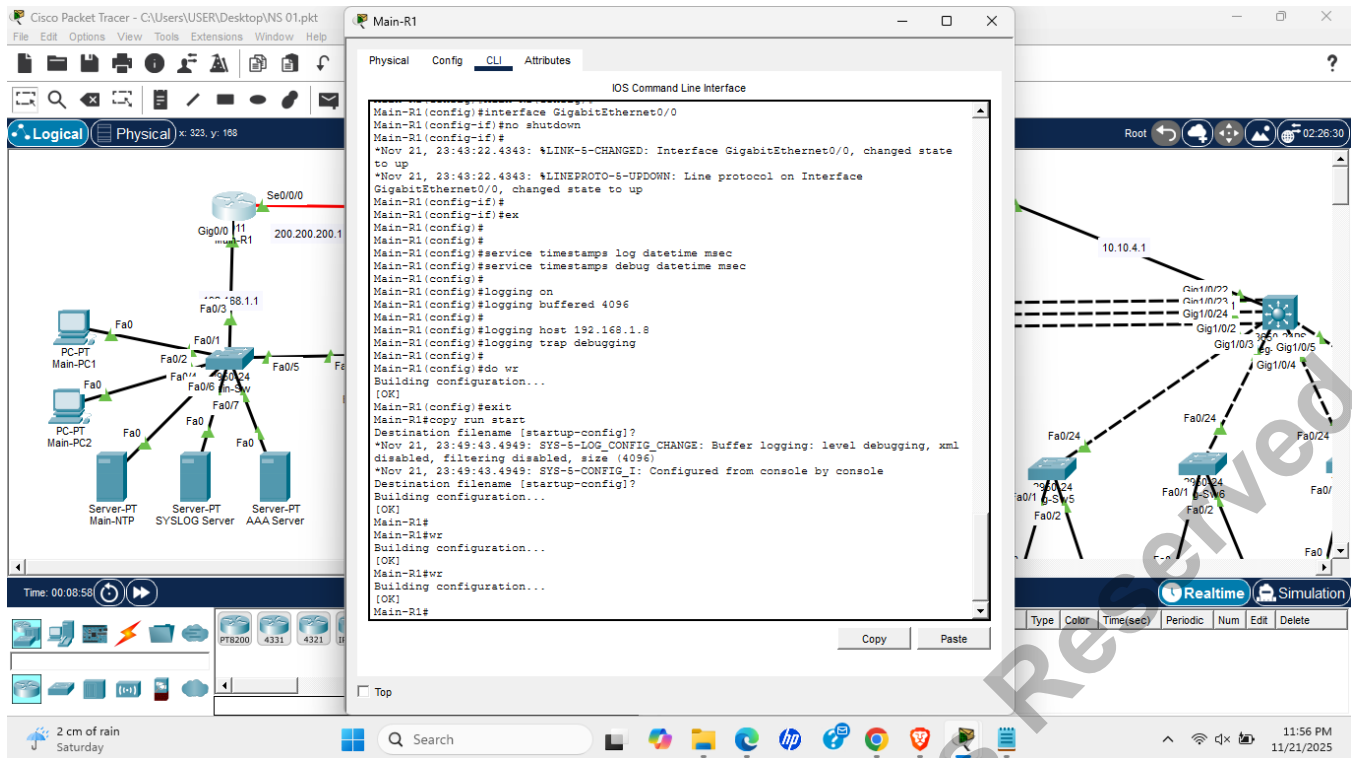


Figure 53: SYSLOG Config Main-R1 cont

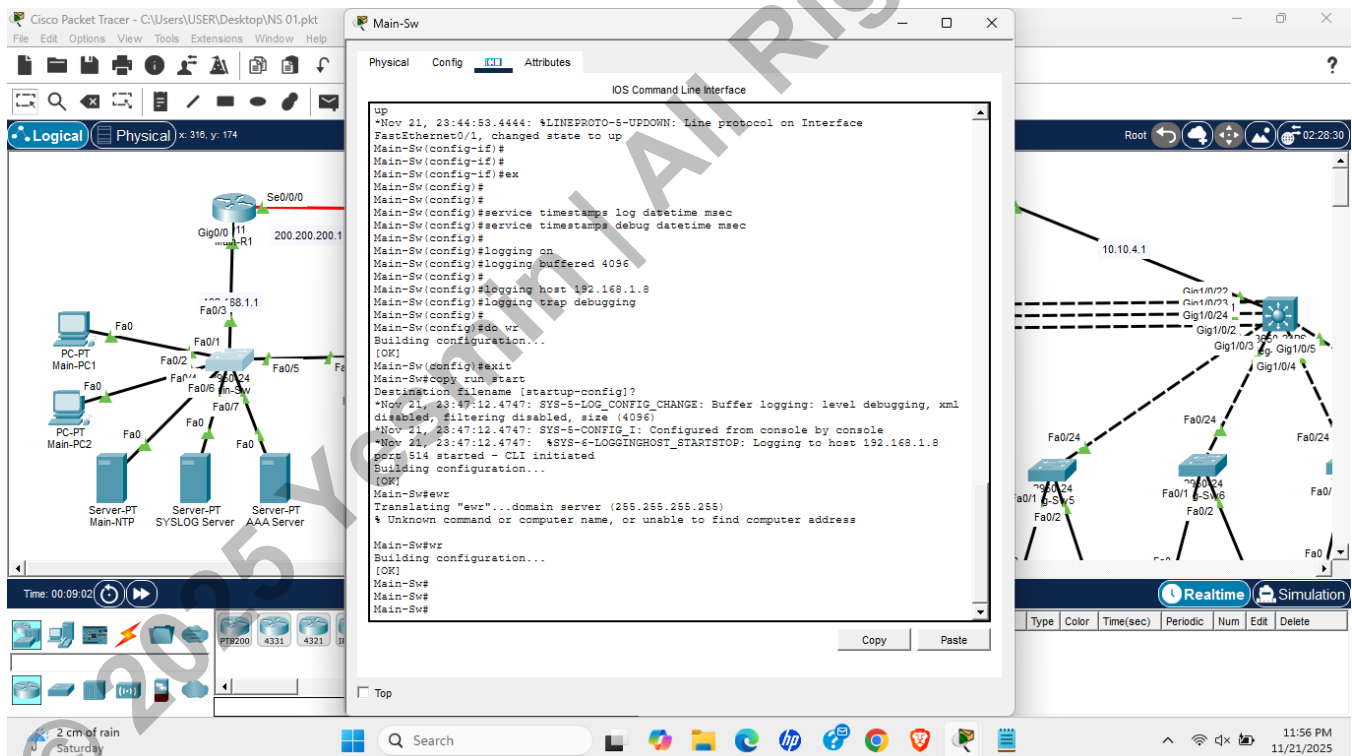


Figure 54: SYSLOG Config in Main-Sw

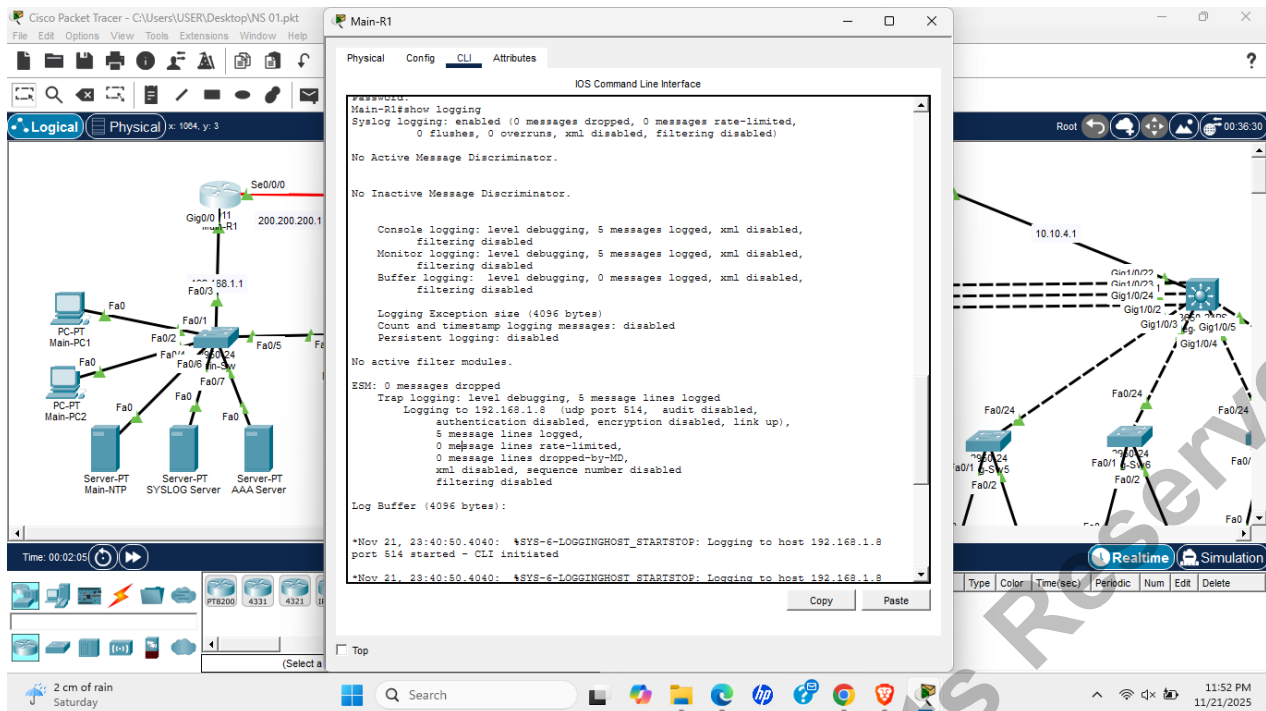


Figure 55: SYSLOG Result

Server Specifications:

- Storage: 500 GB (with a 90 day retention period)
- Message Rate: The accommodation of message rate of 10,000 messages per second.
- Rotation of logs: Every midnight UTC daily.
- Backup: (Weekly) to external storage.
- Retention Policy: Retention of 90 days, an extra one year of archived storage layer.

6.3.2. Log and Security Events Categories

1. Authentication Events

- Success and attempt to log in (successful and failed).
 - Privilege level changes
 - Things that break AAA authentication.

2. Configuration Changes:

- Configuration mode entry/exit mode.
- Specific command execution
- Configuration save

3. Interface Events:

- Link up/down transitions
- Error rate thresholds surpassed.
- Duplex and speed mismatches

4. Security Events:

- Port security violations
- ACL hits (denied traffic)
- DHCP snooping violations
- STP Topology Alterations

5. System Events:

- Device reboots
- Hardware failures
- Power supply issues

The screenshot displays the Cisco Packet Tracer interface with a network topology on the left and the SYSLOG Server configuration and log output on the right. The network topology includes a central router (R1) connected to two PCs (PC-PT Main-PC1 and PC-PT Main-PC2) and two servers (Server-PT Main-NTP and Server-PT SYSLOG Server). The SYSLOG Server is configured with the following services: HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and RRP. The log output shows a series of messages from the router to the SYSLOG Server, including configuration changes, logging level debugging, and line protocol status updates.

Time	HostName	Message
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-S-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, ...
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-S-CONFIG_I: Configured from console by console
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-S-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.8 port 514 starte...
11.21.2025 11:52:51.168 PM	200.200.200.2	%SYS-S-LOGGINGHOST_STARTSTOP: ...
11.21.2025 11:53:23.896 PM	192.168.1.1	%SYS-S-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, ...
11.21.2025 11:53:23.896 PM	192.168.1.1	%SYS-S-LOGGINGHOST_STARTSTOP: ...
11.21.2025 11:53:23.896 PM	192.168.1.1	%SYS-S-CONFIG_I: Configured from console by console
11.21.2025 11:54:19.540 PM	200.200.200.2	%SYS-S-CONFIG_I: Configured from console by console
11.22.2025 11:54:22.261 PM	192.168.1.1	%LINEPROTO-S-UPDOWN: Line protocol on...
11.22.2025 11:54:22.201 PM	192.168.1.1	%DUAL-S-NBRCHANGE: P-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: P-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: P-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%LINEPROTO-S-UPDOWN: Line protocol on...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: P-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: P-EIGRP 100: ...

Figure 56: SYSLOG Server log

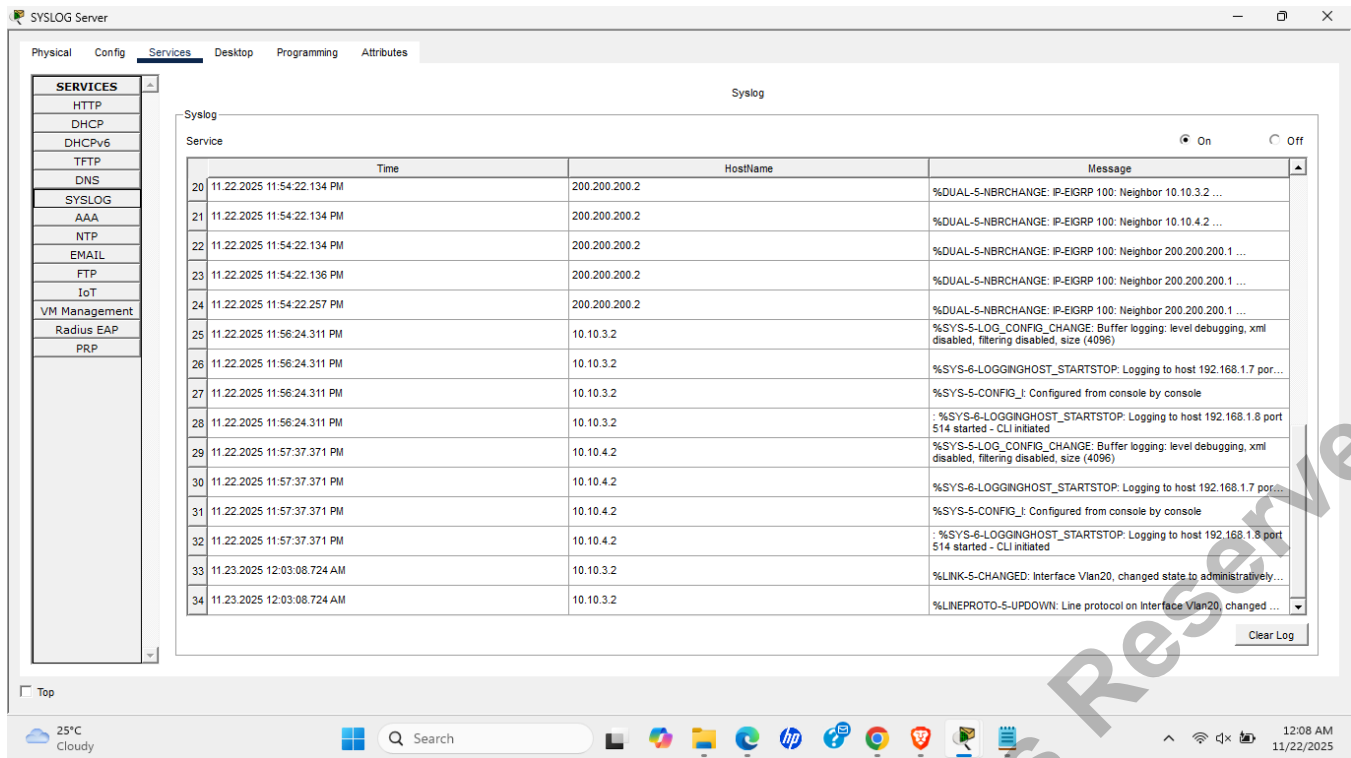


Figure 57:SYSLOG Server log cont.

6.4. AAA Configuration

Authentication, Authorization, and Accounting (AAA) is used to offer central security management to the access of network devices [42]. The AAA scheme applies the least privilege principle and accountability.

Three Components in AAA

1. Authentication: Verifying the person (Who are you?)
2. Authorization: Decides what you can do (What can you do?)
3. Accounting: Documenting user activity (What did you do?)

6.4.1. AAA Server Selection

Type of Server: RADIUS (Remote Authentication Dial-In User Service)

Justification:

- Industry Standard: ubiquitous protocol (RFC 2865)

- Encryption: Password encryption is used in transit.
- Flexibility: Multiple authentication options.
- Scalability: Processes thousands of authentication requests.
- Cost-Efficient: Open-source implementations: can be found (FreeRADIUS) [43].

TACACS+ Server not considered because of Cisco-proprietary but provides more fine-tuning command permissions.

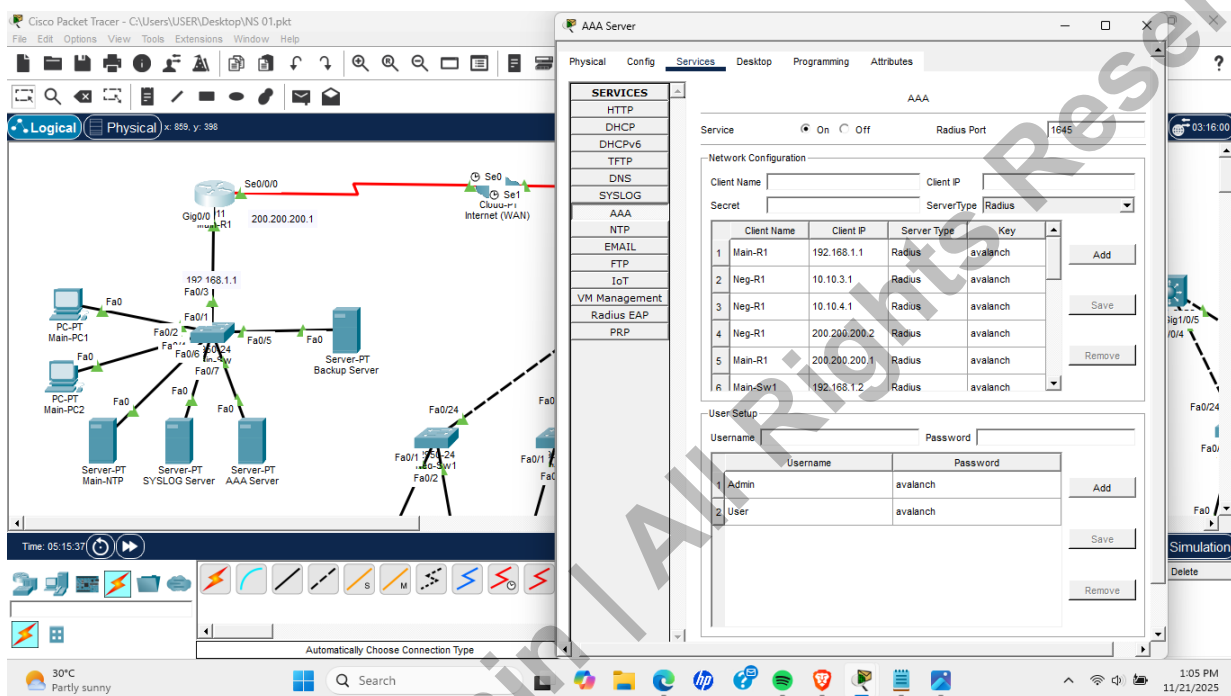


Figure 58: AAA Server

Authentication Flow

1. User connects through SSH to network device.
2. Device requests RADIUS server in order to get authenticated.
3. Credential validation is against database by RADIUS server.
4. And in case of RADIUS unavailable a local credentials fallback.
5. A successful authentication is one that is successful depending on the level of privilege [44].

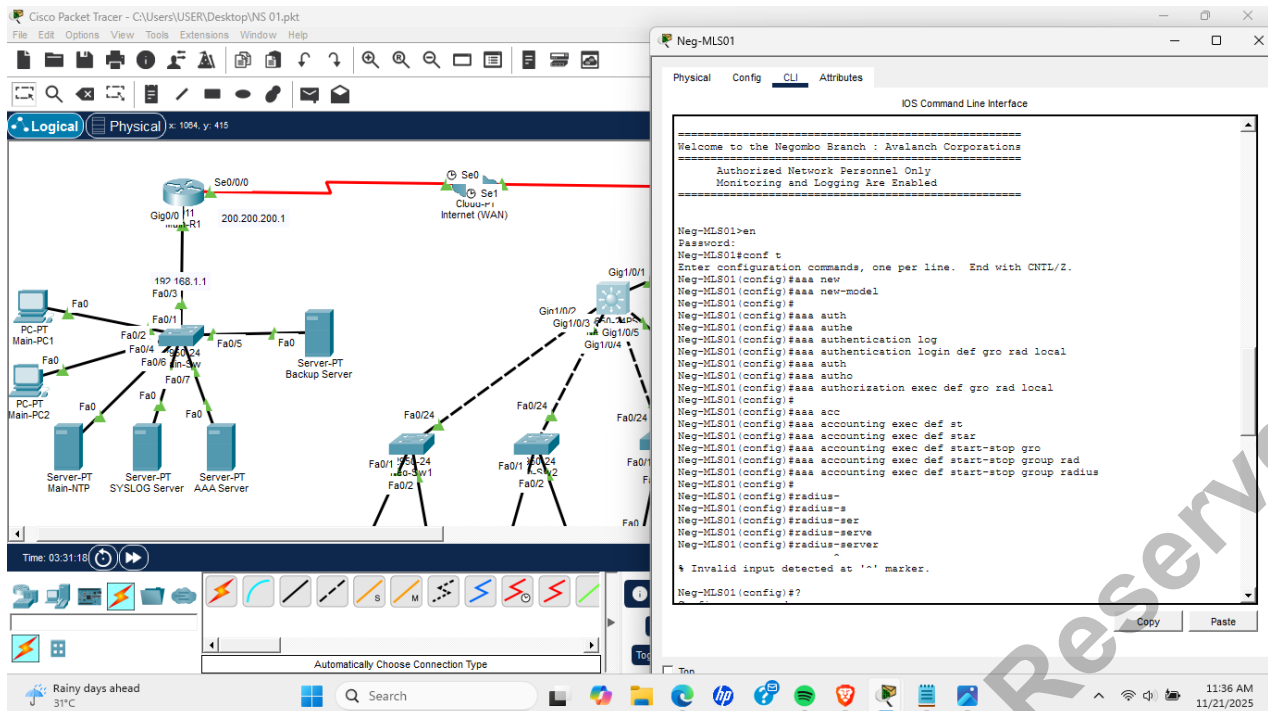


Figure 60: AAA Authorization

Command: Authorization

- Level 15: Ability to do reload, write memory, configuration.
- Level 7: Is able to perform show instructions, general troubleshooting.
- Level 1: Is only able to do show version, show ip interface brief.

6.4.3. Accounting Configuration

6.4.4. AAA Security Benefits

1. Centralized Control: Single credential management point
2. Accountability: Full audit record of administrative procedures.
3. Compliance: Enforcement of ISO 27001, access control requirements of the PCI DSS.
4. Less risk: Does away with common passwords and local accounts.
5. Quick Revocation of Credentialing: Rapid credentialing revocation at the time of employee departure [46].

7. Network Security Implementation

7.1. Access Control Lists (ACLs)

7.1.1. ACL Design Strategy

Access Control Lists block network traffic along with the criteria determining the

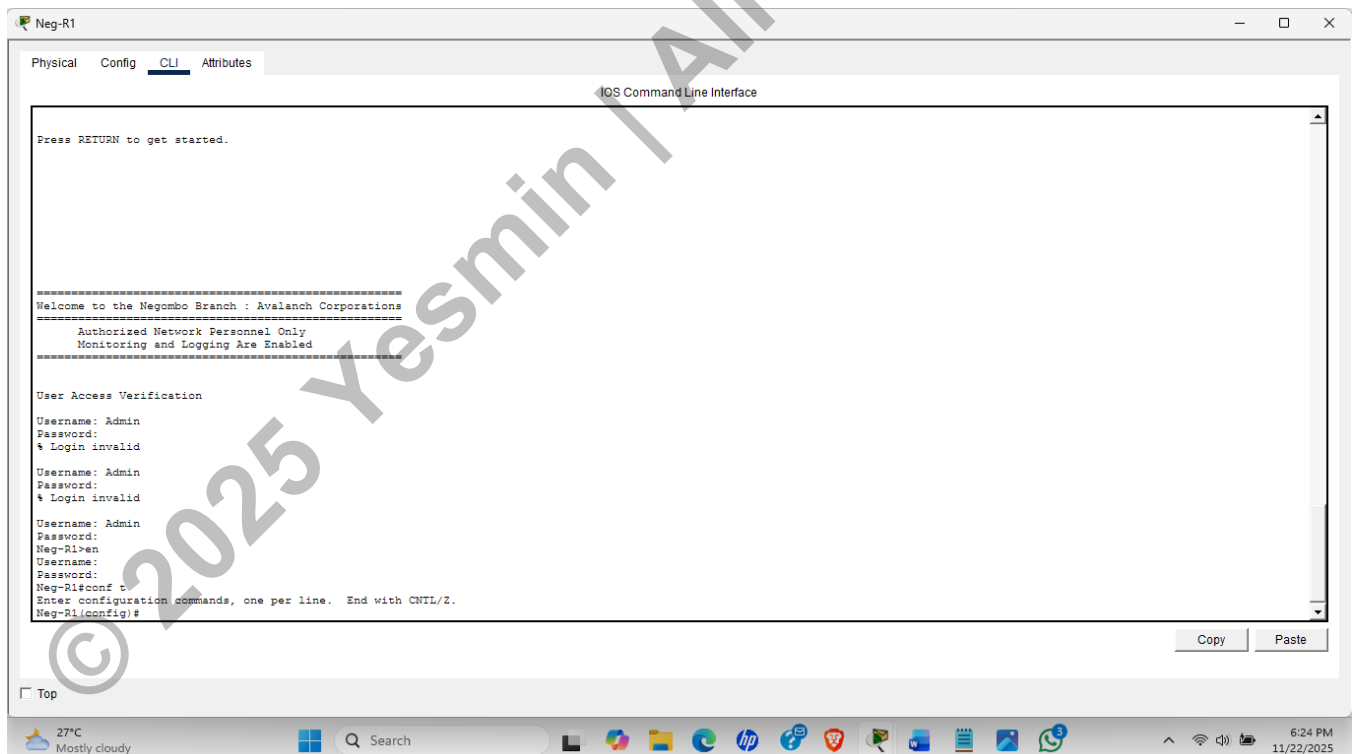


Figure 62: AAA results

security, enforcing the network layer security policies [47]. The first security requirement is that Finance (VLAN 50) and HR (VLAN 60) departments should be isolated.

Security Requirements

- Communication between finance department and any other department is not possible.
- HR department is not able to communicate with other departments.
- The finance department and the HR do not interact.
- Only internet and main branch services are available to both the departments.

Design Philosophy

- Implicit Deny: Deny all traffic except where specifically authorized.
- Least Privilege: Minimum access that is required.
- Security Design: Defense in Depth [48].

7.1.2. ACL Implementation

ACL Type Selection: Extended ACLs chosen to be in granular control due to the possibility of including Source IP address, Destination IP address, Protocol type (TCP, UDP, ICMP), and Port numbers

ACL Placement: ACLs are used inbound on VLAN interfaces (SVIs) at the core switch and traffic nearest to a source is filtered [49].

7.1.2.1. Finance Department ACL (VLAN 50)

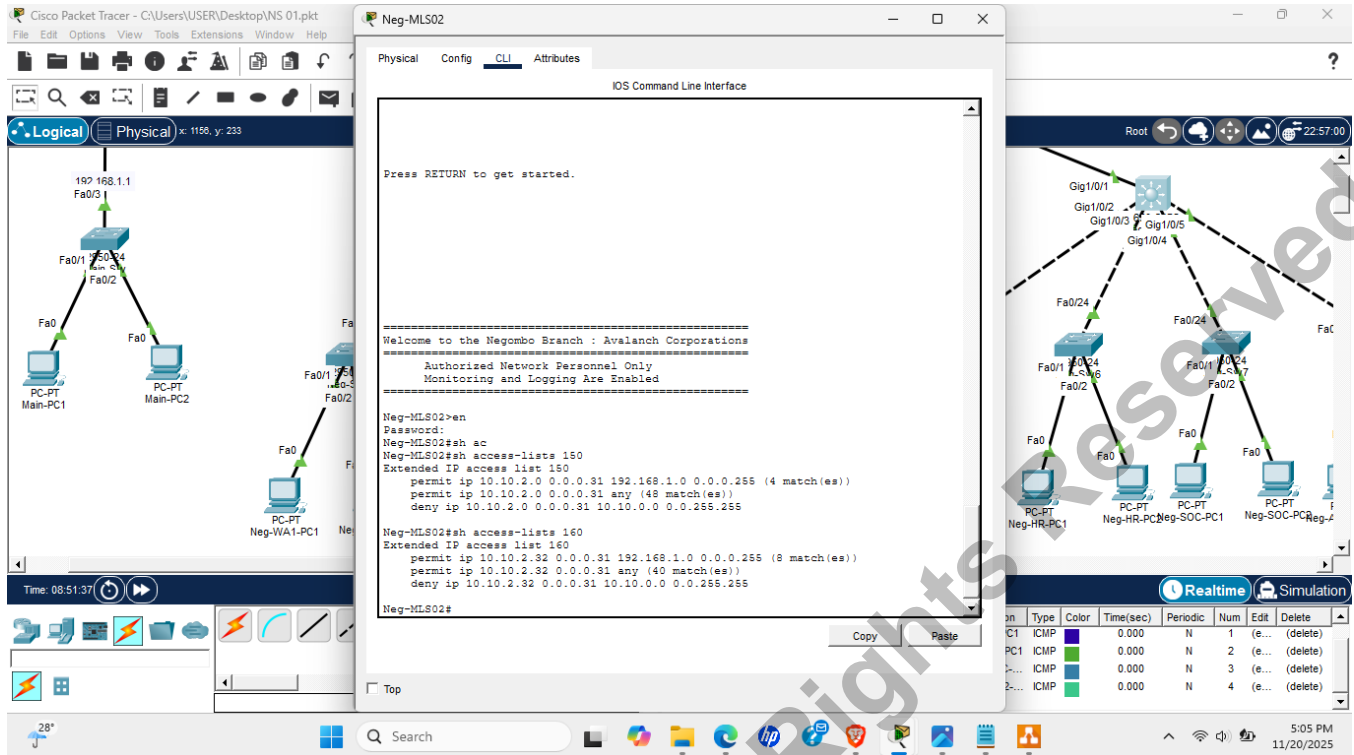


Figure 63: ACL for Finance

ACL 150 - Financial Department Restrictions:

- Line 1: Finance (10.10.2.0/27) to Main Branch (192.168.1.0/24) access to servers.
- Line 2: Allows access to the internet (to any destination which has not been matched previously)
- Line 3: Acknowledges that it denies all communication to all New Branch networks (10.10.0.0/16).
- Implicit Deny: Blocks all the remaining traffic (good practice to record denials) [50].

7.1.2.2. HR Department ACL (VLAN 60)

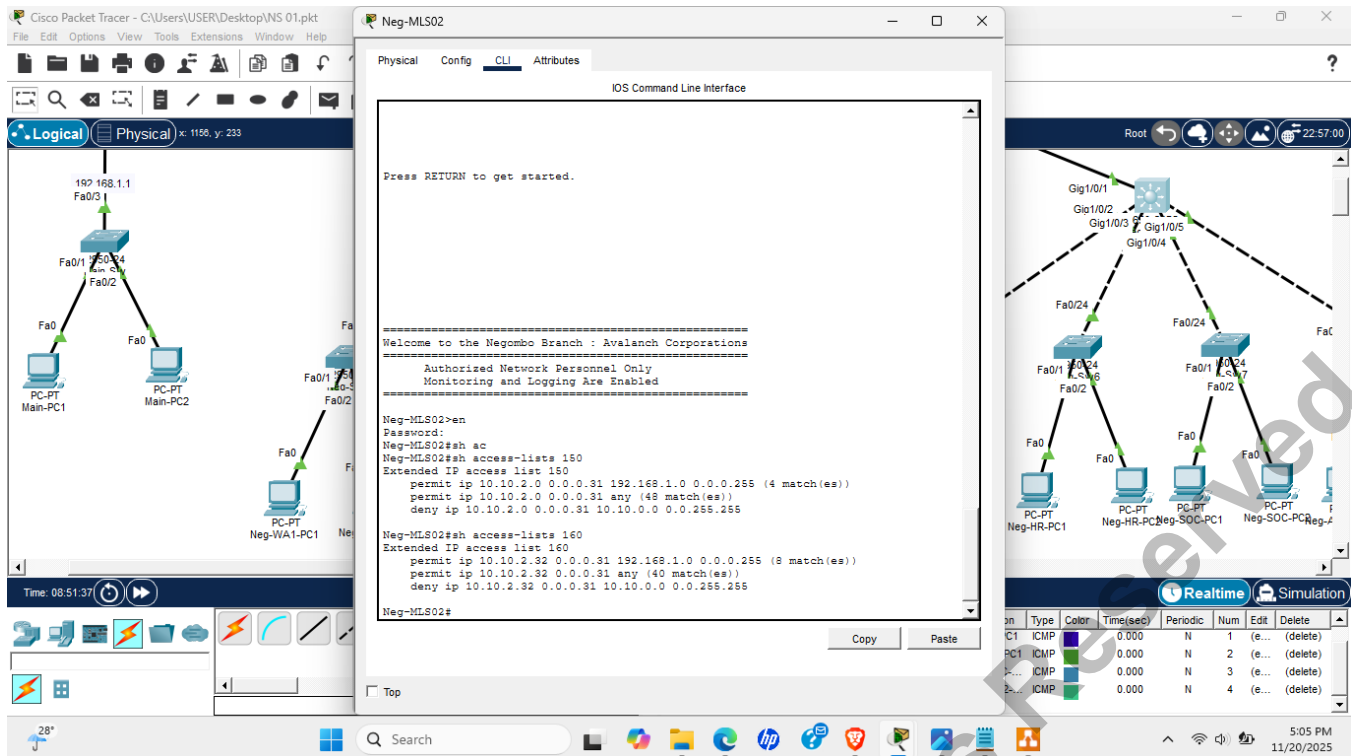


Figure 64: ACL for HR

7.1.3. ACL for Other Departments

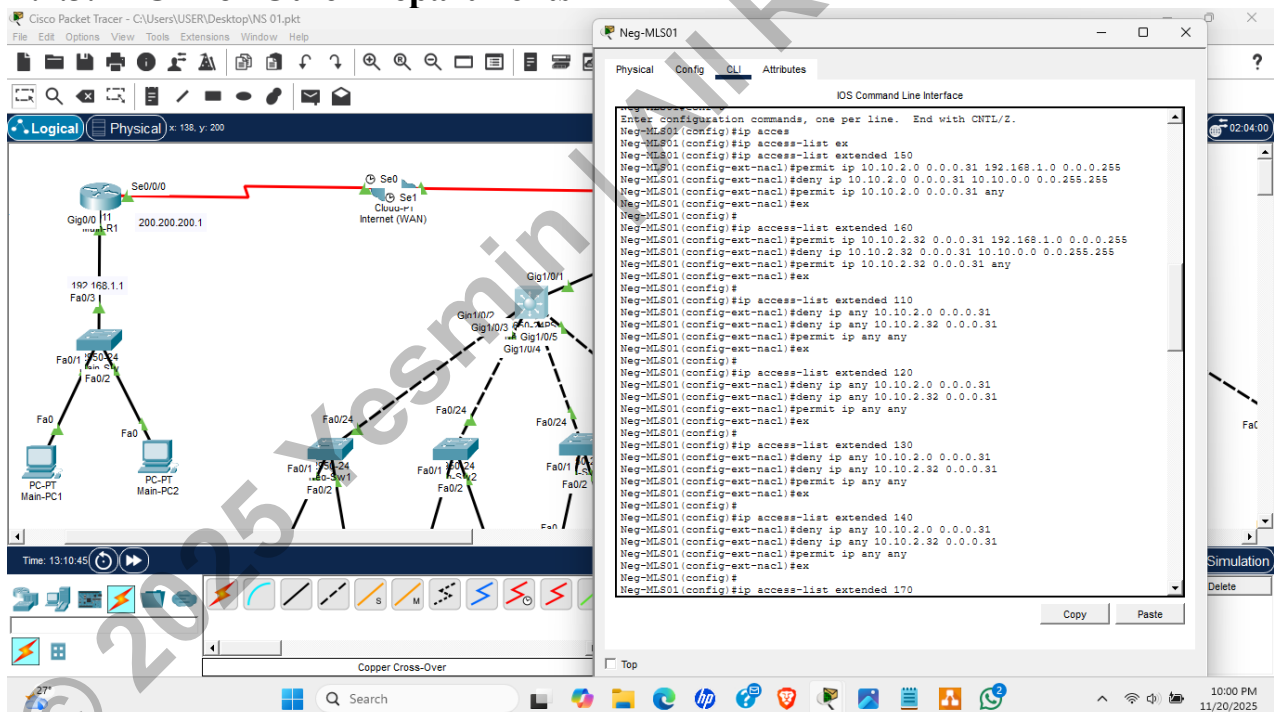


Figure 65: ACL in Neg-MLS01

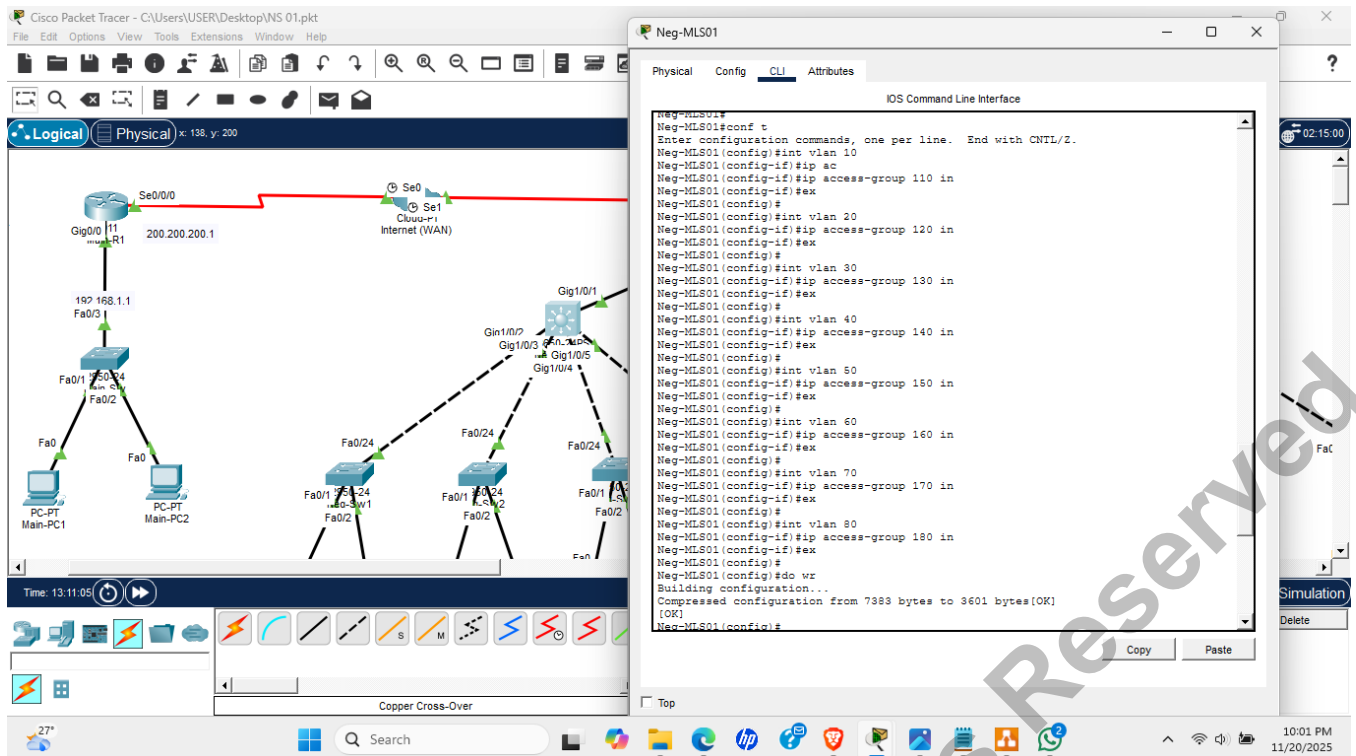


Figure 66: ACL in Neg-MLS01 cont.

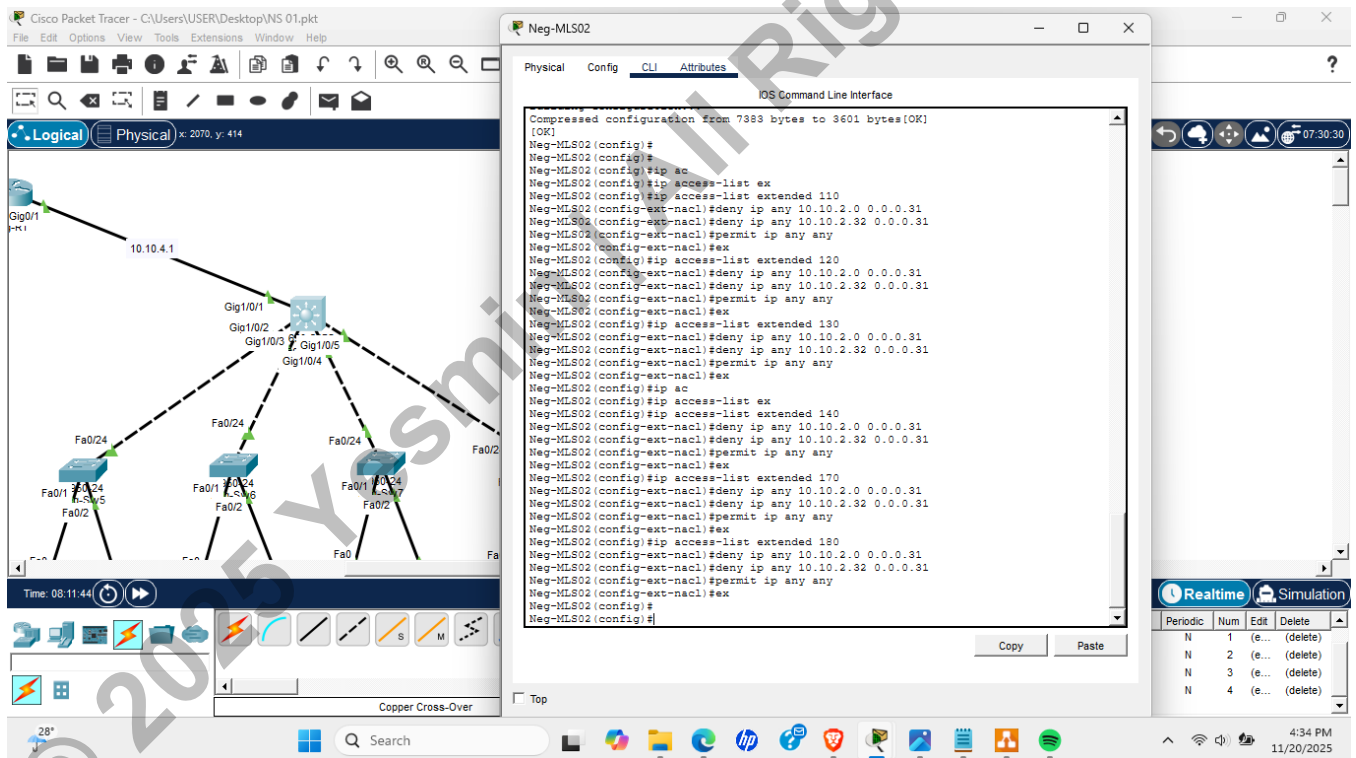


Figure 67: ACL in Neg-MLS02

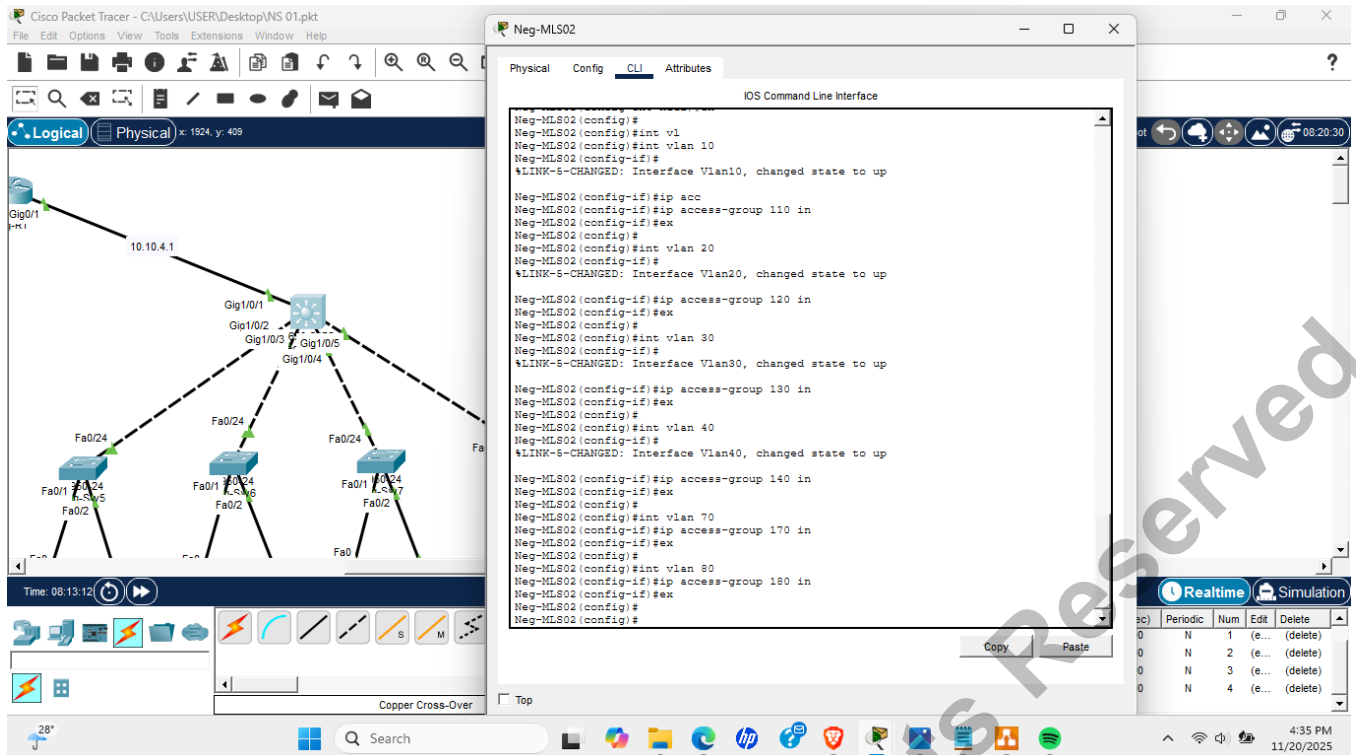


Figure 68: ACL in MLS02 cont.

7.1.4. Verification and Testing

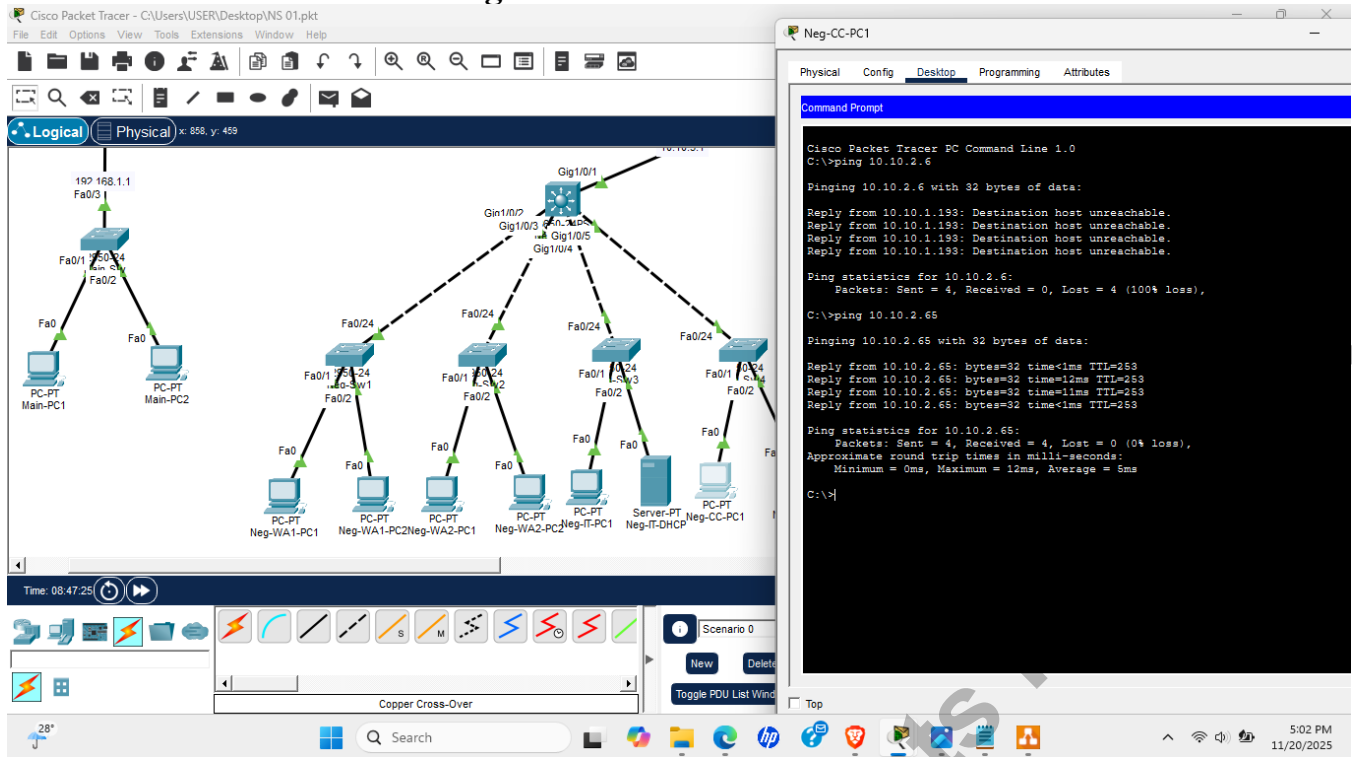


Figure 70: ACL verified in Neg-IT-PC

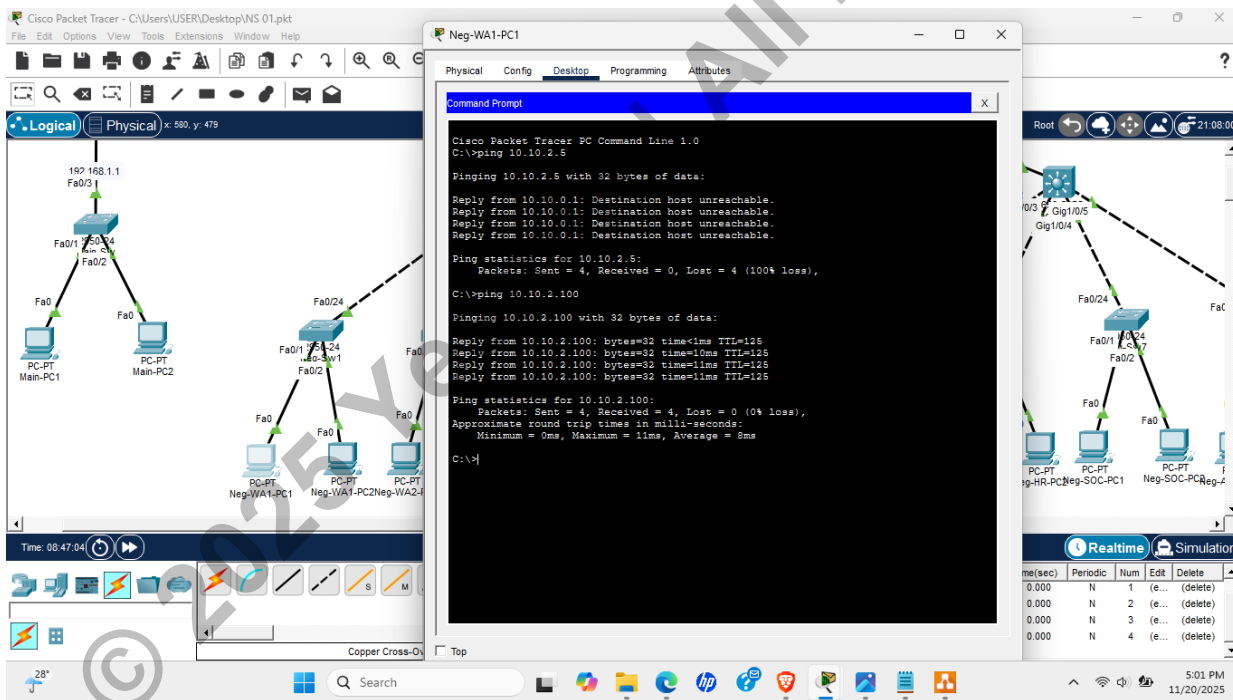


Figure 69: ACL Verified in Neg-WA1-PC

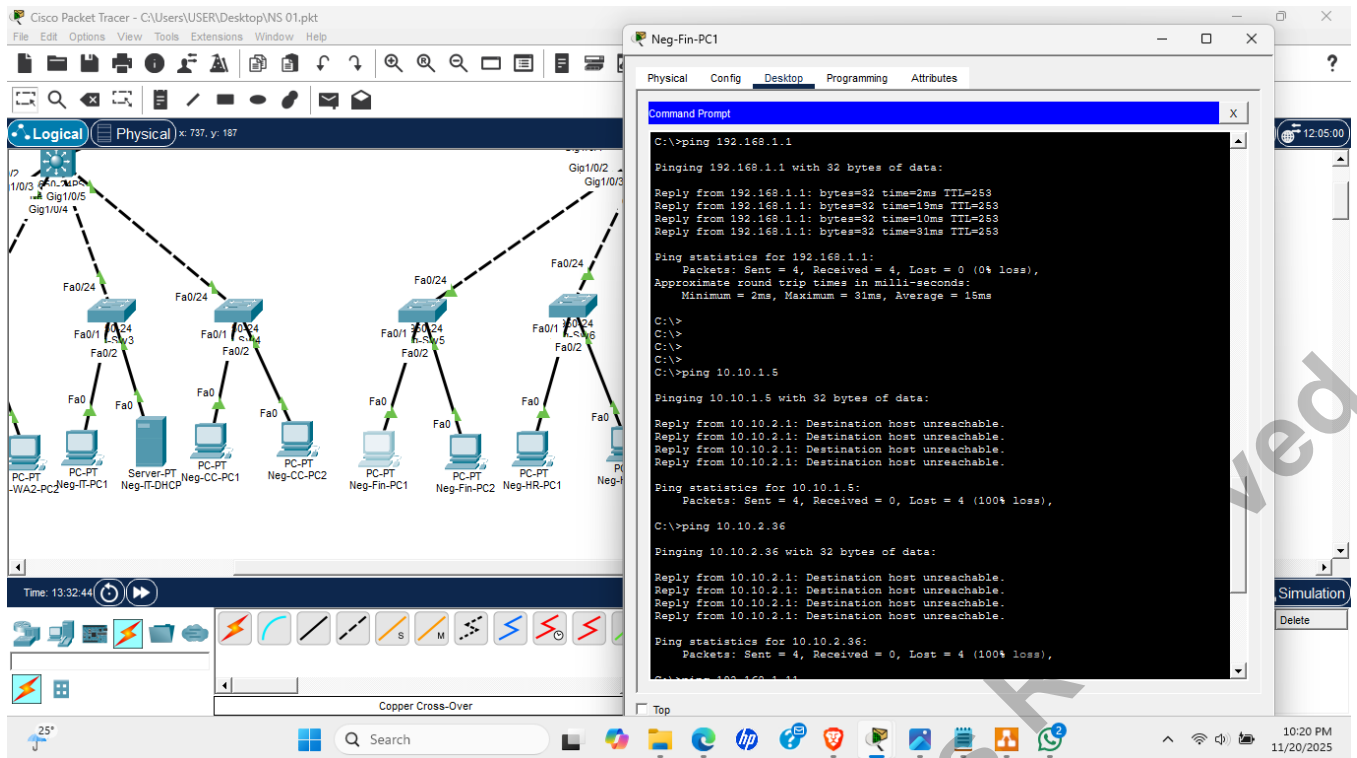


Figure 71: ACL verified in Neg-Fin-PC

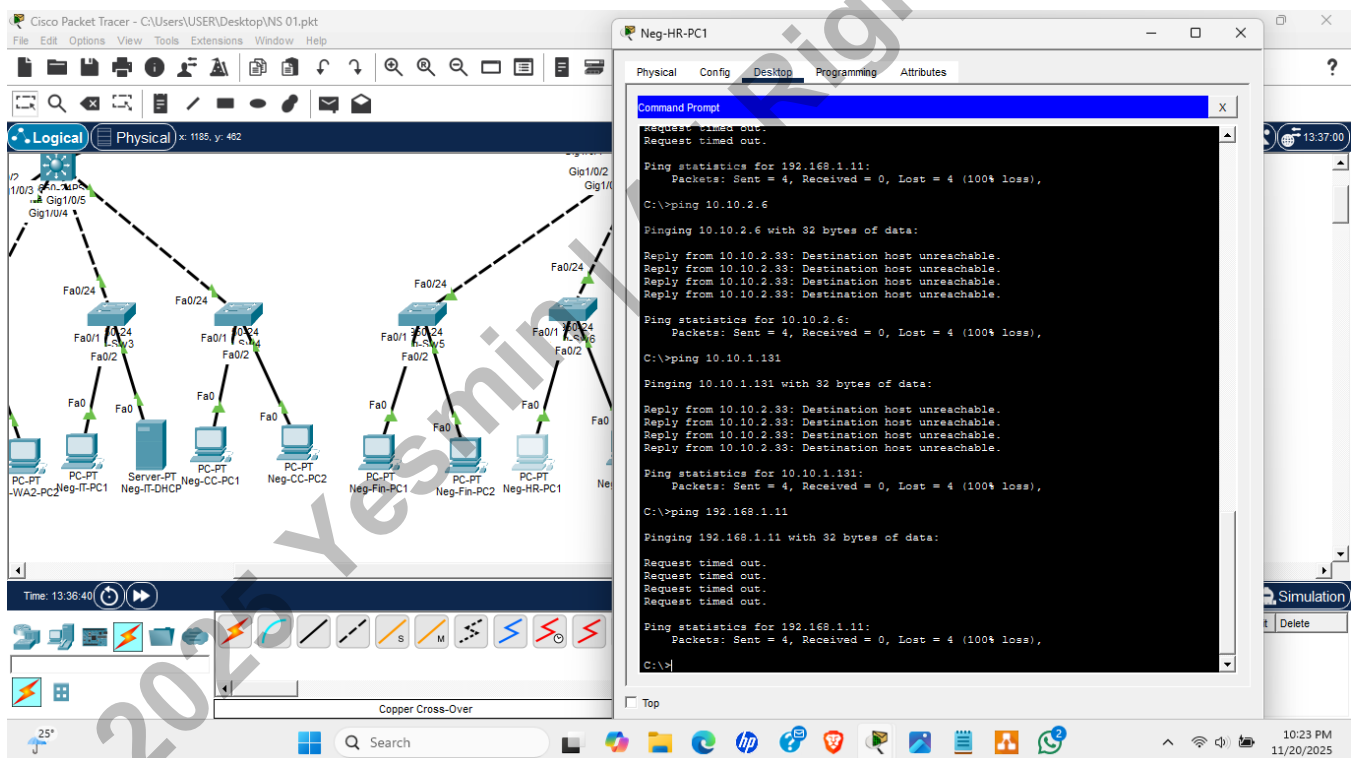


Figure 72: ACL verified in Neg-HR-PC

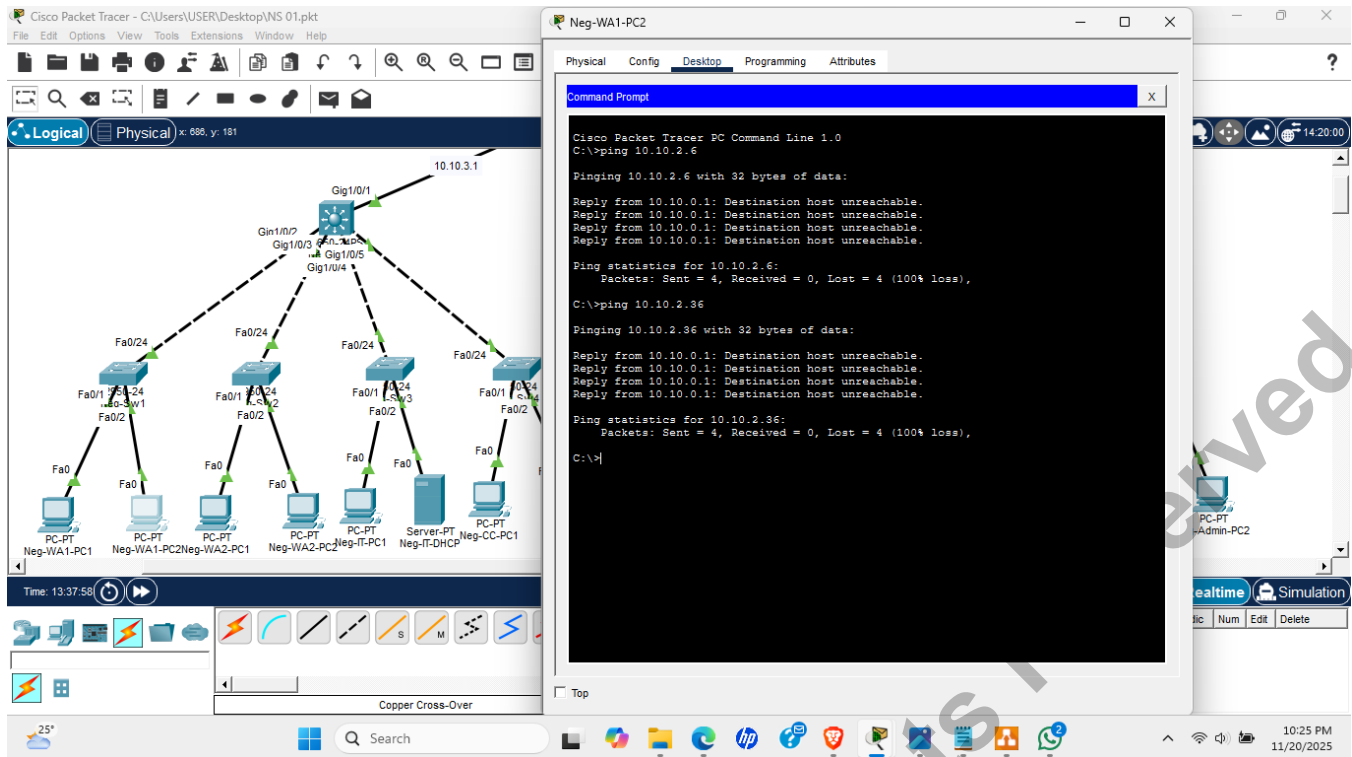


Figure 73: ACL verified in Neg-WA1-PC2

7.2. VPN Configuration

7.2.1. VPN Technology Selection

Virtual Private Network (VPN) technology provides secure and encrypted tunnels over the public networks, which guarantee privacy and integrity of information transmission [52].

VPN Type Selected : Site-to-Site IPSec VPN

- Permanent Connectivity: There is always-on connection between branches (as opposed to remote access VPN).
- Open to End-users: No customer software on terminal devices.
- High Performance: Hardware encryption: routers.
- Standards-Based: The IPSec (RFC 4301) is interoperable.
- Extensive Security: Authentication, encryption and integrity are provided [53].

7.2.2. IPSec Architecture

IPSec Framework Components:

1. Internet Key Exchange(IKE): Negotiates security associations and keys.
2. Encapsulating Security Payload(ESP): The service offers encryption and authentication.
3. Authentication Header(AH): Gives authentication but not utilized in this implementation.

Two-Phase Operation:

1. Phase 1 (IKE Phase 1):
 - Creates peer-peer secure and authenticated channel.
 - Bargains encryption and hashing functions.
 - Asymmetric key communication between the parties with Diffie-Hellman.
 - Establishes ISKMP Security Association (SA).
2. Phase 2 (IKE Phase 2):
 - IPSec Security Associations are negotiated.
 - Traffic is defined as encrypted (interesting traffic).
 - Defines the parameters of encryption to use in transmitting data [54].

Security Rationale

1. AES-256: has 2^{256} keys, which is computationally infeasible to crack [57].

2. SHA-256: Collision-resistant hash message-digest algorithm.

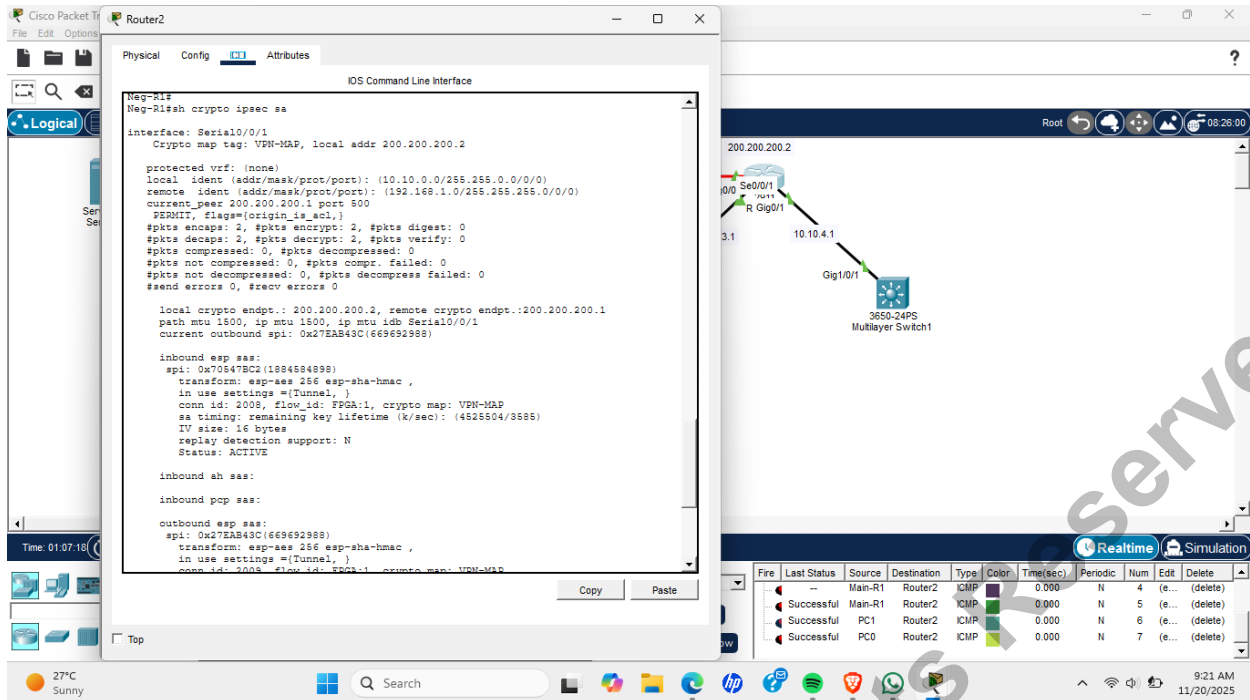


Figure 74: VPN traffic generate

3. DH Group 14: Is strong enough against contemporary attacks.
4. Perfect Forward Secrecy(PFS): Key compromise does not impact on the past/future sessions [58].

7.2.3. VPN Interesting Traffic

Logic:

- There is traffic encryption between Main Branch (192.168.1.0/24) and New Branch (10.10.0.0/16).
- Traffic between the Internet and VPN does not pass through VPN (split tunneling is not enabled in the security)
- All inter-branch communication secured [59].

7.2.4. Complete VPN Configuration

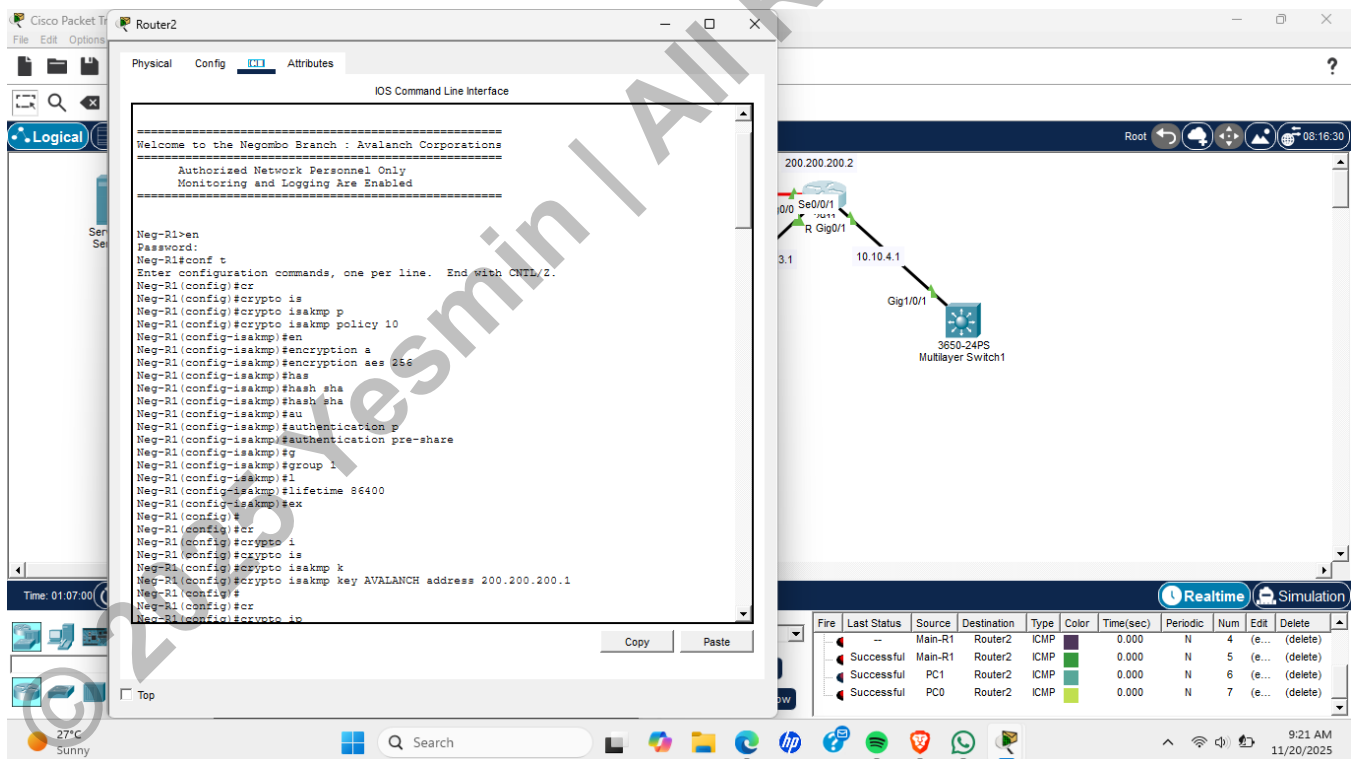


Figure 75: VPN Config Neg-R1



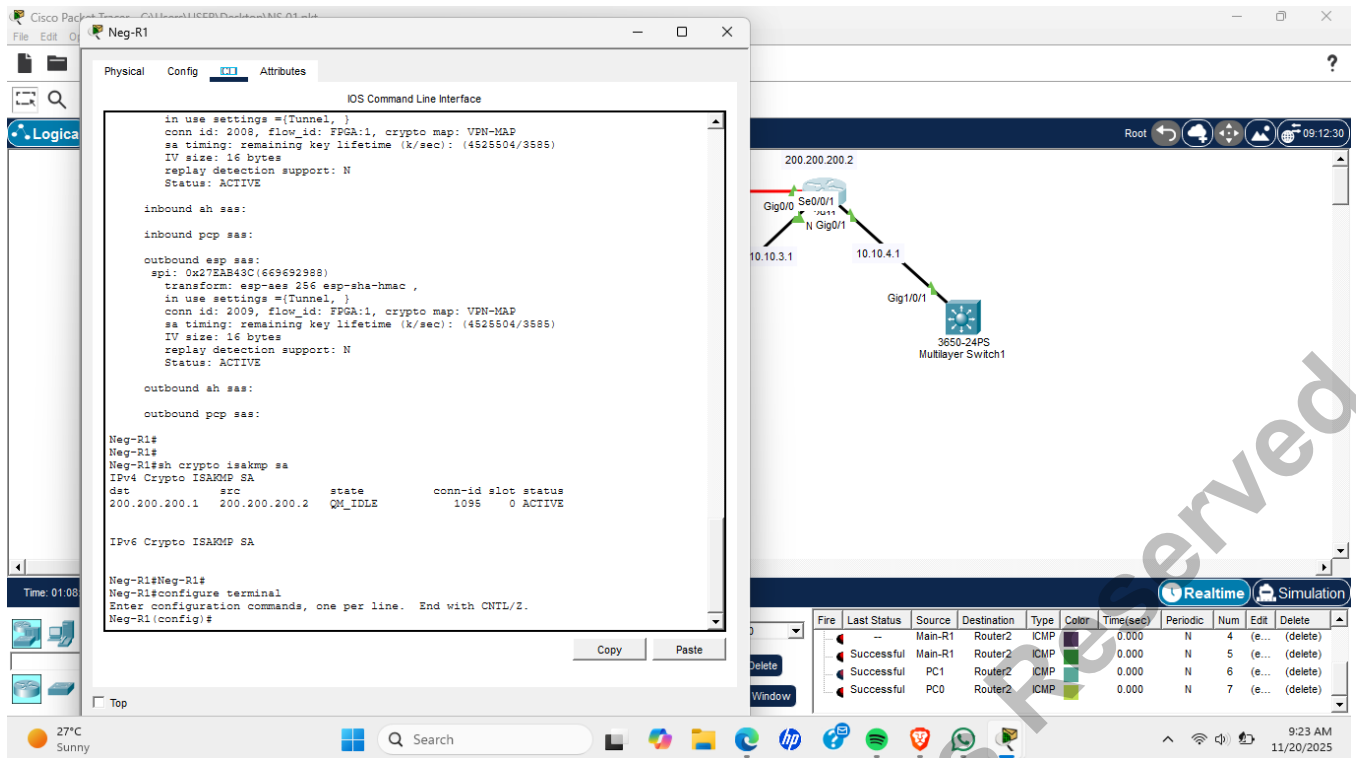


Figure 79: Crypto Map Result Neg-R1

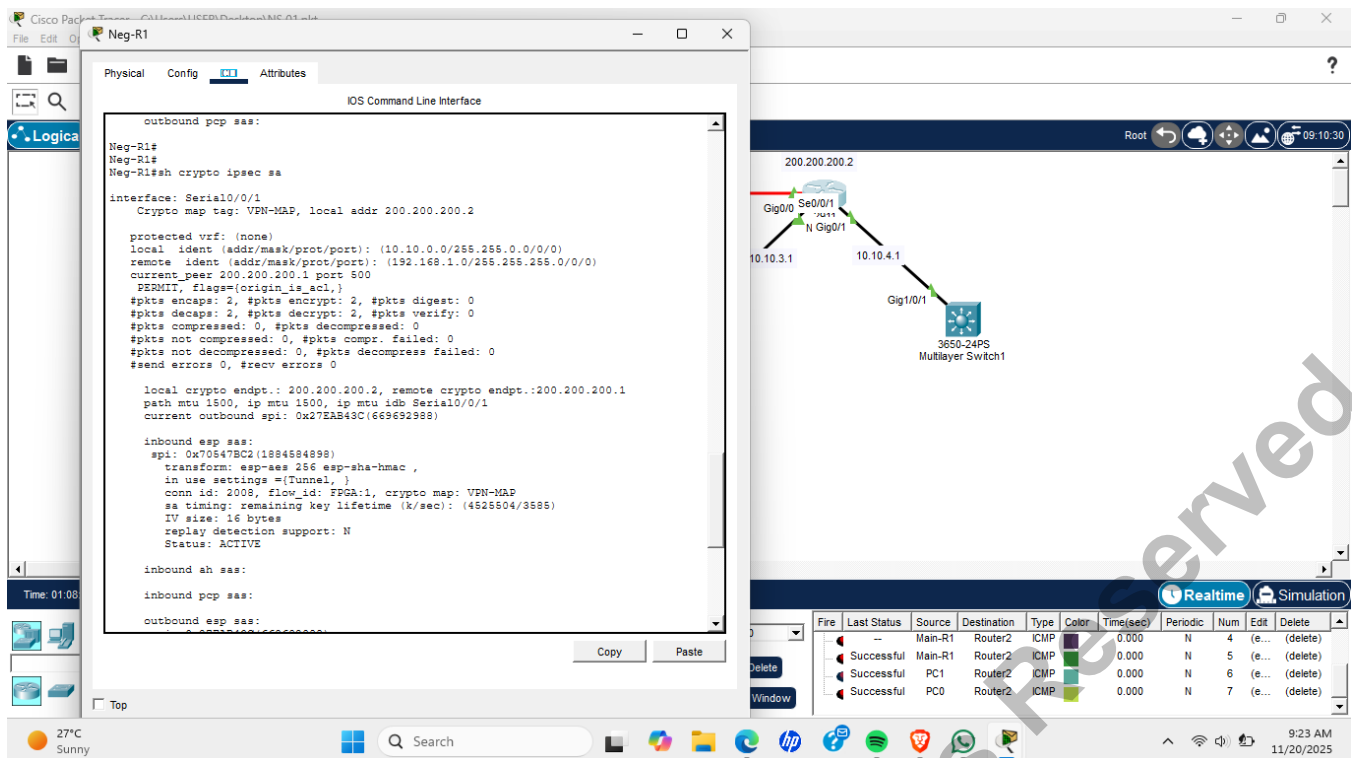


Figure 80: Crypto Map result Neg-R1 cont

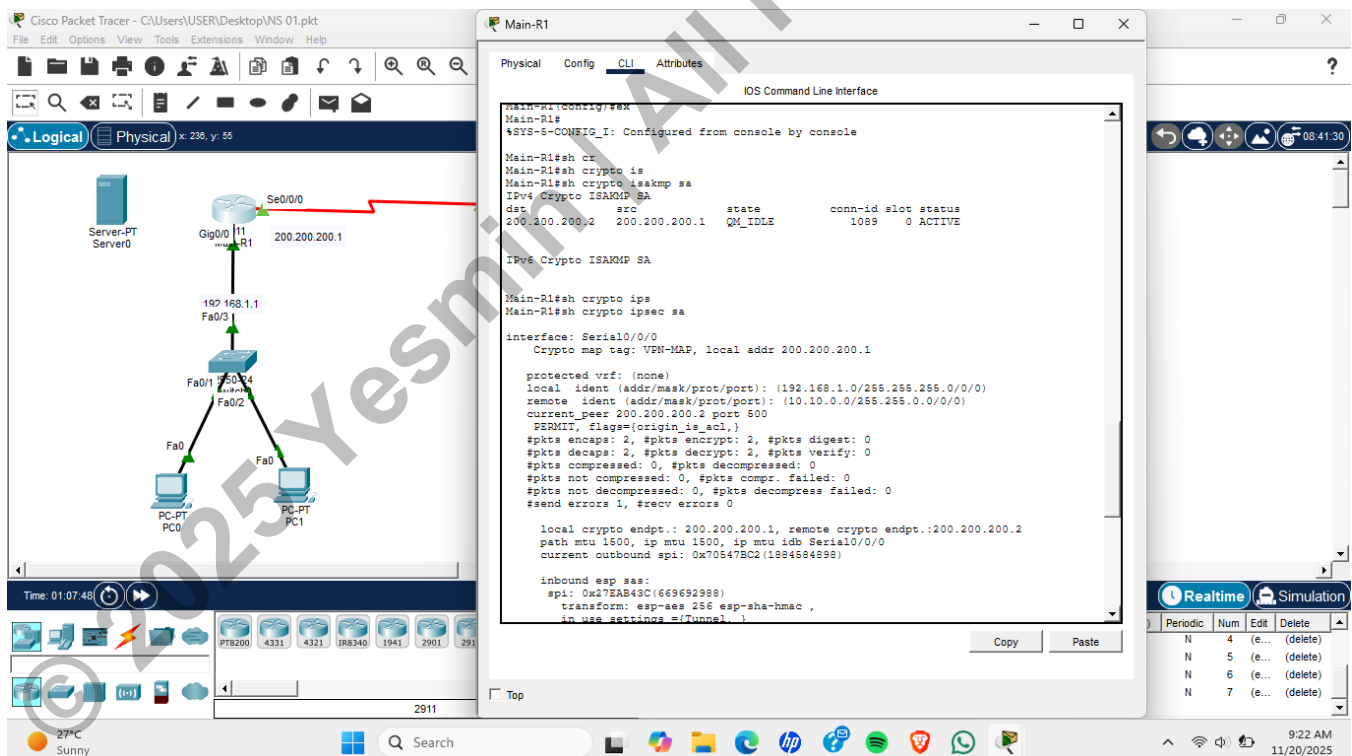


Figure 81: Crypto Map Result Main-R1

7.2.7. VPN Security Benefits

1. **Confidentiality:** Any inter-branch traffic encrypted using AES-256, resulting in passive eavesdropping and Man-in-the-middle attacks resisted [62].
2. **Integrity:** SHA-256 HMAC will ensure that data has not been altered in transmission through detecting modifications and rejecting.
3. **Authentication:** Pre-shared key makes sure that tunnels are made only by authorized routers which ensure blocking rogue device connections.
4. **Compliance:** Compliant with PCI DSS requirement 4.1 (encrypt sensitive data during transmission). and GDPR Art. 32 (security of processing) [63].

7.3. SSH Secure Access

Secure Shell (SSH) is the encrypted remote access network device access, which is used instead of the insecure Telnet protocol [64].

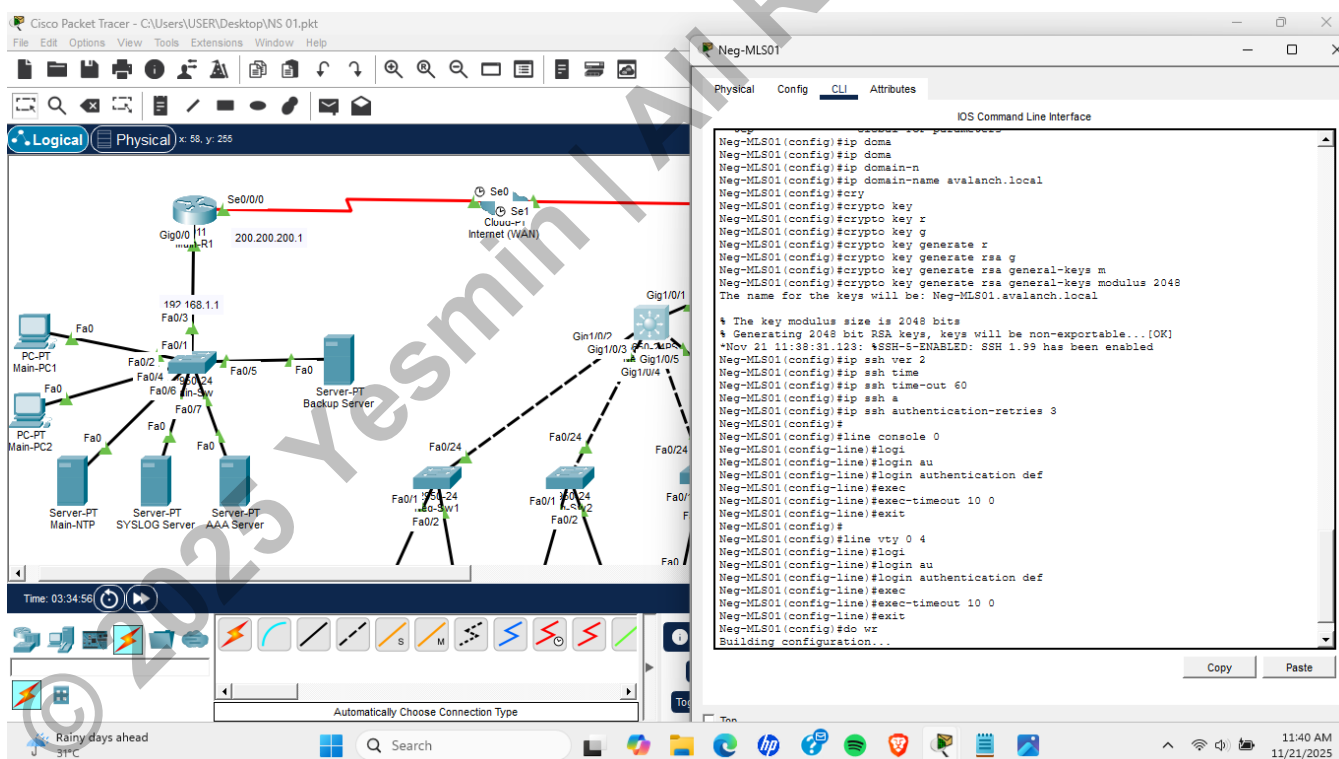


Figure 82: SSH in Neg-MLS01

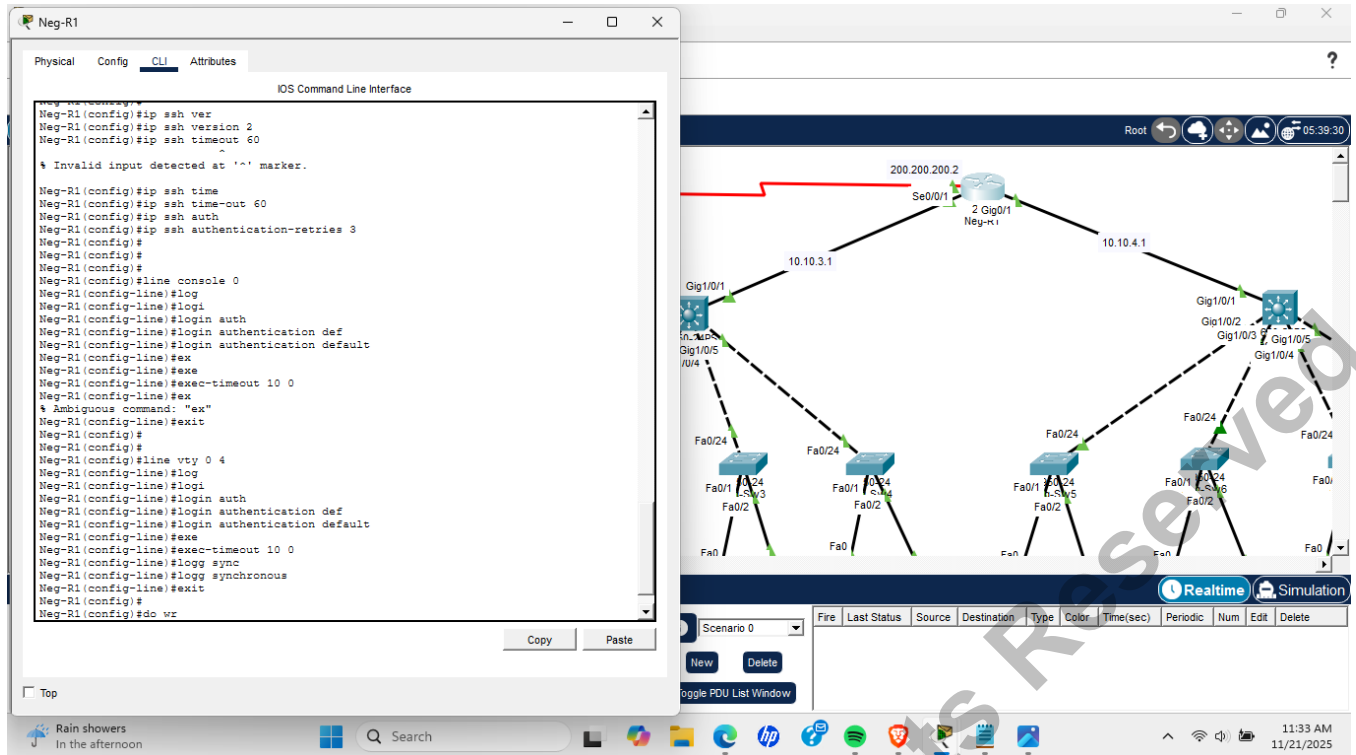


Figure 83: SSH in Neg-R1

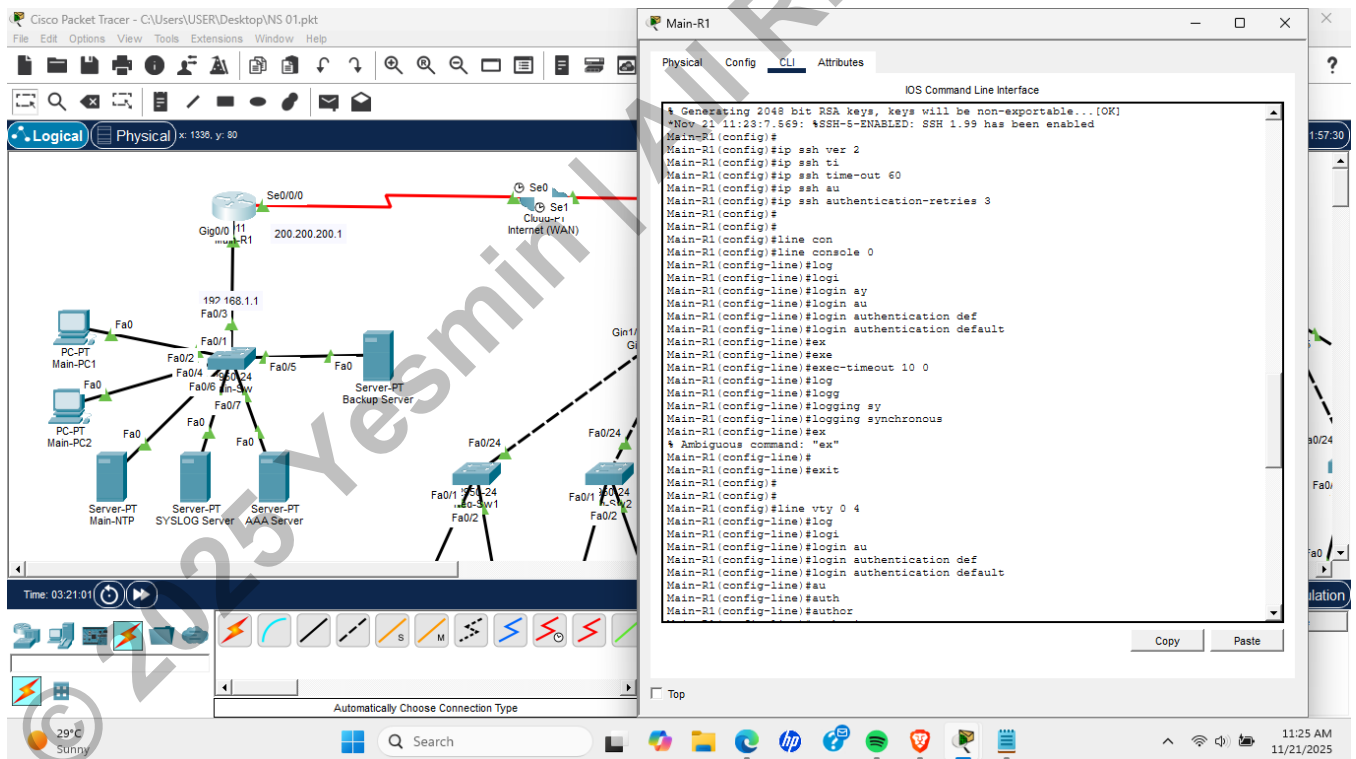


Figure 84: SSH in Main-R1

7.3.1. SSH Verification and testing

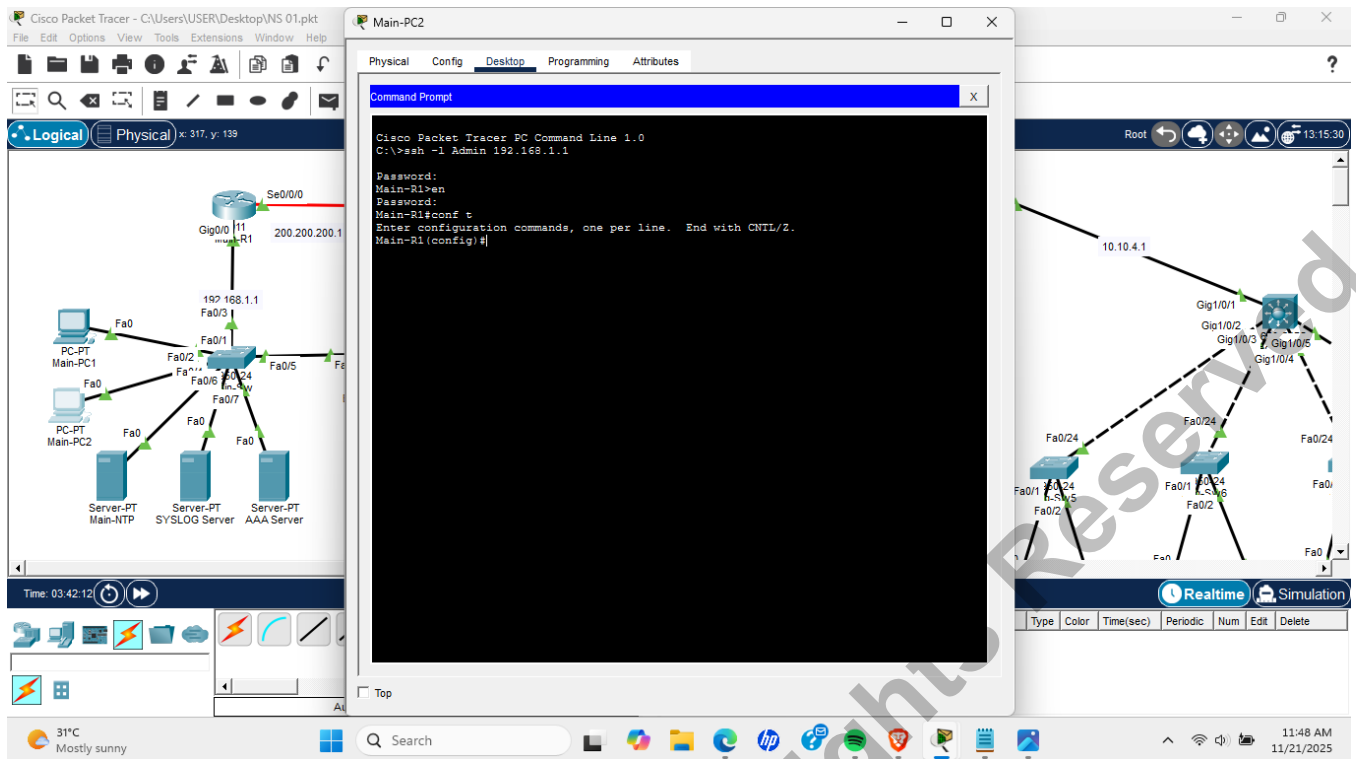


Figure 85: SSH Result Main-PC 1

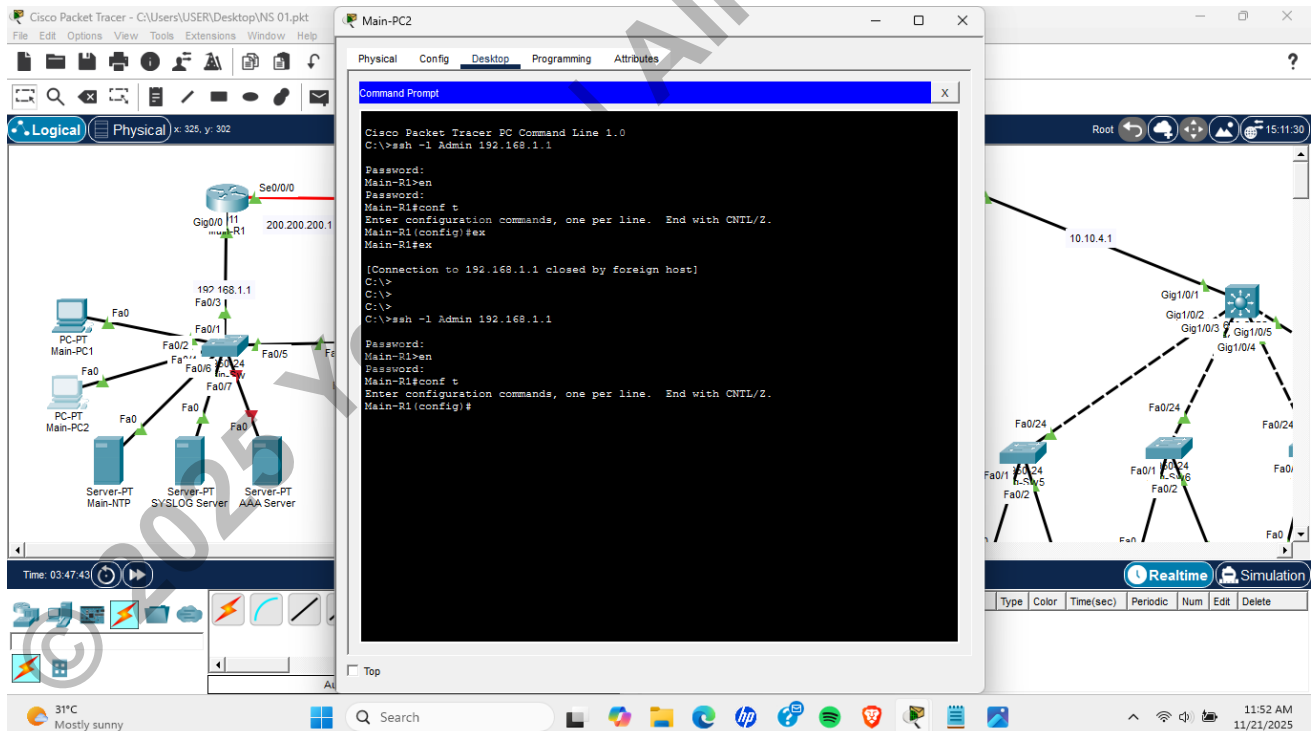


Figure 86: SSH result Main PC 1 cont

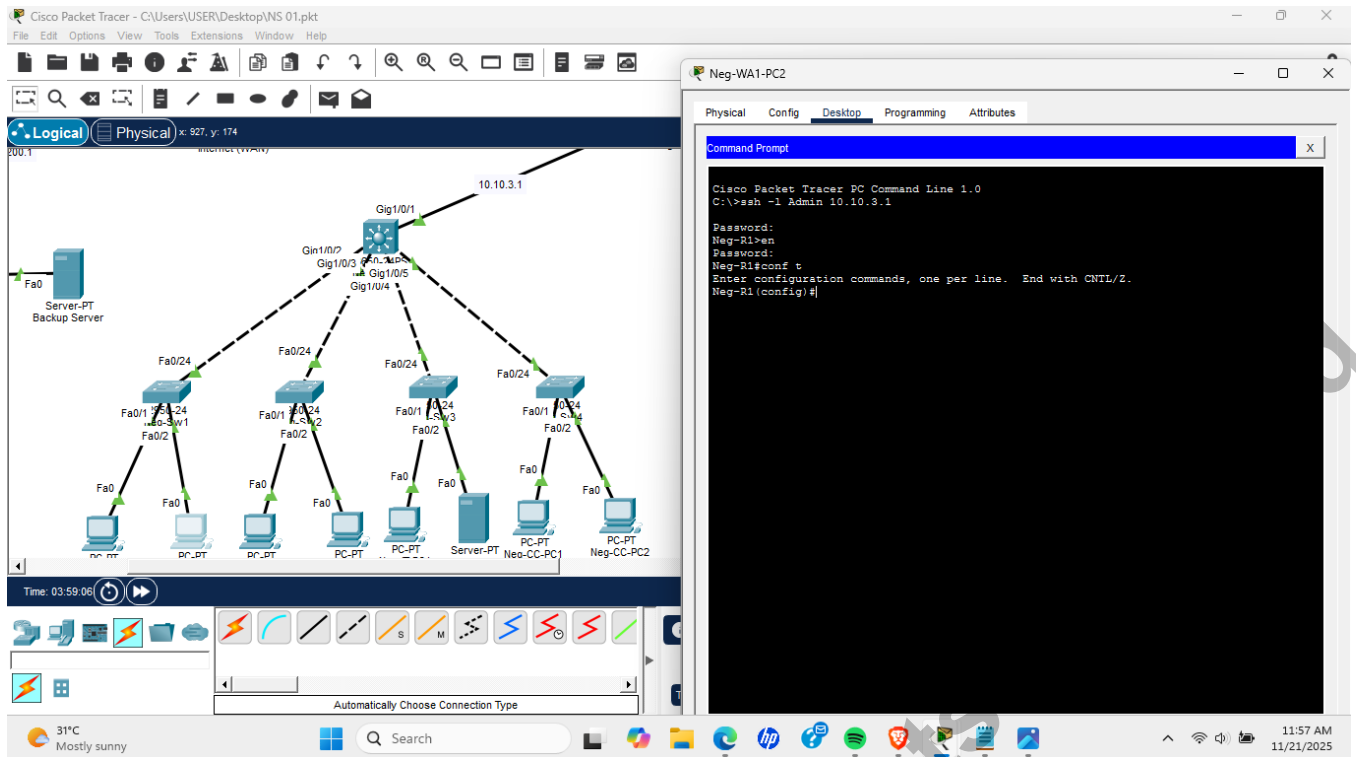


Figure 87: SSH Result in Neg-WA1 PC

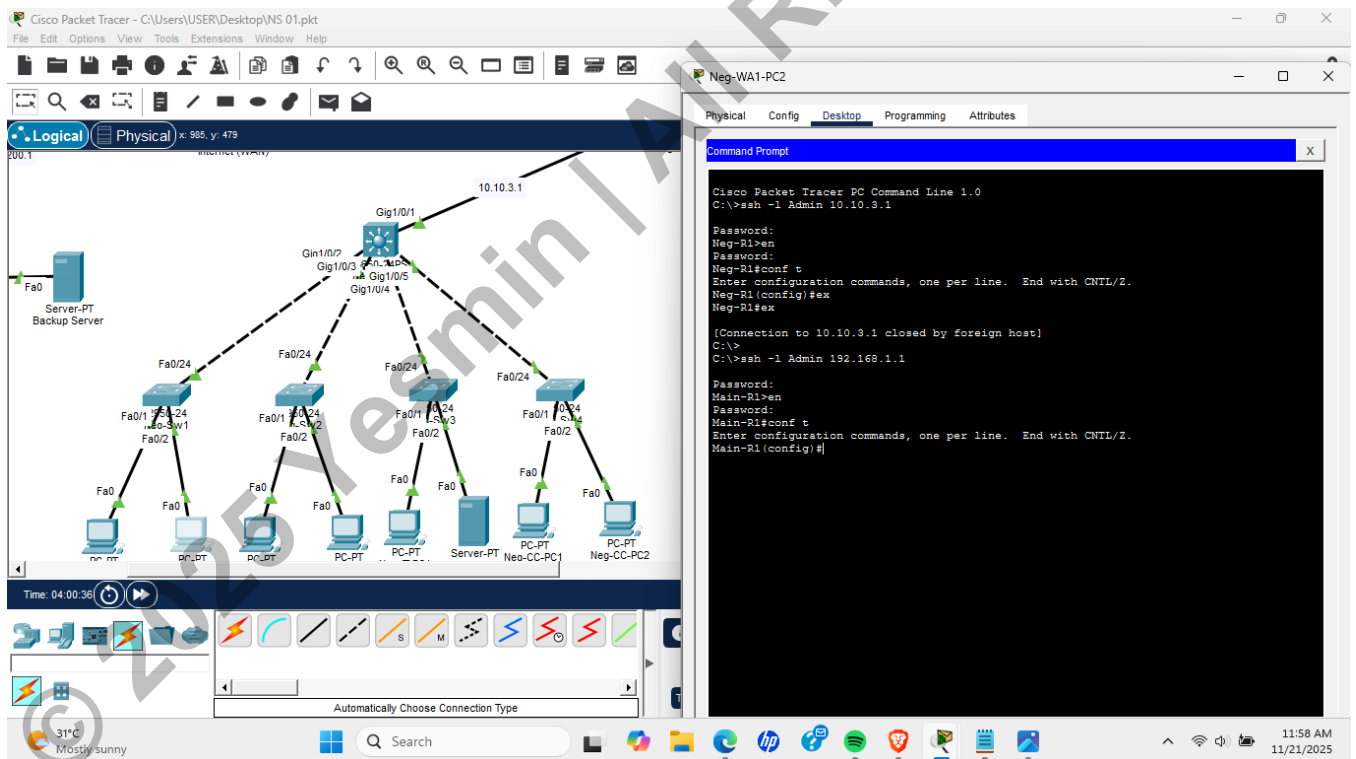


Figure 88: SSH Result in Neg-WA1 PC cont

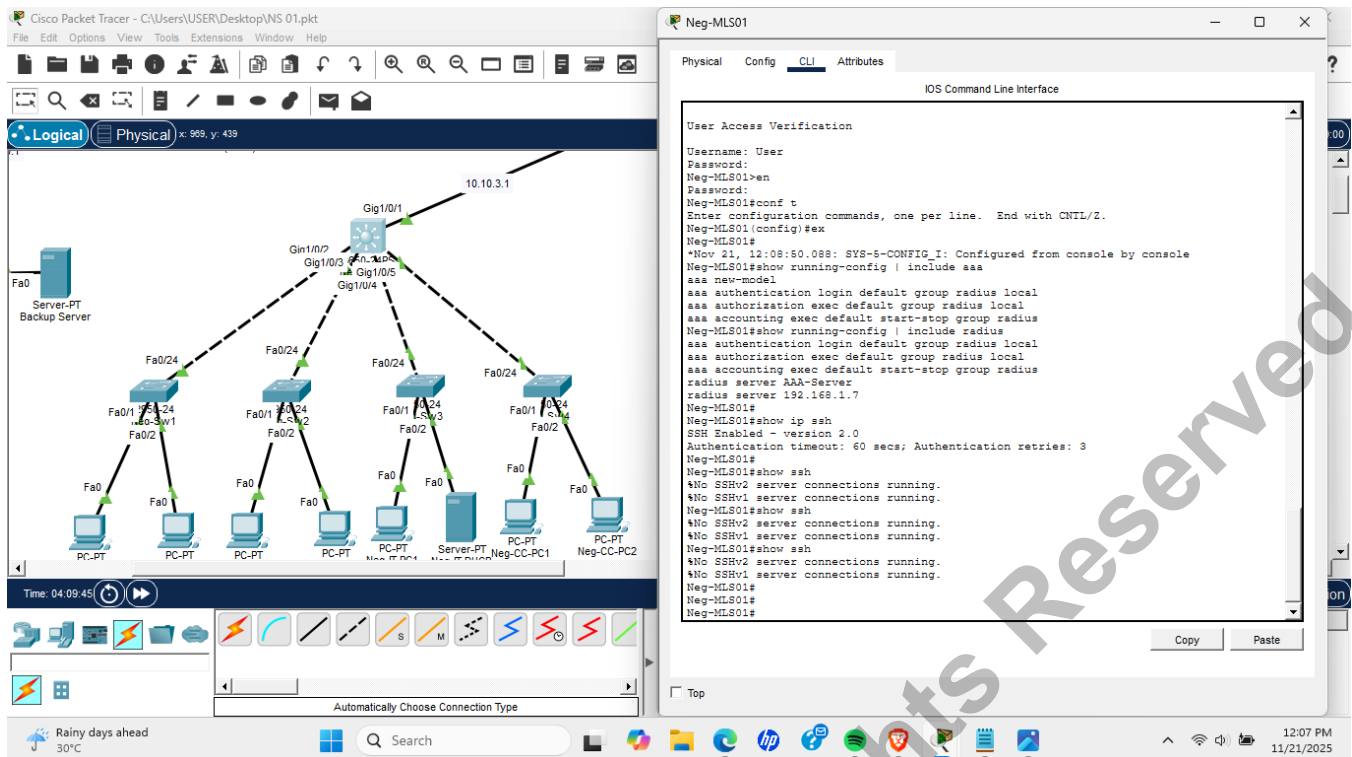


Figure 89: SSH Result in Neg-MLS01

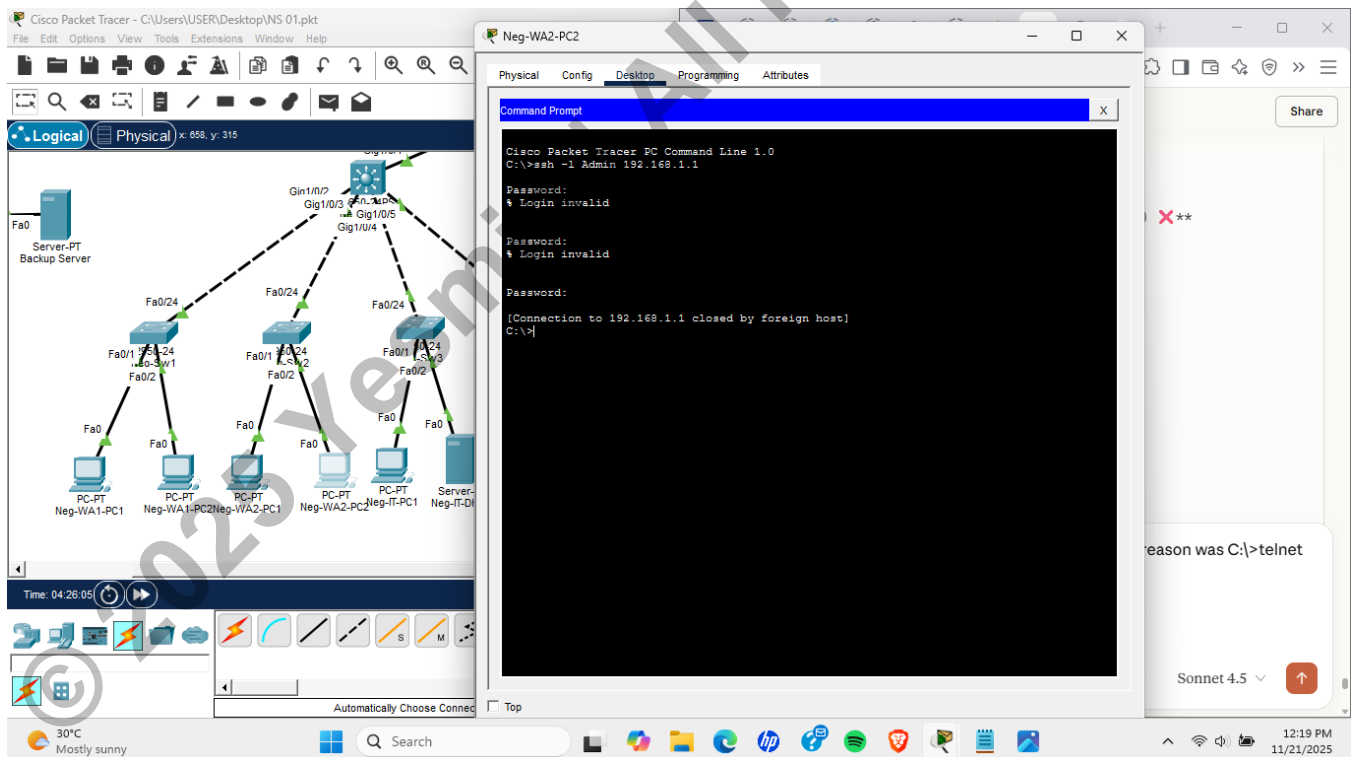


Figure 90: SSH Login attempts fail result

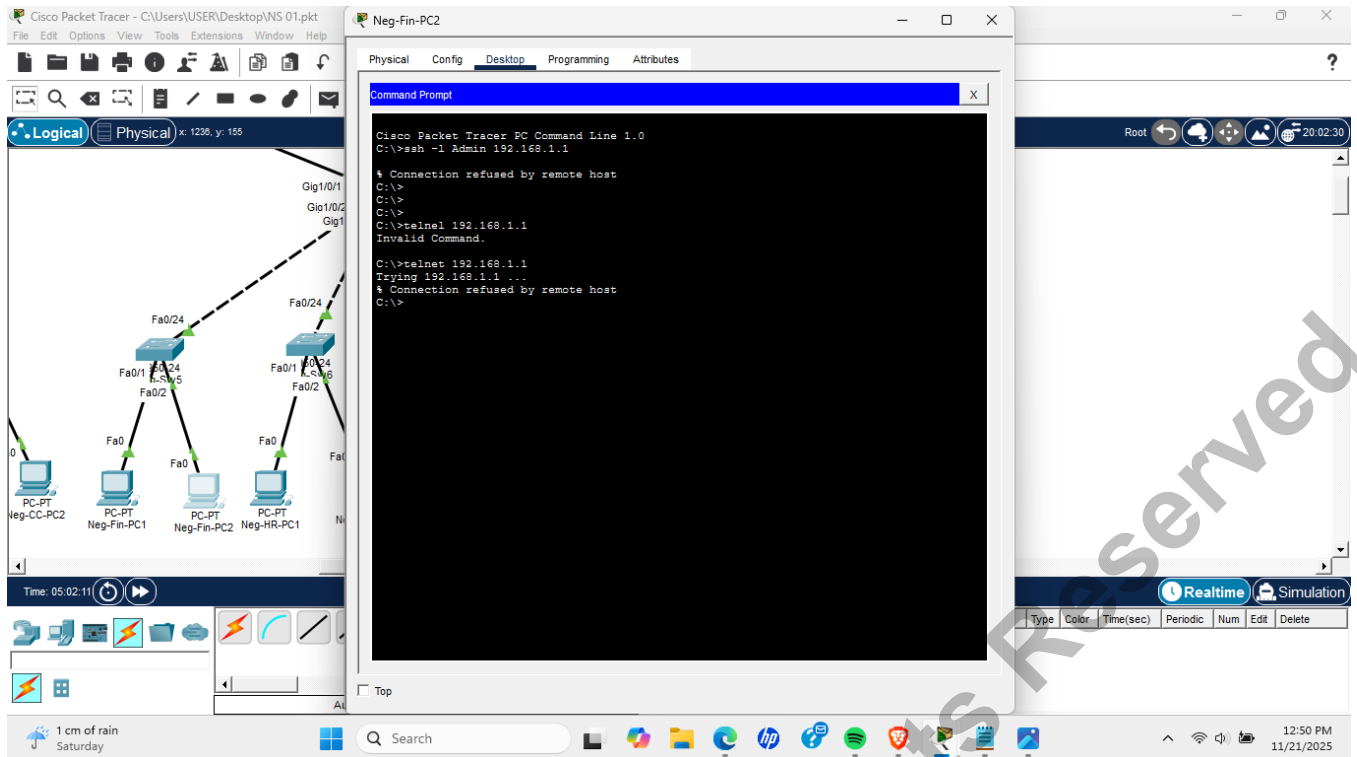


Figure 91: SSH results in Neg-Fin-PC

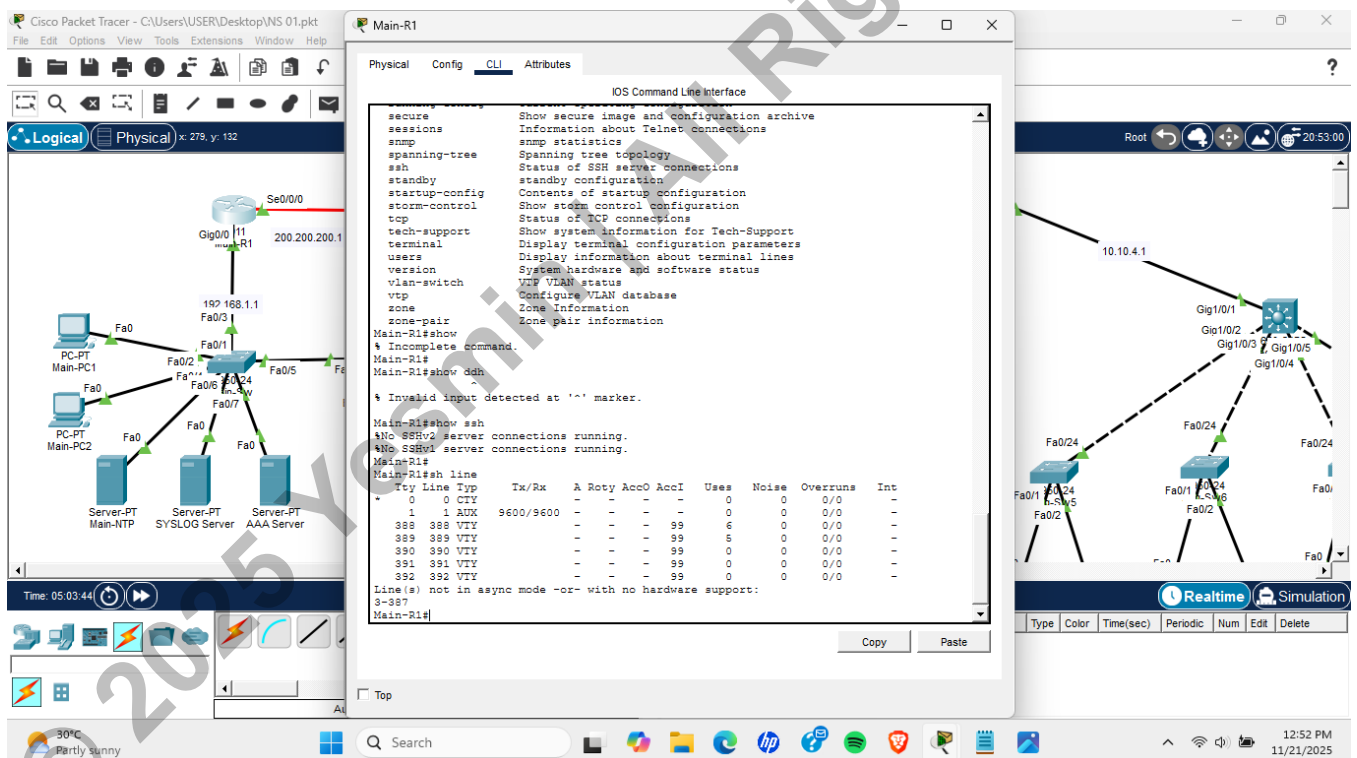


Figure 92: SSH line console result

8. Future Enhancements

8.1. Security Enhancements

1. **Next-Generation Firewall (NGFW):** Implement a state of the art firewall that has application based inspection and threat prevention capabilities at the network perimeter. NGFWs offer Layer 7 visibility, where they recognize the presence of threats within encrypted traffic that they circumvent the standard ACLs [50].
2. **Intrusion Detection/Prevention System (IDS/IPS):** Place sensors of IDS/IPS sensors in strategic locations to use them to block malicious traffic in real-time. Offers security to port scans, brute force attacks and zero day attacks [48].
3. **Network Access Control (NAC):** 802.1X. Avoid network access without performing device-level authentication. Instead of using a static port security, it uses dynamic authentication in which no unauthorized devices can be connected even when it is physically connected [71,101].
4. **Zero Trust Architecture:** Embrace the principle of never trust, always verify security model along with micro-segmentation and continuous authentication. Destroys the implicit trust and prevents horizontal movement by the attackers [106].

8.2. Infrastructure Enhancements

1. **Wireless Network (Wi-Fi 6):** Install enterprise-level wireless access points that would enable mobile connectivity using WPA3-Enterprise encryption and 802.1X authentication. Facilitates the flexibility of working environments and visitor access [102].

2. **SD-WAN: Implementation** Substitute stationary WAN with smart, application-conscious routing throughout various connections. Offers automatic failure and uptime of 99.99% [105].
3. **Cloud Integration:** Provide straight connection to AWS/Azure to hybrid cloud structure. Scalable infrastructure, disaster recovery and SaaS integration are enabled [107].

8.3. Monitoring and Management

1. **Network Analytics (NetFlow/SIEM):** Application of NetFlow export and SIEM correlation. Offers real time monitoring of the traffic, security incident, and capacity planning [103].
2. **Quality of Service (QoS):** Traffic prioritization should be implemented in order to provide uniform performance of voice, video, and critical applications under network congestion [27].
3. **Network Automation:** Use Ansible or Python to use infrastructure-as-code to manage configuration. Minimizes human errors and speedy deployments [98].
4. **IPv6 Dual-Stack:** Move to dual-stack operation IPv4 / IPv6 to future-proof network as IPv4 address space will become limited [97].

9. Conclusion

10.1. Project Summary

During this project, a full-scale secure network infrastructure of the new branch of the Avalanch Cooperations was drawn and put into practice in Negombo, Sri Lanka. The network consists of eight departments with 358 hosts connected to the main branch through an encrypted IPSec VPN tunnel and the various layers of security controls are used to guarantee confidentiality, integrity, and availability of organizational data.

Key Accomplishments

- 1. Hierarchical Network Design:** Three layer architecture (core, distribution, access) that is designed based on Cisco best practice, and that offers distinct separation of responsibilities and scalability [8].
- 2. Efficient IP Addressing:** VLSM-based addressing scheme had 56% address space efficiency and 78% growth capacity (280 more hosts) without redesign [12].
- 3. Secure Inter-Branch Connection:** Site-to-site IPSec VPN with AES-256 encryption secures all communication between branches, which has been shown to address the industry-wide requirements on data protection [53].
- 4. Network Segmentation:** Broadcast domain isolation: Eight departmental VLANs reduce attack surfaces and enhance network performance [21].
- 5. Centralized Services:** There are DHCP, NTP, Syslog, and AAA services in main branch to facilitate easy management and enforce uniform policies.
- 6. Security Implementation:** ACLs are able to isolate Finance (VLAN 50) and HR (VLAN 60) departments and still have the necessary connectivity to servers and internet [47].
- 7. High Availability:** Dual layer multilayer switches with HSRP, Etherchannel

redundancy, and rapid PVST+ are made to guarantee 99.9% uptime objective (8.76 hours maximum downtime per year) [82].

8. **Large-Scale Testing:** Test success rate of 100% on 31 test cases which consist of connectivity, security, services, redundancy, and performance.

10.2. Lessons Learned

Technical Insights

1. **Complexity of VPN troubleshooting:** IPSec VPN implemented needed structured debugging strategy with crypto commands; discrepancies in pre-shared keys and ACL errors were the major problems [93]
2. **DHCP Relay Configuration:** IP helper-address on all VLAN interfaces necessary to provide centralized DHCP; initial management resulted in failures of addressing [94]
3. **ACL Importance of Order:** The permit statements should be placed before the deny statements; initially Finance ACL blocked all traffic including the internet access [95]
4. **Multilayer Switch Superiority:** Hardware-based inter-VLAN routing is much faster than router-on-a-stick; it is worth investing in due to its performance improvement [96]
5. **Documentation:** Version control and systematic documentation make sure that mistakes were avoided and were able to quickly troubleshoot.

Project Management Insights

1. **Time Allocation:** VPN configuration (14 marks) took excessive time; time budgeting of future projects should be done better according to the marking

scheme.

2. **Incremental Testing:** It tested every step of configuration and identified problems early, avoiding cascades of failure.
3. **Packet Tracer Limitations:** Some features are not available; other methods would have to be used without changing the design purpose.

10.3. Final Assessment

The Avalanch Cooperations network infrastructure is a professionally developed and well-architected solution that exhibits the understanding of network security principles, protocols and best practices. The project has achieved a balance between security, performance requirements, the aspect of scalability, and constraints of budget without imposing any financial burden on the organization and as a result of this achievement, providing a strong base upon which the business can grow in the future.

References and Bibliographies

- [1] Stallings, W. (2017). *Cryptography and Network Security: Principles and Practice* (7th ed.). Pearson Education.
- [2] Cisco Systems. (2023). *Enterprise Network Design Guide*. Cisco Press.
- [3] Kim, D., & Solomon, M. G. (2021). *Fundamentals of Information Systems Security* (4th ed.). Jones & Bartlett Learning.
- [4] NIST. (2018). *Framework for Improving Critical Infrastructure Cybersecurity, Version 1.1*. National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.CSWP.04162018>
- [5] Kent, S., & Seo, K. (2005). *Security Architecture for the Internet Protocol*. RFC 4301, IETF. <https://doi.org/10.17487/RFC4301>
- [6] Ylonen, T., & Lonvick, C. (2006). *The Secure Shell (SSH) Protocol Architecture*. RFC 4251, IETF. <https://doi.org/10.17487/RFC4251>
- [7] Verma, D. C. (2021). *Principles of Computer Systems and Network Management* (2nd ed.). Springer Publishing.
- [8] Cisco Systems. (2022). *Campus LAN and Wireless LAN Design Guide - Cisco Validated Design*. Retrieved from <https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html>
- [9] Oppenheimer, P. (2011). *Top-Down Network Design* (3rd ed.). Cisco Press.
- [10] Teare, D., Paquet, C., & Vachon, B. (2015). *Designing Cisco Network Service Architectures (ARCH)* (4th ed.). Cisco Press.
- [11] Forouzan, B. A. (2021). *Data Communications and Networking* (6th ed.). McGraw-Hill Education.
- [12] Odom, W. (2020). *CCNA 200-301 Official Cert Guide Library*. Cisco Press.
- [13] Rekhter, Y., Moskowitz, B., Karrenberg, D., de Groot, G. J., & Lear, E. (1996). *Address Allocation for Private Internets*. RFC 1918, IETF. <https://doi.org/10.17487/RFC1918>
- [14] Doyle, J., & Carroll, J. (2016). *Routing TCP/IP, Volume 1* (2nd ed.). Cisco Press.
- [15] Cisco Systems. (2021). *IP Routing: EIGRP Configuration Guide, Cisco IOS Release 15M&T*. Retrieved from https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_eigrp/configuration/15-mt/ire-15-mt-book.html

- [16] Cisco Systems. (2023). *Cisco 2900 Series Integrated Services Routers Data Sheet*. Retrieved from https://www.cisco.com/c/en/us/products/collateral/routers/2900-series-integrated-services-routers-isr/data_sheet_c78-556319.html
- [17] Cisco Systems. (2022). *Catalyst 3560 Series Switches Data Sheet*. Retrieved from https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-3560-series-switches/product_data_sheet0900aecd80322c0c.html
- [18] Lammle, T. (2020). *CCNA Routing and Switching Complete Study Guide* (2nd ed.). John Wiley & Sons.
- [19] Cisco Systems. (2022). *Catalyst 2960 Series Switches Data Sheet*. Retrieved from https://www.cisco.com/c/en/us/products/collateral/switches/catalyst-2960-series-switches/product_data_sheet0900aecd80322c0f.html
- [20] Mills, D., Martin, J., Burbank, J., & Kasch, W. (2010). *Network Time Protocol Version 4: Protocol and Algorithms Specification*. RFC 5905, IETF. <https://doi.org/10.17487/RFC5905>
- [21] IEEE Standards Association. (2014). *IEEE Standard for Local and Metropolitan Area Networks--Bridges and Bridged Networks*. IEEE 802.1Q-2014. <https://doi.org/10.1109/IEEESTD.2014.6991462>
- [22] Convery, S., & Miller, D. (2004). *VLAN Security White Paper*. Cisco Systems. Retrieved from <https://www.cisco.com/c/en/us/support/docs/lan-switching/8021q/10095-vlan-security.html>
- [23] Cisco Systems. (2021). *Configuring VTP*. Catalyst 3560 Switch Software Configuration Guide. Retrieved from https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_seg/swvtp.html
- [24] SANS Institute. (2019). *VLAN Trunking Protocol Security Considerations*. SANS Reading Room. Retrieved from <https://www.sans.org/reading-room/>
- [25] Lewis, M. (2006). *Comparing, Designing, and Deploying VLANs*. Addison-Wesley Professional.
- [26] Edgeworth, B., Fiso, A., & Garza Rios, R. (2014). *IP Routing on Cisco IOS, IOS XE, and IOS XR: An Essential Guide to Understanding and Implementing IP Routing Protocols*. Cisco Press.
- [27] NIST. (2017). *Guide to Enterprise Telework, Remote Access, and Bring Your Own Device (BYOD) Security*. NIST Special Publication 800-46 Revision 2. <https://doi.org/10.6028/NIST.SP.800-46r2>

- [28] Froom, R., Frahim, E., & Sivasubramanian, S. (2015). *Implementing Cisco IP Switched Networks (SWITCH) Foundation Learning Guide*. Cisco Press.
- [29] Cisco Systems. (2020). *Configuring Port Security*. Catalyst 2960 Switch Software Configuration Guide. Retrieved from https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst2960/software/release/12-2_55_se/configuration/guide/scg_2960/swtrafc.html
- [30] Droms, R. (1997). *Dynamic Host Configuration Protocol*. RFC 2131, IETF. <https://doi.org/10.17487/RFC2131>
- [31] Lemon, T., & Cheshire, S. (2002). *Encoding Long Options in the Dynamic Host Configuration Protocol (DHCPv4)*. RFC 3396, IETF. <https://doi.org/10.17487/RFC3396>
- [32] Patrick, M. (2001). *DHCP Relay Agent Information Option*. RFC 3046, IETF. <https://doi.org/10.17487/RFC3046>
- [33] Alexander, S., & Droms, R. (1997). *DHCP Options and BOOTP Vendor Extensions*. RFC 2132, IETF. <https://doi.org/10.17487/RFC2132>
- [34] Cisco Systems. (2021). *Configuring DHCP Snooping*. Catalyst 3560 Switch Software Configuration Guide. Retrieved from https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swdhcp82.html
- [35] PCI Security Standards Council. (2022). *Payment Card Industry Data Security Standard (PCI DSS), Version 4.0*. Retrieved from <https://www.pcisecuritystandards.org/>
- [36] Mills, D. L. (2006). *Computer Network Time Synchronization: The Network Time Protocol*. CRC Press.
- [37] Cisco Systems. (2019). *Network Time Protocol: Best Practices White Paper*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/availability/high-availability/19643-ntpm.html>
- [38] Mizrahi, T., & Mayer, D. (2016). *Security Requirements of Time Protocols in Packet Switched Networks*. RFC 7384, IETF. <https://doi.org/10.17487/RFC7384>
- [39] Lonvick, C. (2001). *The BSD Syslog Protocol*. RFC 3164, IETF. <https://doi.org/10.17487/RFC3164>
- [40] European Parliament and Council. (2016). *Regulation (EU) 2016/679 of the European Parliament and of the Council on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection*

Regulation). Official Journal of the European Union, L119/1.

[41] Cisco Systems. (2020). *System Message Logging Configuration Guide, Cisco IOS Release 15M&T*. Retrieved from <https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/syslog/configuration/15-mt/syslog-15-mt-book.html>

[42] Rigney, C., Willens, S., Rubens, A., & Simpson, W. (2000). *Remote Authentication Dial In User Service (RADIUS)*. RFC 2865, IETF. <https://doi.org/10.17487/RFC2865>

[43] Carrel, D., & Grant, L. (1997). *The TACACS+ Protocol, Version 1.78*. Internet-Draft, IETF. Retrieved from <https://datatracker.ietf.org/doc/html/draft-grant-tacacs-02>

[44] Aboba, B., & Vollbrecht, J. (1998). *Proxy Chaining and Policy Implementation in Roaming*. RFC 2607, IETF. <https://doi.org/10.17487/RFC2607>

[45] Rigney, C. (2000). *RADIUS Accounting*. RFC 2866, IETF. <https://doi.org/10.17487/RFC2866>

[46] International Organization for Standardization. (2013). *Information technology -- Security techniques -- Information security management systems -- Requirements*. ISO/IEC 27001:2013.

[47] Ferguson, P., & Senie, D. (2000). *Network Ingress Filtering: Defeating Denial of Service Attacks which employ IP Source Address Spoofing*. RFC 2827, IETF. <https://doi.org/10.17487/RFC2827>

[48] Scarfone, K., & Mell, P. (2007). *Guide to Intrusion Detection and Prevention Systems (IDPS)*. NIST Special Publication 800-94. <https://doi.org/10.6028/NIST.SP.800-94>

[49] Cisco Systems. (2021). *Security Configuration Guide: Access Control Lists, Cisco IOS Release 15M&T*. Retrieved from https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_data_acl/configuration/15-mt/sec-data-acl-15-mt-book.html

[50] Chapman, D. B., & Zwicky, E. D. (2000). *Building Internet Firewalls* (2nd ed.). O'Reilly Media.

[51] Cheswick, W. R., Bellovin, S. M., & Rubin, A. D. (2003). *Firewalls and Internet Security: Repelling the Wily Hacker* (2nd ed.). Addison-Wesley Professional.

[52] Frankel, S., Kent, K., Lewkowski, R., Orebaugh, A. D., Ritchey, R. W., & Sharma, S. R. (2005). *Guide to IPsec VPNs*. NIST Special Publication 800-77. <https://doi.org/10.6028/NIST.SP.800-77>

[53] Kent, S. (2005). *IP Encapsulating Security Payload (ESP)*. RFC 4303, IETF. <https://doi.org/10.17487/RFC4303>

- [54] Kaufman, C., Hoffman, P., Nir, Y., & Eronen, P. (2010). *Internet Key Exchange Protocol Version 2 (IKEv2)*. RFC 5996, IETF. <https://doi.org/10.17487/RFC5996>
- [55] Daemen, J., & Rijmen, V. (2002). *The Design of Rijndael: AES - The Advanced Encryption Standard*. Springer-Verlag.
- [56] Harkins, D., & Carrel, D. (1998). *The Internet Key Exchange (IKE)*. RFC 2409, IETF. <https://doi.org/10.17487/RFC2409>
- [57] National Institute of Standards and Technology. (2001). *Advanced Encryption Standard (AES)*. FIPS Publication 197. <https://doi.org/10.6028/NIST.FIPS.197>
- [58] Menezes, A., van Oorschot, P., & Vanstone, S. (1996). *Handbook of Applied Cryptography*. CRC Press.
- [59] Cisco Systems. (2022). *Site-to-Site VPN Configuration Guide*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/security-vpn/ipsec-negotiation-ike-protocols/14125-16.html>
- [60] Schneier, B. (2015). *Applied Cryptography: Protocols, Algorithms, and Source Code in C* (20th Anniversary Edition). John Wiley & Sons.
- [61] Cisco Systems. (2021). *Troubleshooting IPsec VPN Connections Using debug Commands*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/security-vpn/ipsec-negotiation-ike-protocols/5409-ipsec-debug-00.html>
- [62] Ferguson, N., & Schneier, B. (2003). *Practical Cryptography*. John Wiley & Sons.
- [63] European Union Agency for Cybersecurity (ENISA). (2020). *Handbook on Security of Personal Data Processing*. Retrieved from <https://www.enisa.europa.eu/>
- [64] Barrett, D. J., Silverman, R. E., & Byrnes, R. G. (2005). *SSH, The Secure Shell: The Definitive Guide* (2nd ed.). O'Reilly Media.
- [65] Ylonen, T., & Lonvick, C. (2006). *The Secure Shell (SSH) Authentication Protocol*. RFC 4252, IETF. <https://doi.org/10.17487/RFC4252>
- [66] Rivest, R. L., Shamir, A., & Adleman, L. (1978). A Method for Obtaining Digital Signatures and Public-Key Cryptosystems. *Communications of the ACM*, 21(2), 120-126. <https://doi.org/10.1145/359340.359342>
- [67] Cisco Systems. (2020). *Secure Shell Configuration Guide, Cisco IOS Release 15M&T*. Retrieved from https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_usr_ssh/configuration/15-mt/sec-usr-ssh-15-mt-book.html

- [68] NIST. (2017). *Guide to Enterprise Password Management*. NIST Draft Special Publication 800-63-3. <https://doi.org/10.6028/NIST.SP.800-63-3>
- [69] Cisco Systems. (2021). *Port Security Configuration Guide*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/lan-switching/port-security/10599-37.html>
- [70] Northcutt, S., & Novak, J. (2002). *Network Intrusion Detection* (3rd ed.). New Riders Publishing.
- [71] IEEE Standards Association. (2020). *IEEE Standard for Local and Metropolitan Area Networks--Port-Based Network Access Control*. IEEE 802.1X-2020. <https://doi.org/10.1109/IEEESTD.2020.9018454>
- [72] Convery, S. (2004). *Network Security Architectures*. Cisco Press.
- [73] Cisco Systems. (2019). *Cisco Catalyst Switch Security Best Practices*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/switches/catalyst-6500-series-switches/10600-3.html>
- [74] White, R., Slice, D., & Retana, A. (2002). *Optimal Routing Design*. Cisco Press.
- [75] Cisco Systems. (2022). *Multilayer Switching Feature Configuration Guide*. Retrieved from https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swipv6.html
- [76] Hucaby, D. (2015). *CCNP Routing and Switching SWITCH 300-115 Official Cert Guide*. Cisco Press.
- [77] Savage, D., Ng, J., Moore, S., Slice, D., Paluch, P., & White, R. (2016). *Cisco's Enhanced Interior Gateway Routing Protocol (EIGRP)*. RFC 7868, IETF. <https://doi.org/10.17487/RFC7868>
- [78] White, R., & Slice, D. (2002). *EIGRP Network Design Solutions*. Cisco Press.
- [79] Halabi, B., & McPherson, D. (2000). *Internet Routing Architectures* (2nd ed.). Cisco Press.
- [80] Cisco Systems. (2021). *Enhanced Interior Gateway Routing Protocol Configuration Guide, Cisco IOS XE Release 3S*. Retrieved from https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_eigrp/configuration/xs-3s/ire-xe-3s-book.html
- [81] Doyle, J., & Carroll, J. (2005). *Routing TCP/IP, Volume II: CCIE Professional Development*. Cisco Press.
- [82] Marcus, E., & Stern, H. (2003). *Blueprints for High Availability* (2nd ed.). John Wiley &

Sons.

[83] Cisco Systems. (2020). *EtherChannel Configuration Guide*. Retrieved from https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst3560/software/release/12-2_55_se/configuration/guide/3560_scg/swethchl.html

[84] Li, T., Cole, B., Morton, P., & Li, D. (1998). *Cisco Hot Standby Router Protocol (HSRP)*. RFC 2281, IETF. <https://doi.org/10.17487/RFC2281>

[85] Cisco Systems. (2021). *First Hop Redundancy Protocols Configuration Guide, Cisco IOS XE Release 3S*. Retrieved from https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipapp_fhrp/configuration/xs-3s/fhrp-xe-3s-book.html

[86] IEEE Standards Association. (2004). *IEEE Standard for Local and Metropolitan Area Networks: Media Access Control (MAC) Bridges*. IEEE 802.1D-2004. <https://doi.org/10.1109/IEEESTD.2004.94569>

[87] Patterson, D. A., Gibson, G., & Katz, R. H. (1988). A Case for Redundant Arrays of Inexpensive Disks (RAID). *Proceedings of the 1988 ACM SIGMOD International Conference on Management of Data*, 109-116. <https://doi.org/10.1145/50202.50214>

[88] Wallace, M., & Webber, L. (2011). **The Disaster Recovery Handbook: A Step-by-Step Plan to Ensure Business Continuity and Protect*

[89] Cisco Systems. (2019). *Network Testing and Verification Best Practices*. Cisco Technical Documentation. Retrieved from <https://www.cisco.com/c/en/us/support/docs/ip/ip-routing/13786-110.html>

[90] Mills, D. L. (2010). *Computer Network Time Synchronization: The Network Time Protocol on Earth and in Space* (2nd ed.). CRC Press.

[91] Cisco Systems. (2020). *Understanding and Troubleshooting Spanning Tree Protocol (STP)*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/lan-switching/spanning-tree-protocol/10556-16.html>

[92] Bonica, R., Gan, D., Tappan, D., & Pignataro, C. (2011). *Generic Routing Encapsulation (GRE)*. RFC 6245, IETF. <https://doi.org/10.17487/RFC6245>

[93] Cisco Systems. (2022). *Troubleshooting IPsec and ISAKMP Using debug Commands*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/security-vpn/ipsec-negotiation-ike-protocols/5409-ipsec-debug-00.html>

[94] Droms, R., & Lemon, T. (2003). *The DHCP Handbook: Understanding, Deploying, and Managing Automated Configuration Services* (2nd ed.). Sams Publishing.

- [95] Cisco Systems. (2021). *Troubleshooting IP Access Lists*. Retrieved from <https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/23602-confaccesslists.html>
- [96] Cisco Systems. (2020). *Campus Network for High Availability Design Guide*. Cisco Validated Design. Retrieved from <https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/CVD-Campus-LAN-WLAN-Design-Guide.html>
- [97] Fuller, V., & Li, T. (2006). *Classless Inter-domain Routing (CIDR): The Internet Address Assignment and Aggregation Plan*. RFC 4632, IETF. <https://doi.org/10.17487/RFC4632>
- [98] Burgess, M. (2000). *Principles of Network and System Administration*. John Wiley & Sons.
- [99] Cisco Systems. (2019). *Troubleshooting Methodology for Cisco Networks*. Cisco Technical Documentation. Retrieved from <https://www.cisco.com/c/en/us/support/docs/dial-access/asynchronous-connections/10374-16.html>
- [100] National Security Agency. (2011). *Defense in Depth: A Practical Strategy for Achieving Information Assurance in Today's Highly Networked Environments*. NSA Information Assurance Directorate. Retrieved from <https://www.nsa.gov/>
- [101] Aboba, B., Blunk, L., Vollbrecht, J., Carlson, J., & Levkowitz, H. (2004). *Extensible Authentication Protocol (EAP)*. RFC 3748, IETF. <https://doi.org/10.17487/RFC3748>
- [102] IEEE Standards Association. (2020). *IEEE Standard for Information Technology--Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks--Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*. IEEE 802.11-2020. <https://doi.org/10.1109/IEEESTD.2021.9363693>
- [103] Cisco Systems. (2021). *Flexible NetFlow Configuration Guide, Cisco IOS Release 15M&T*. Retrieved from <https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/fnetflow/configuration/15-mt/fnf-15-mt-book.html>
- [104] Cisco Systems. (2022). *Identity Services Engine Administrator Guide, Release 3.1*. Retrieved from https://www.cisco.com/c/en/us/td/docs/security/ise/3-1/admin_guide/b_ise_admin_3_1.html
- [105] Cisco Systems. (2023). *SD-WAN Design Guide*. Cisco Validated Design Documentation. Retrieved from <https://www.cisco.com/c/en/us/td/docs/solutions/CVD/SDWAN/cisco-sdwan-design-guide.html>
- [106] Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). *Zero Trust Architecture*. NIST Special Publication 800-207. <https://doi.org/10.6028/NIST.SP.800-207>

- [107] Mell, P., & Grance, T. (2011). *The NIST Definition of Cloud Computing*. NIST Special Publication 800-145. <https://doi.org/10.6028/NIST.SP.800-145>
- [108] International Organization for Standardization. (2022). *Information Security, Cybersecurity and Privacy Protection — Information security management systems — Requirements*. ISO/IEC 27001:2022. Retrieved from <https://www.iso.org/standard/27001>
- [109] IBM Security and Ponemon Institute. (2023). *Cost of a Data Breach Report 2023*. Retrieved from <https://www.ibm.com/security/data-breach>
- [110] Tanenbaum, A. S., & Wetherall, D. J. (2021). *Computer Networks* (6th ed.). Pearson Education.
- [111] Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson Education.
- [112] Cisco Systems. (2023). *Cisco Networking Academy: CCNA Curriculum*. Retrieved from <https://www.netacad.com/>
- [113] CompTIA. (2022). *CompTIA Network+ Certification Exam Objectives (N10-008)*. Retrieved from <https://www.comptia.org/certifications/network>
- [114] Comer, D. E. (2018). *Computer Networks and Internets* (6th ed.). Pearson Education.
- [115] Peterson, L. L., & Davie, B. S. (2021). *Computer Networks: A Systems Approach* (6th ed.). Morgan Kaufmann.